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(54) **DISTAL RADIUS STABILIZATION SYSTEM**

(71) Applicant: **GLOBUS MEDICAL, INC.**, Audubon, PA (US)

(72) Inventors: **Gabrielle Zingalis**, Philadelphia, PA (US); **Peter Evans**, Lafayette Hill, PA (US); **Evan Langdale**, Philadelphia, PA (US); **Joseph DePaul**, Lansdale, PA (US); **Barclay Davis**, Glenmoore, PA (US); **Andrew Davison**, Downingtown, PA (US); **Michelle Gray**, Oley, PA (US); **David Laird, Jr.**, Brandamore, PA (US); **Stephanie Wolfe**, Perkasie, PA (US)

(73) Assignee: **Globus Medical, Inc.**, Audubon, PA (US)

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11,141,204, which is a continuation-in-part of application No. 15/456,642, filed on Mar. 13, 2017, (Continued)

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(52) **U.S. Cl.**

CPC **A61B 17/8061** (2013.01); **A61B 17/1615** (2013.01); **A61B 17/1782** (2016.11)

(58) **Field of Classification Search**

CPC **A61B 17/1728**; **A61B 17/1615–1617**
See application file for complete search history.

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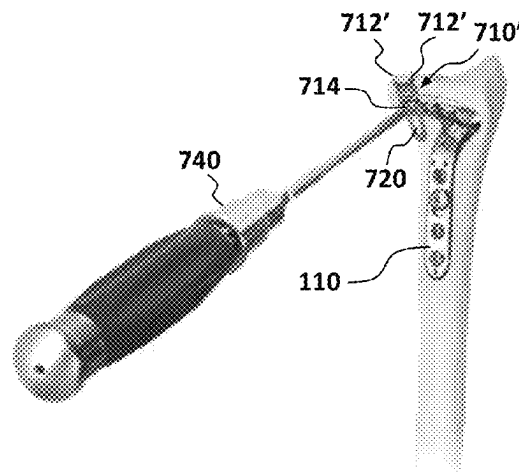
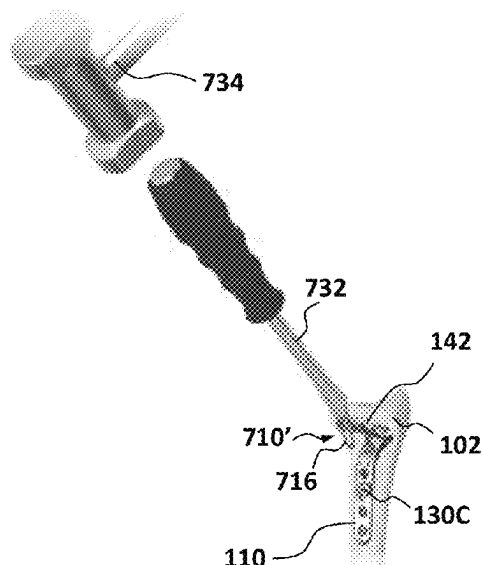
Primary Examiner — Olivia C Chang

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ABSTRACT

Devices, systems, and methods for bone stabilization, especially ulna head stabilization. The stabilization system may include a bone plate having an elongated portion extending along a longitudinal axis between a proximal end and a distal end. The bone plate defines a plurality of through holes extending through the elongated portion. A plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone. The proximal end of the elongate portion has an arcuate configuration.

15 Claims, 39 Drawing Sheets



Related U.S. Application Data

now Pat. No. 10,420,596, which is a continuation-in-part of application No. 15/238,773, filed on Aug. 17, 2016, now Pat. No. 10,383,668.

- (60) Provisional application No. 62/554,700, filed on Sep. 6, 2017.

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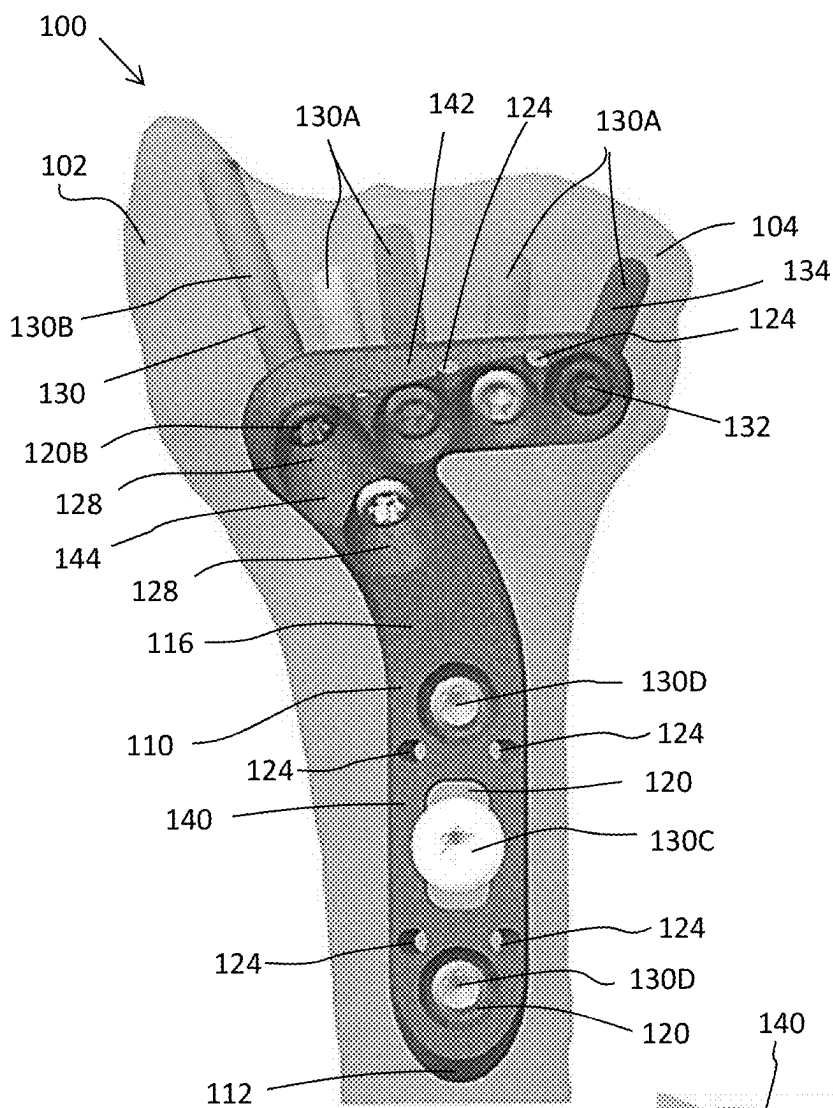


FIG. 1A

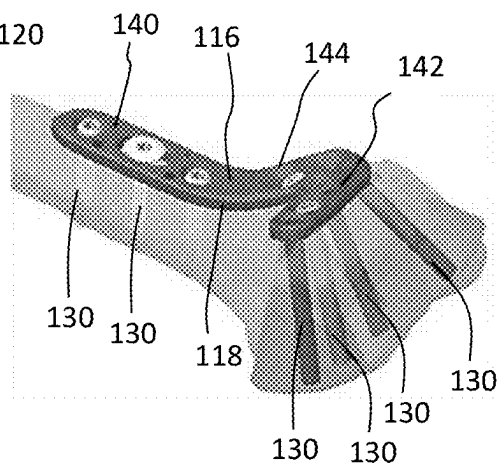


FIG. 1B

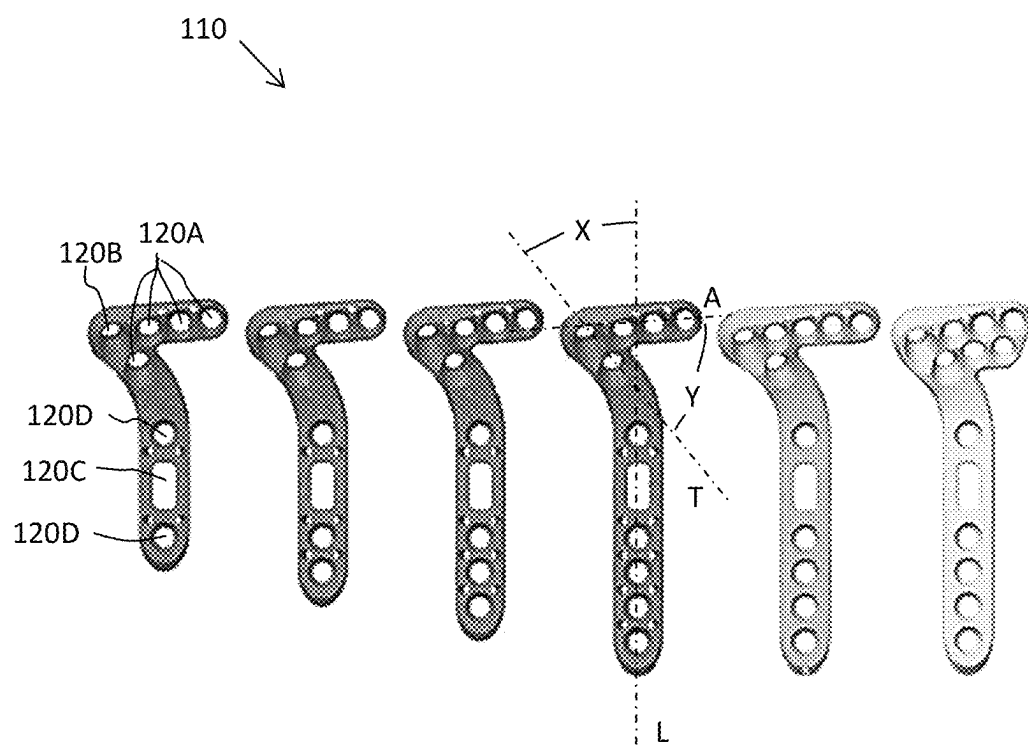


FIG. 1C

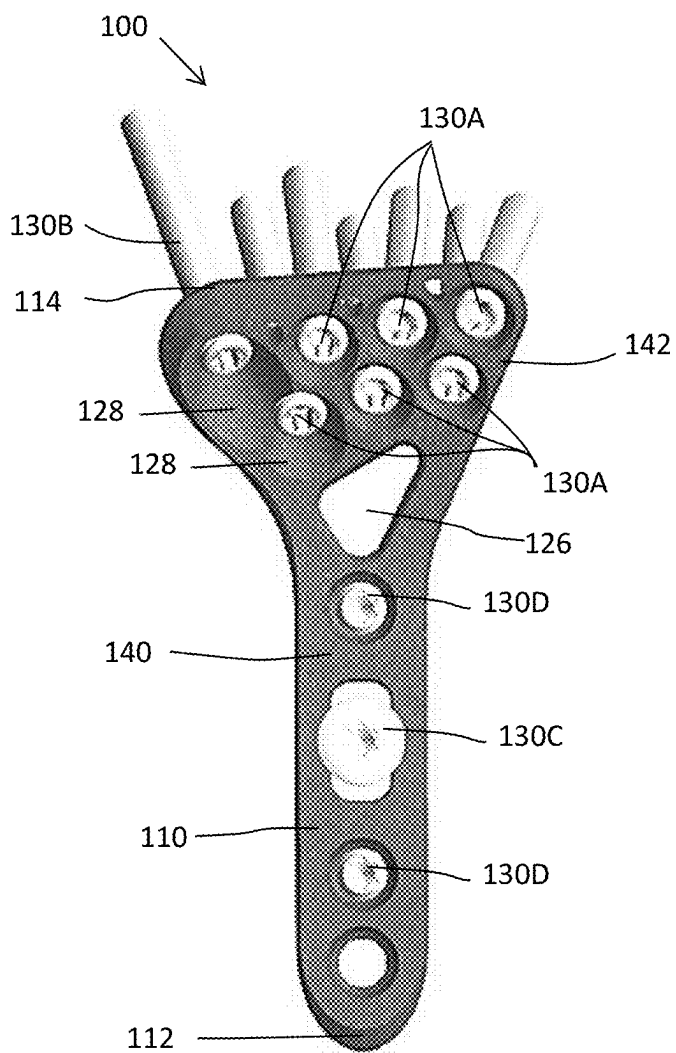


FIG. 1D

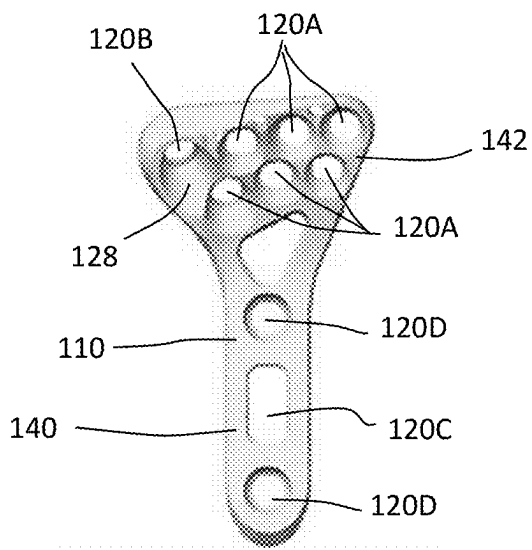


FIG. 1E

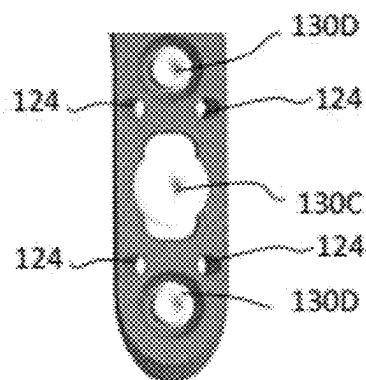


FIG. 1F

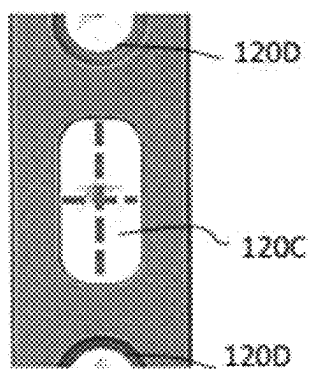


FIG. 1G

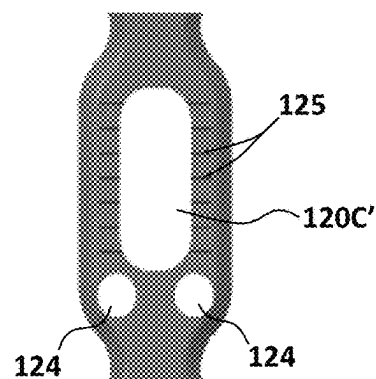


FIG. 1H

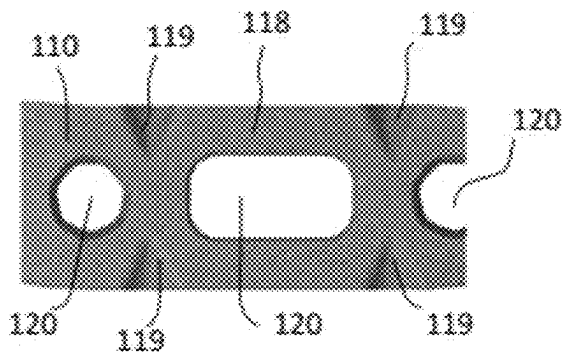


FIG. 1J

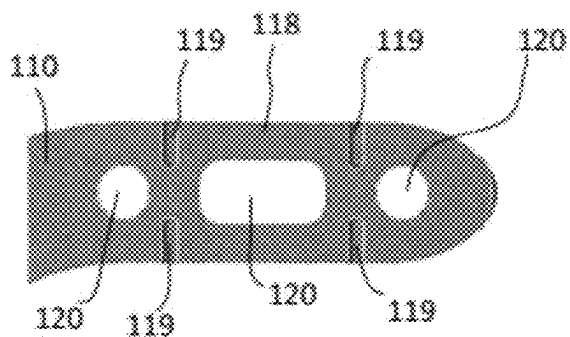


FIG. 1I

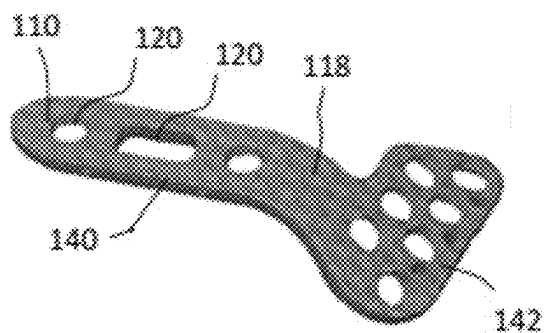


FIG. 1K

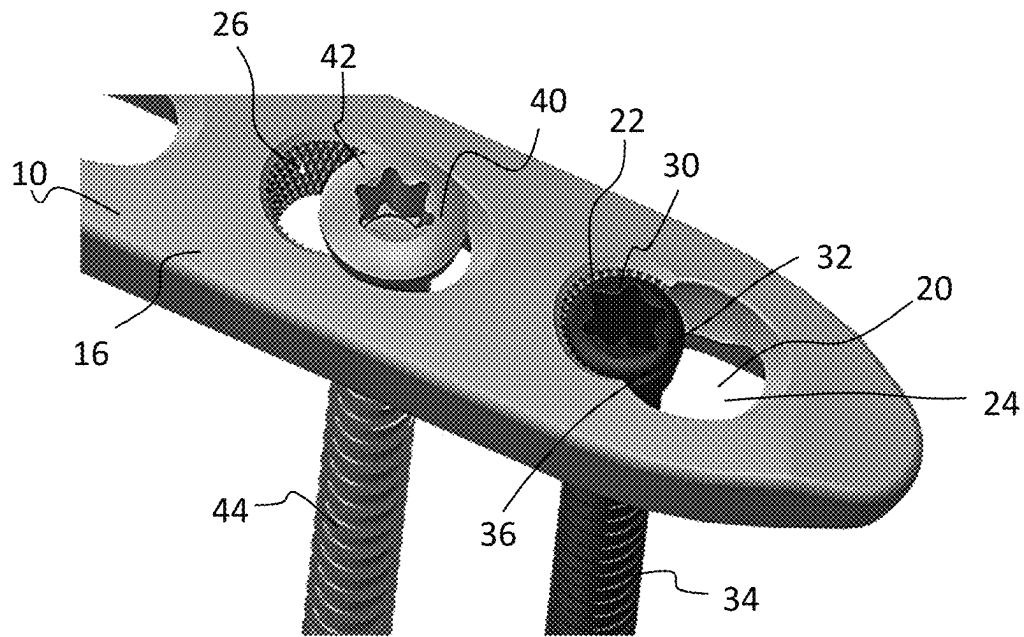


FIG. 2

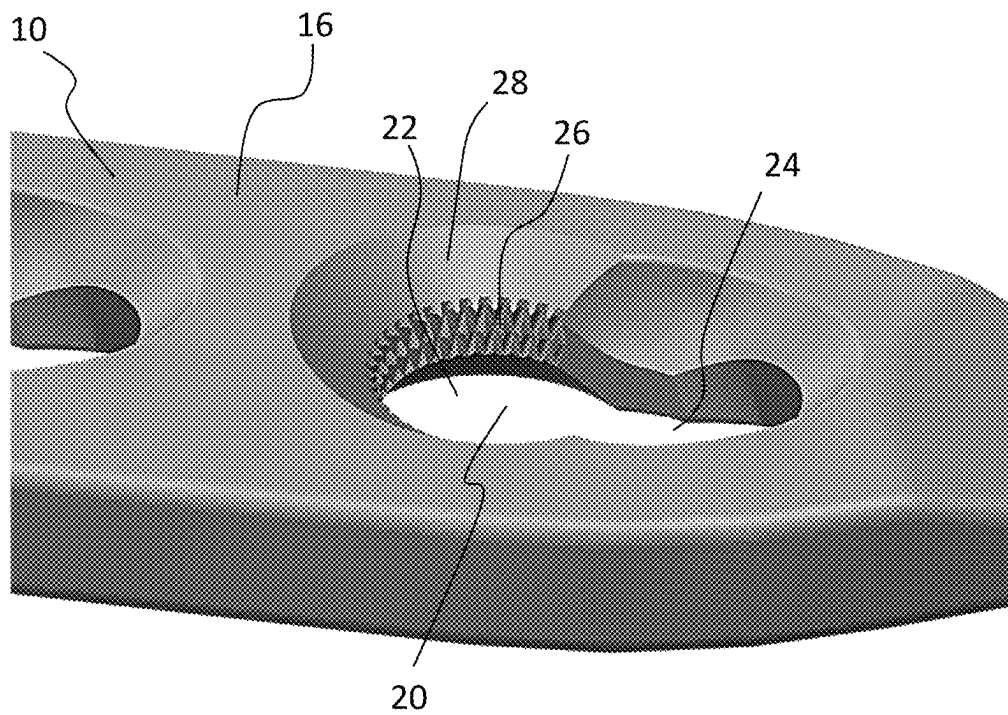


FIG. 3

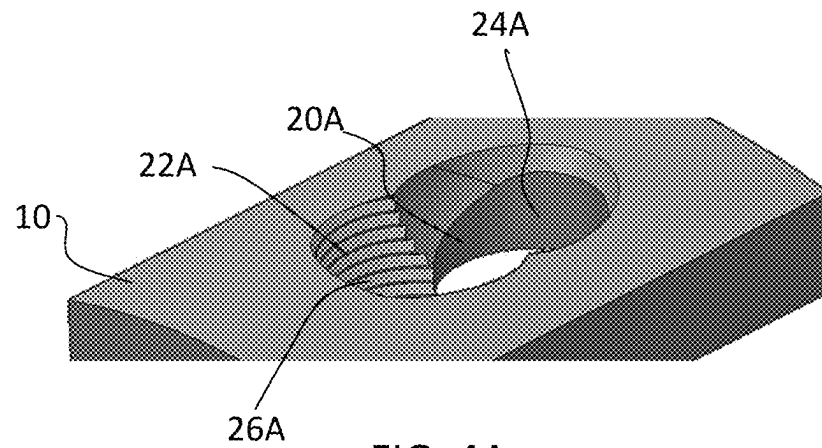


FIG. 4A

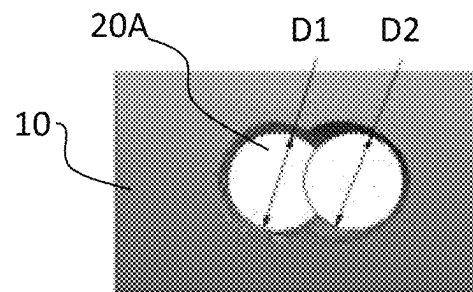


FIG. 4B

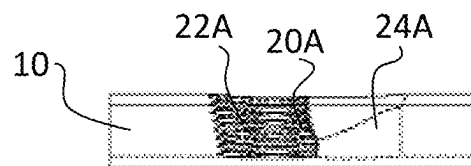


FIG. 4C

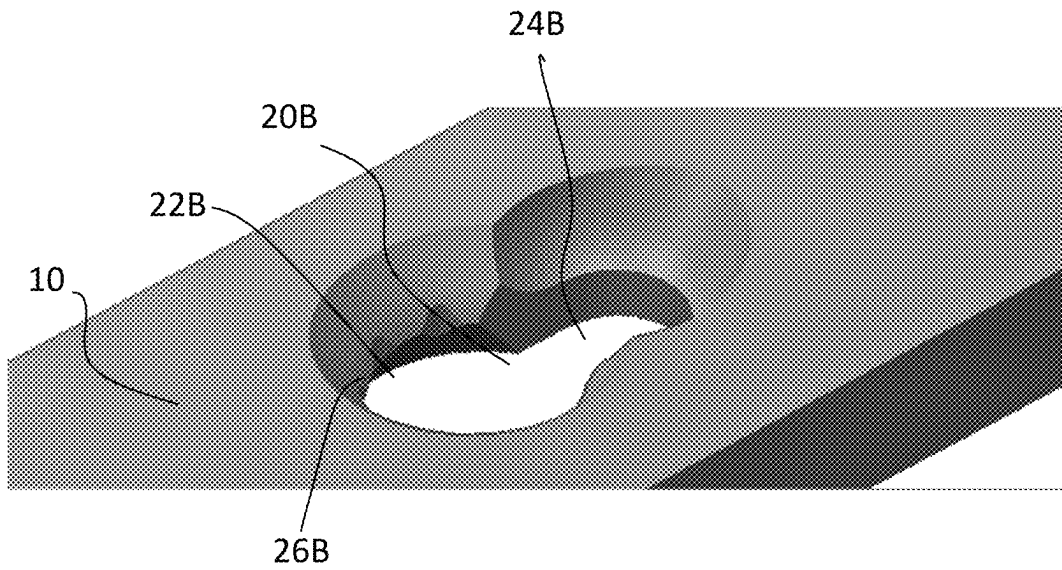


FIG. 5A

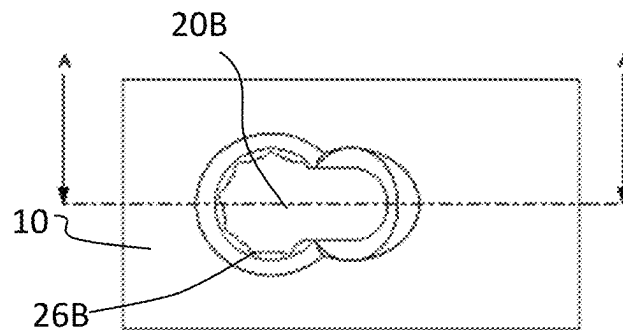


FIG. 5B

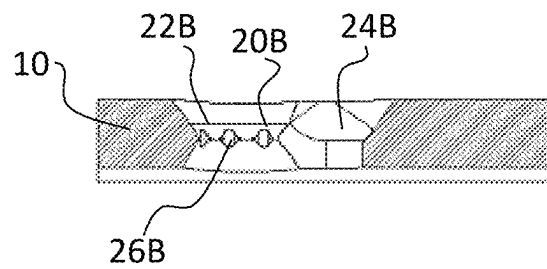


FIG. 5C

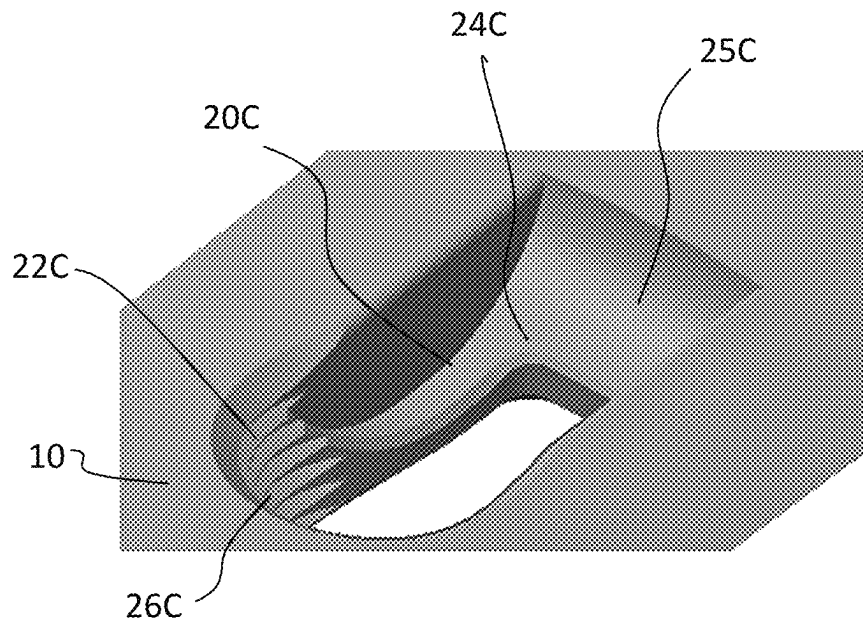


FIG. 6A

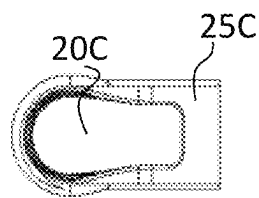


FIG. 6B

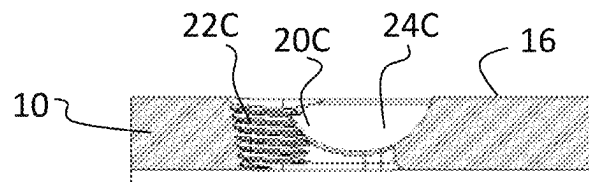


FIG. 6C

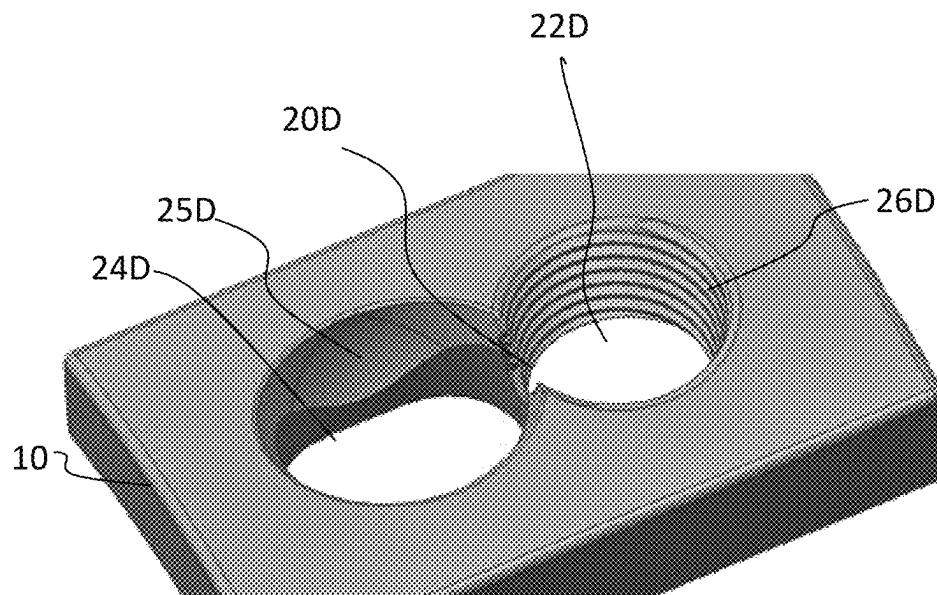


FIG. 7A

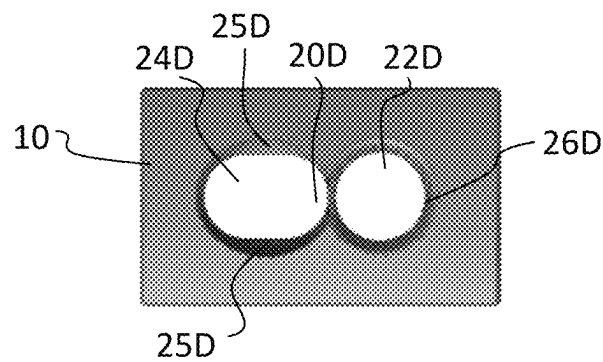


FIG. 7B

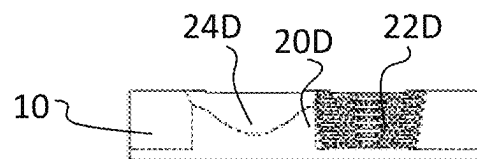


FIG. 7C

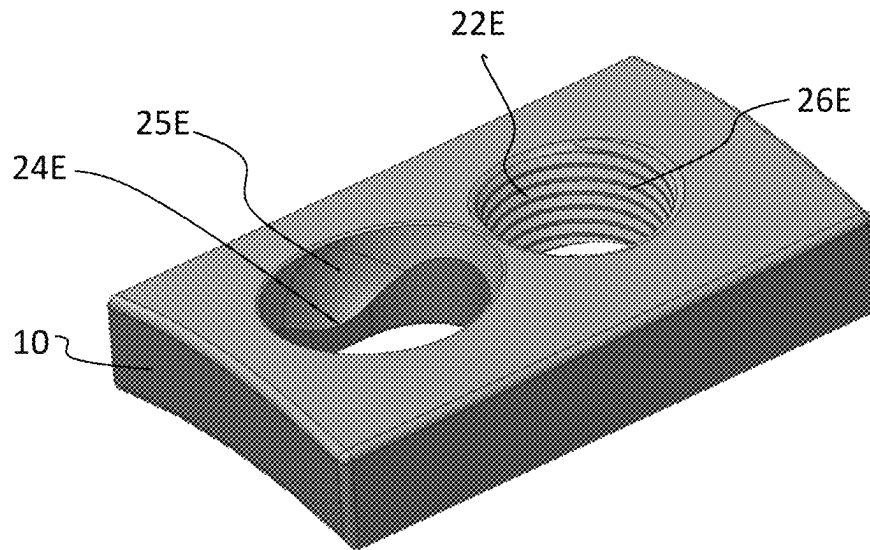


FIG. 8A

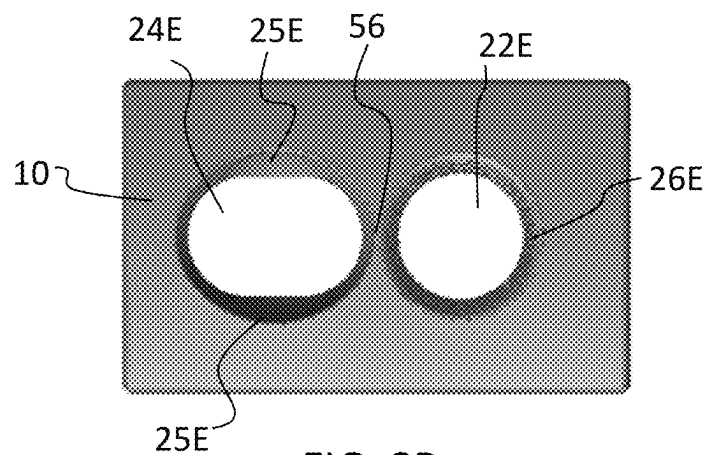


FIG. 8B

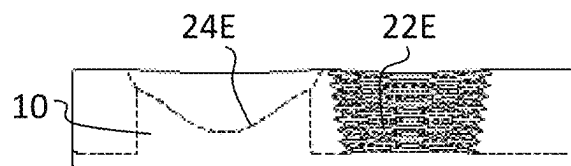


FIG. 8C

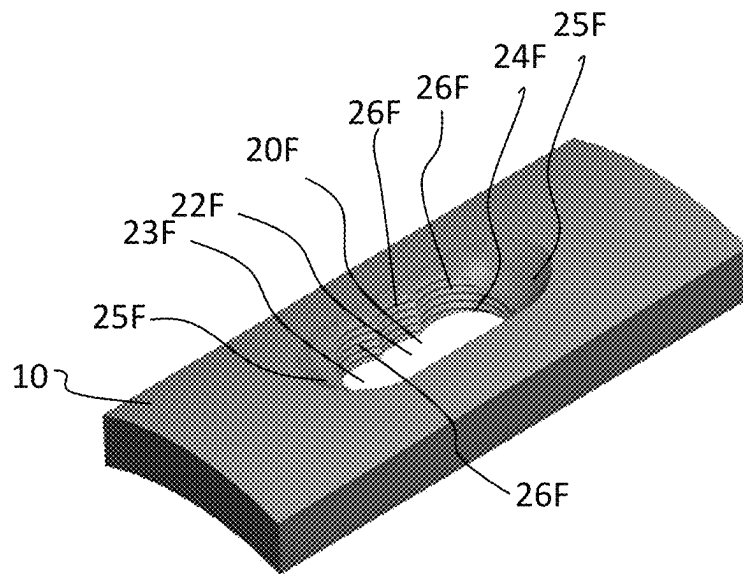


FIG. 9A

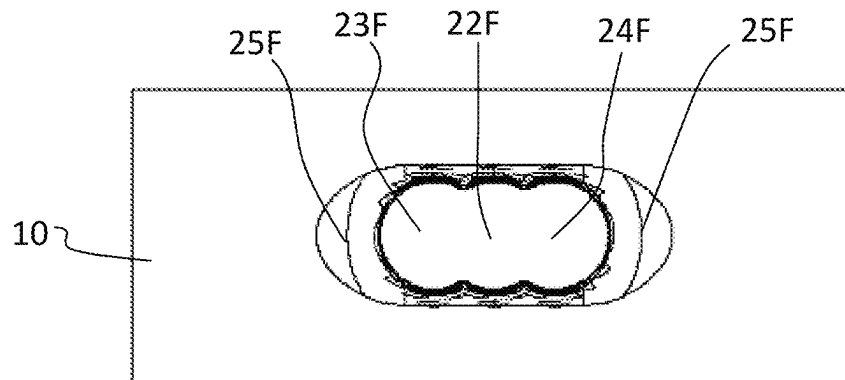


FIG. 9B

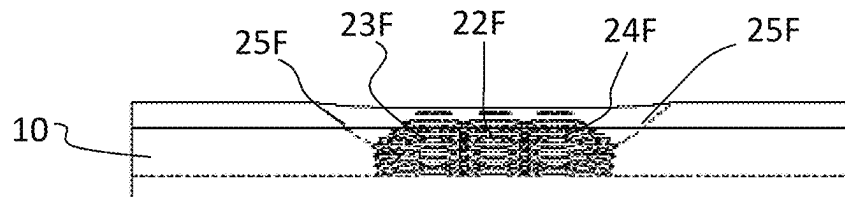


FIG. 9C

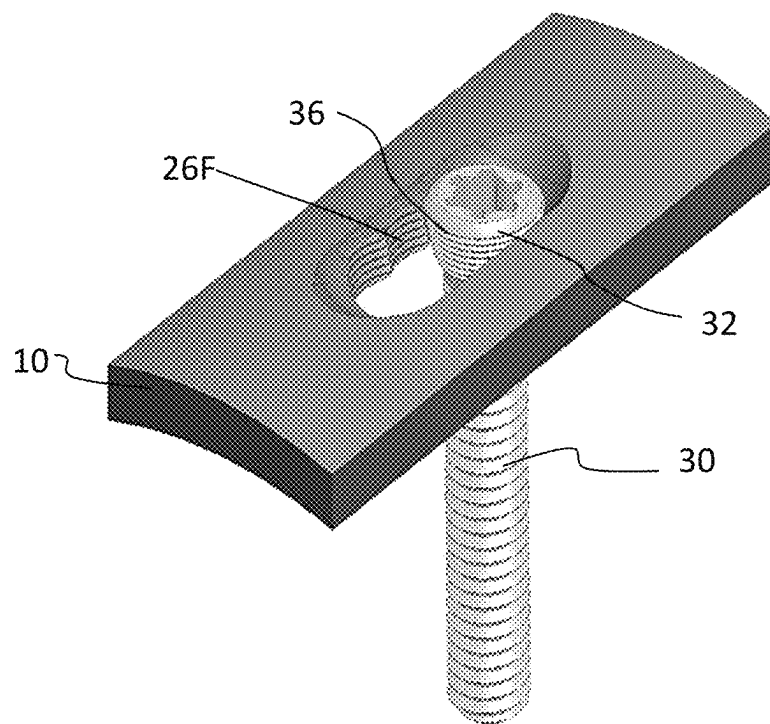


FIG. 9D

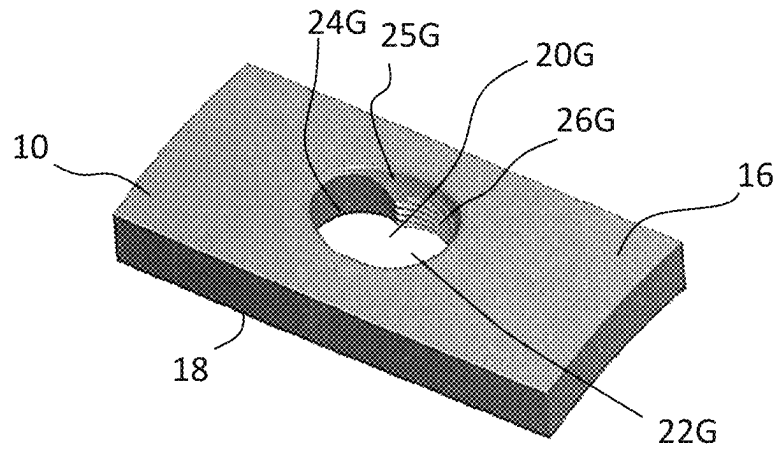


FIG. 10A

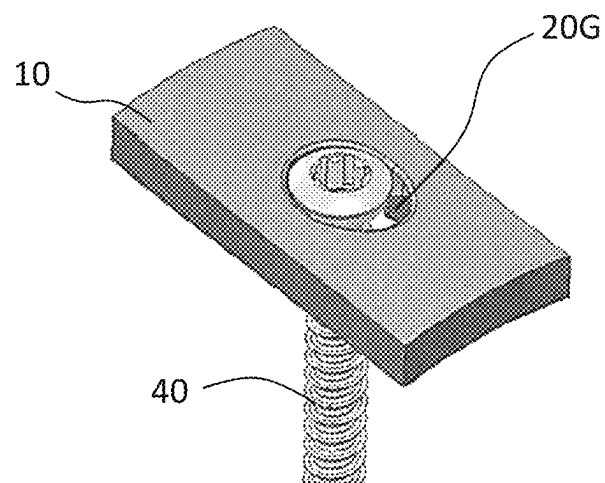


FIG. 10B

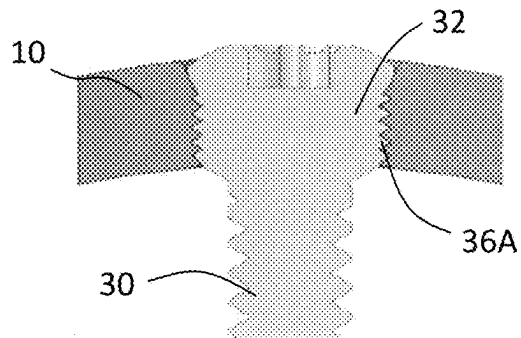


FIG. 11A

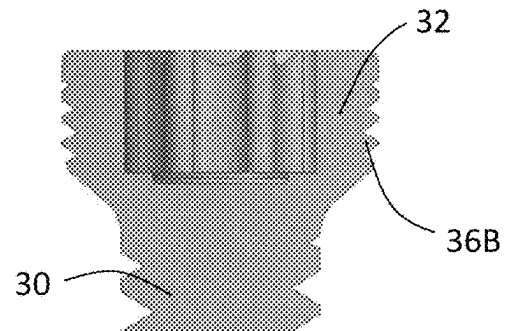


FIG. 11B

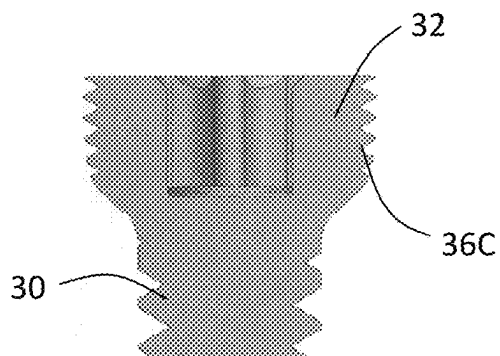


FIG. 11C

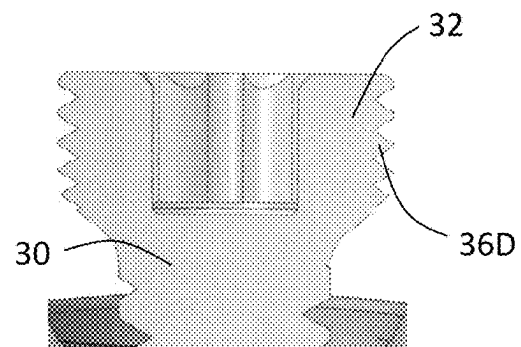


FIG. 11D

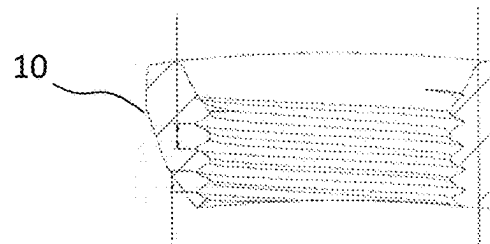


FIG. 11E

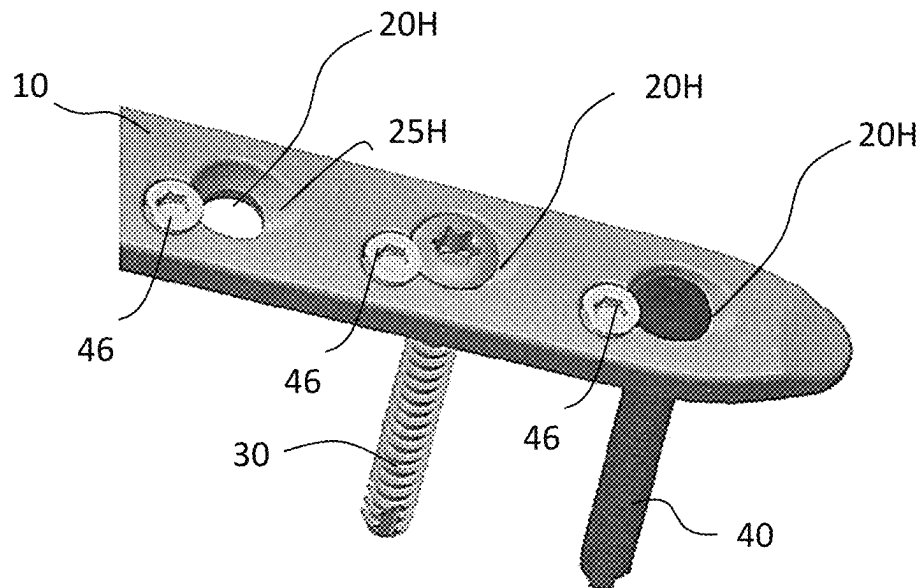


FIG. 12A

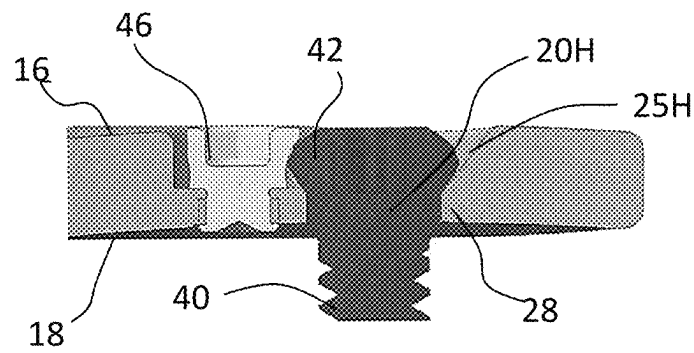


FIG. 12B

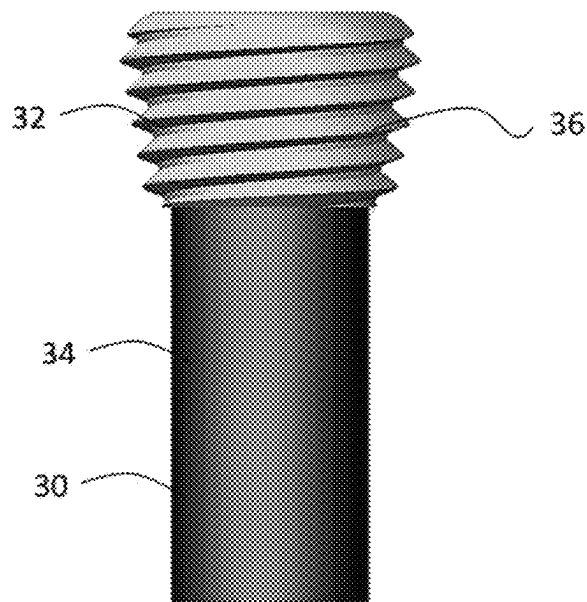


FIG. 13A

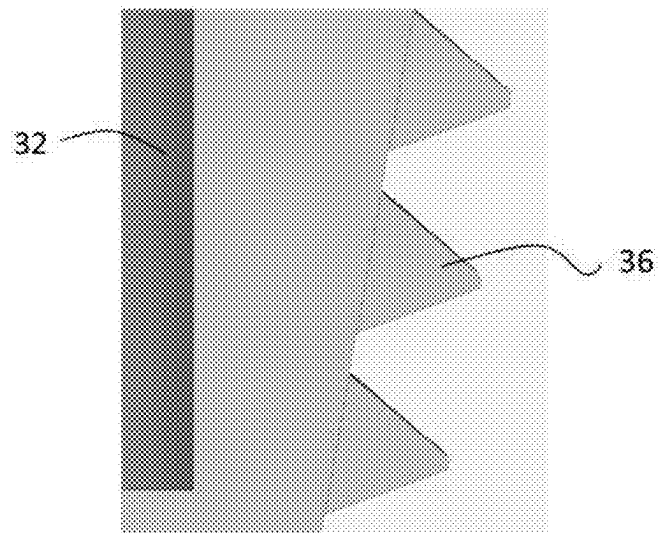


FIG. 13B

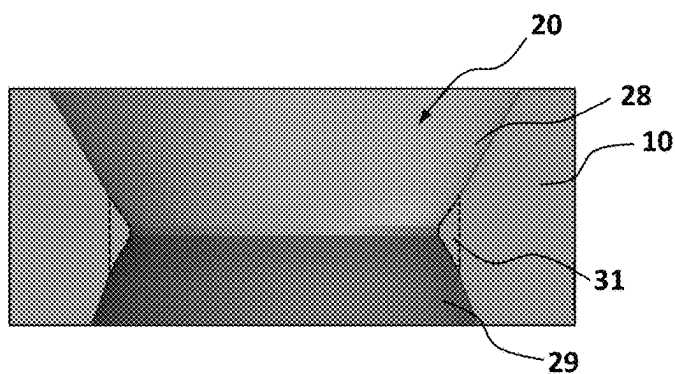


FIG. 13C

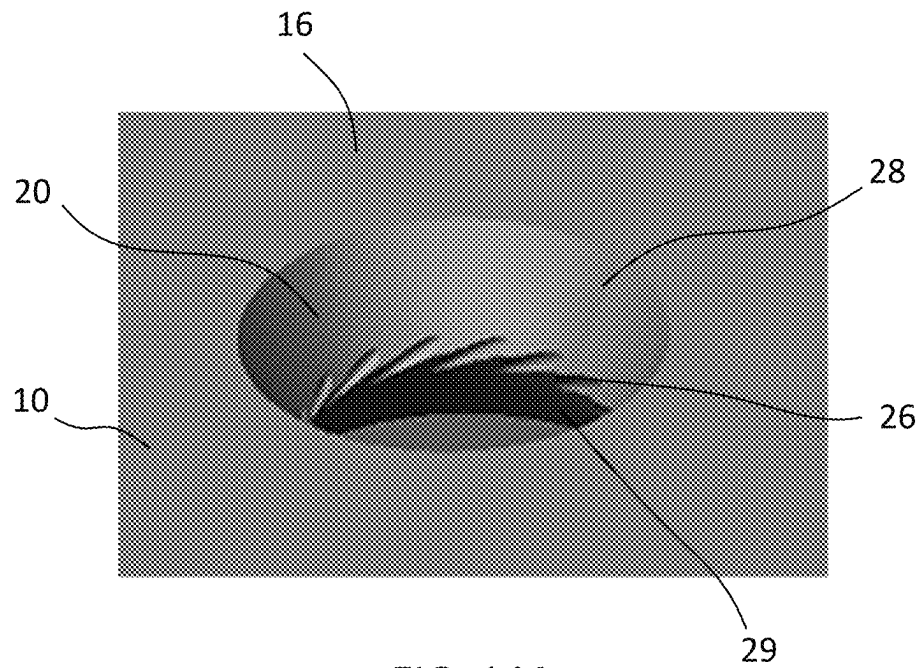


FIG. 14A

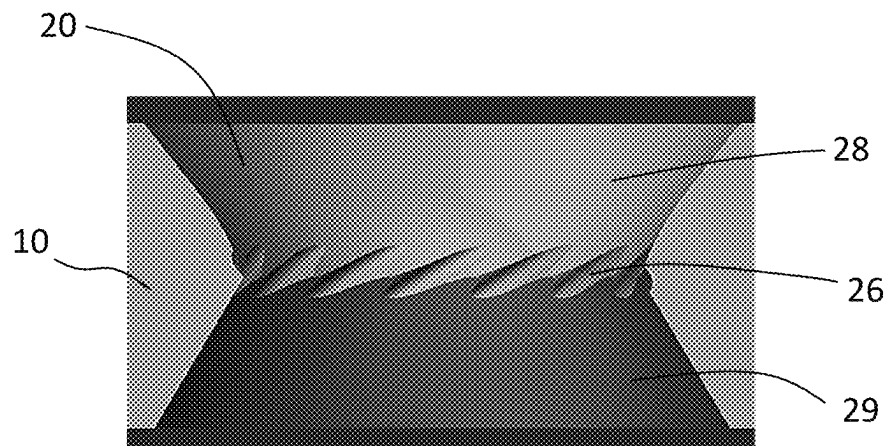


FIG. 14B

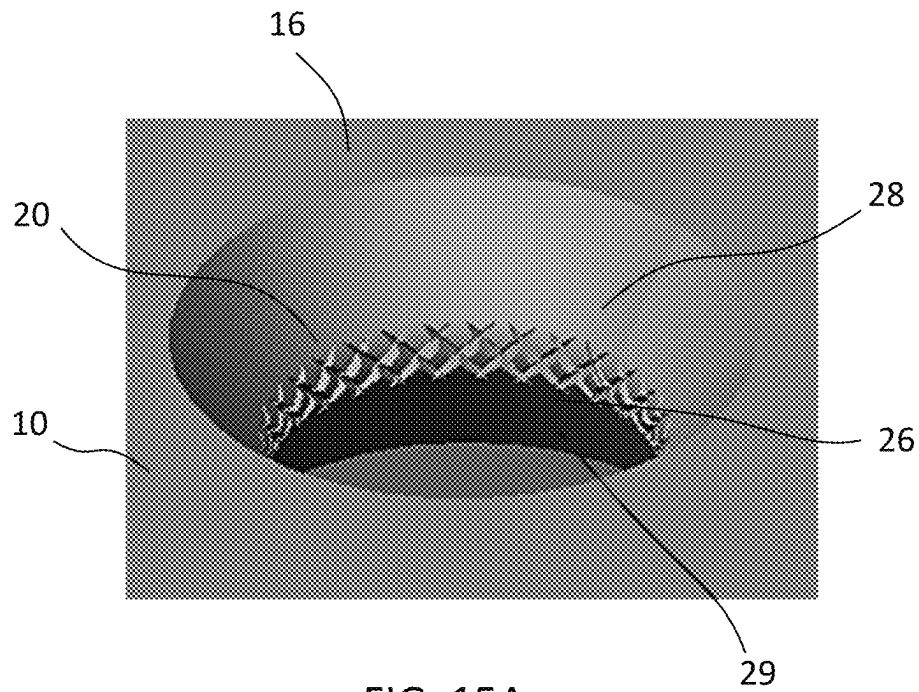


FIG. 15A

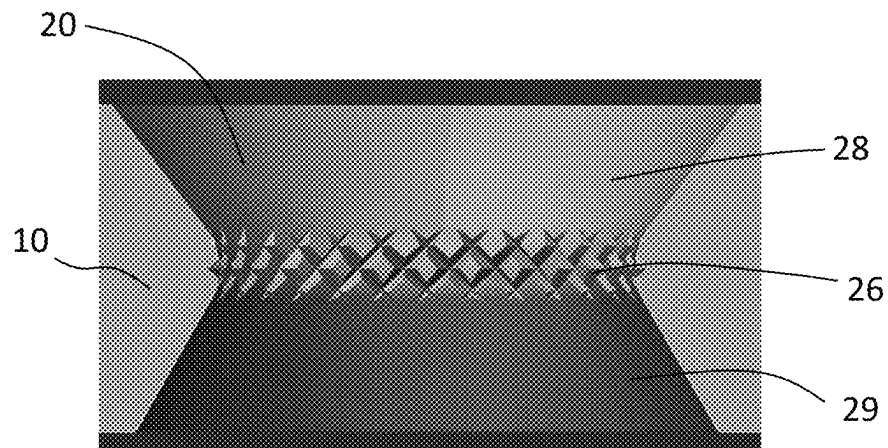


FIG. 15B

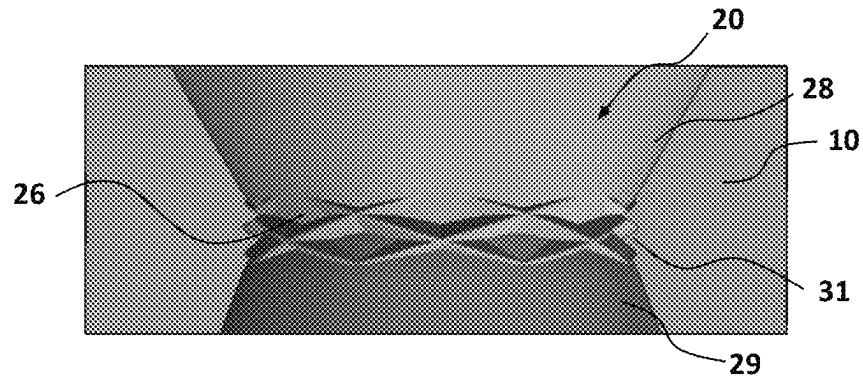


FIG. 15C

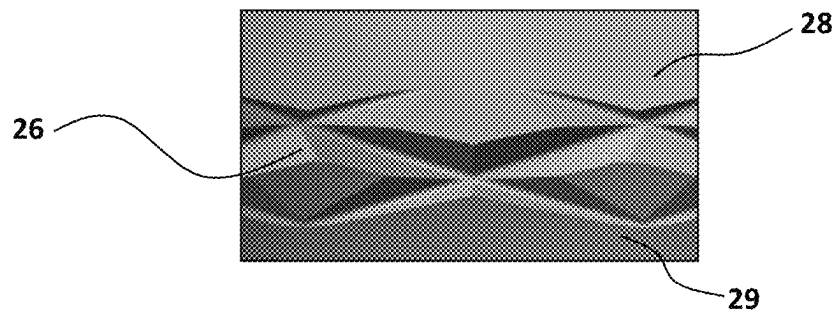


FIG. 15D

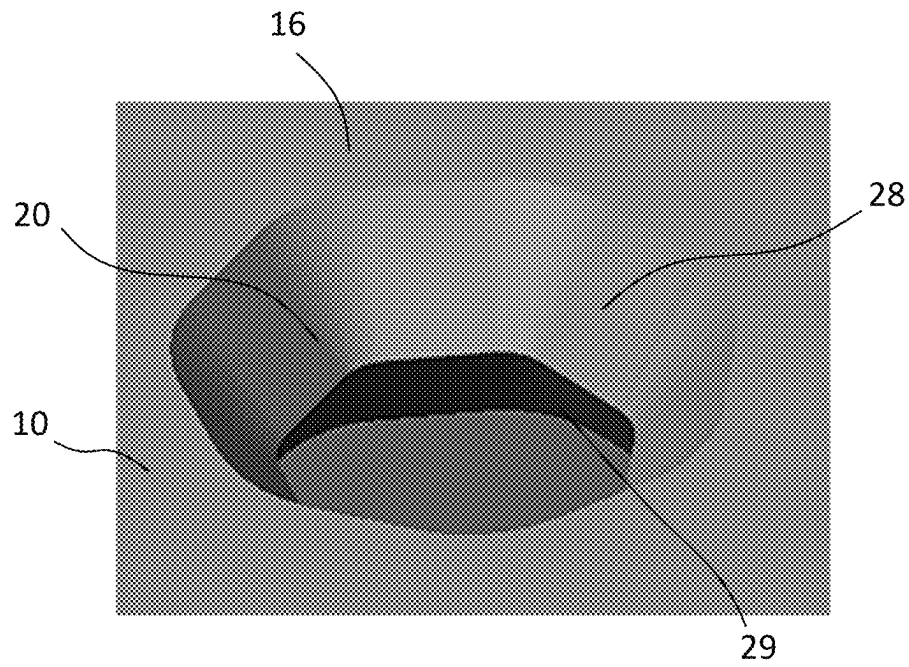


FIG. 16A

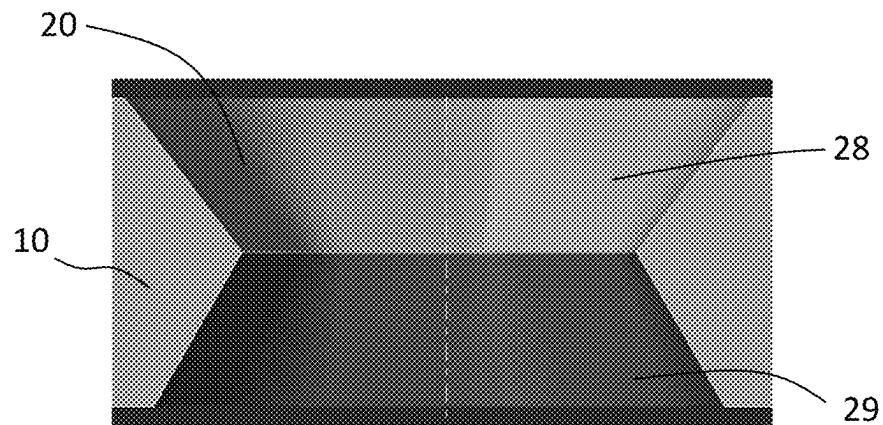


FIG. 16B

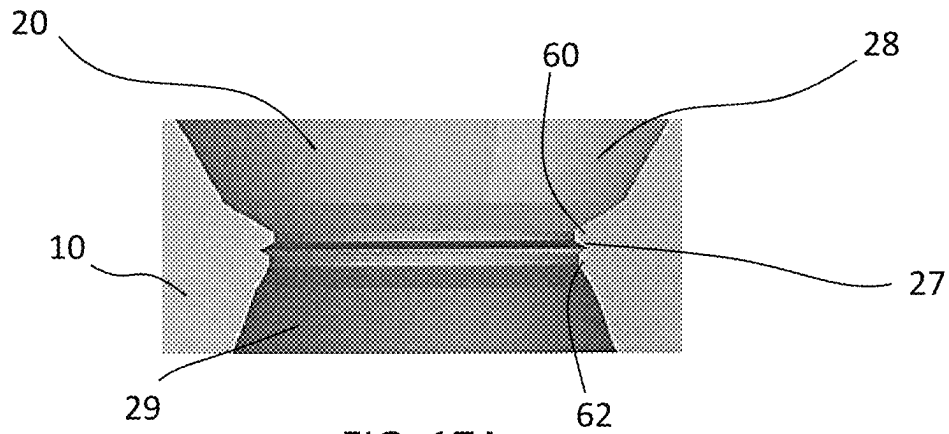


FIG. 17A

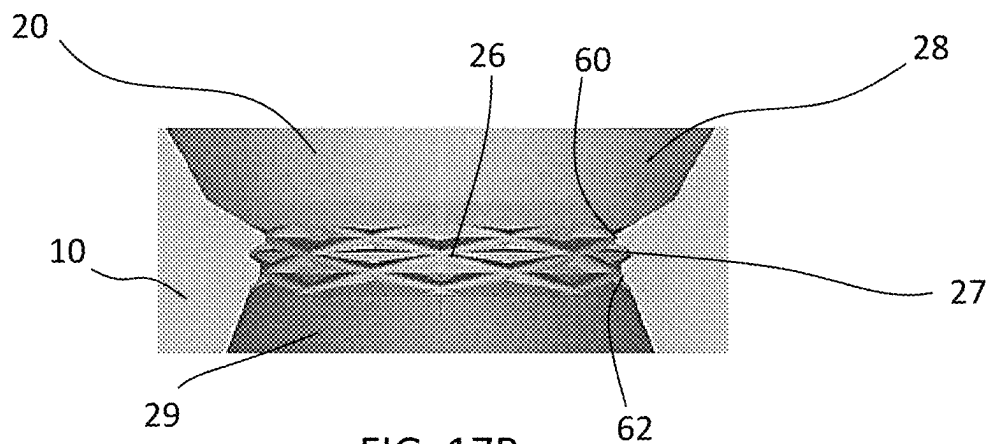


FIG. 17B

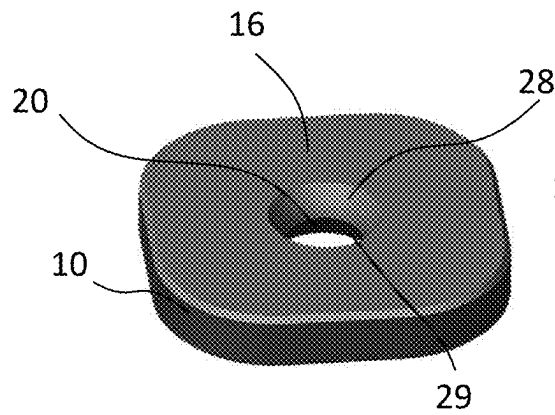


FIG. 18A

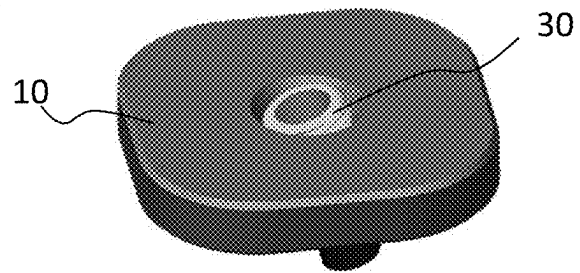


FIG. 18B

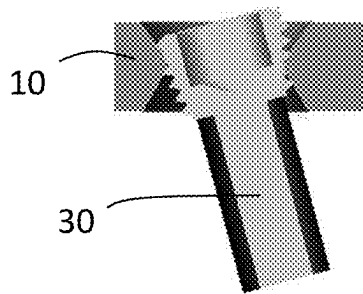


FIG. 18C

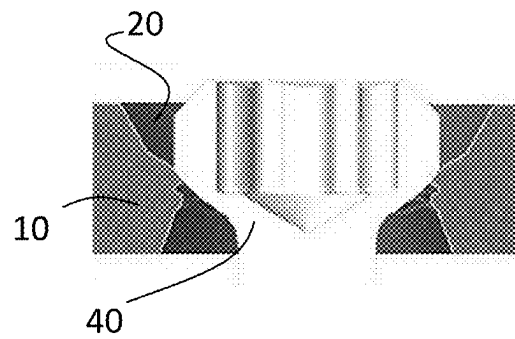


FIG. 18D

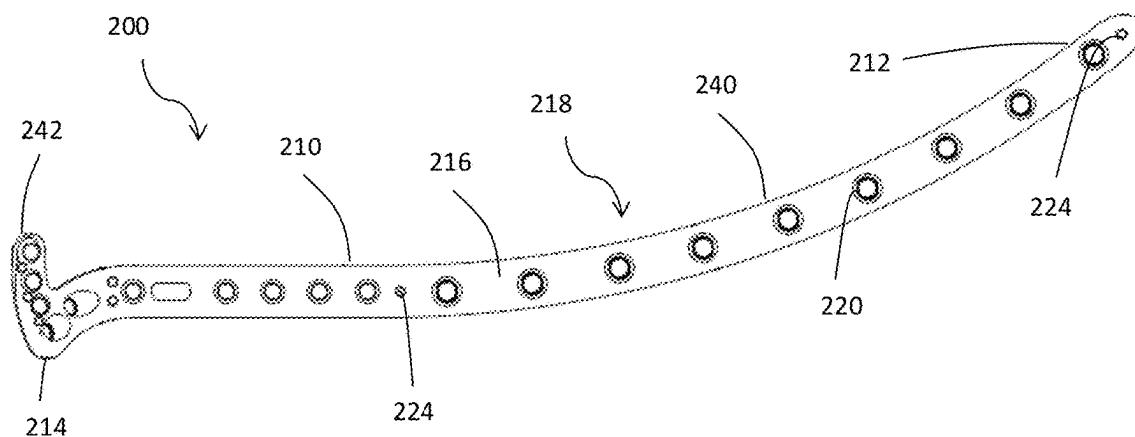


FIG. 19

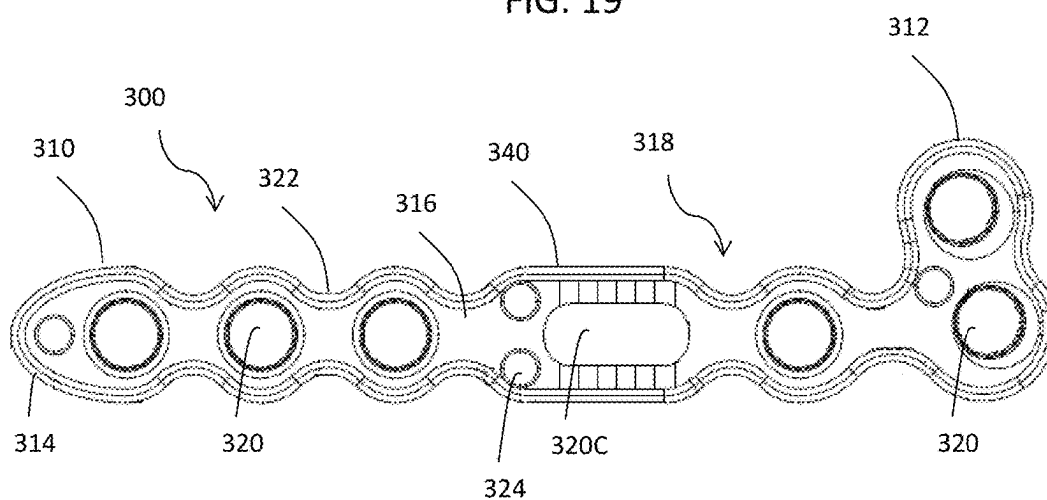


FIG. 20

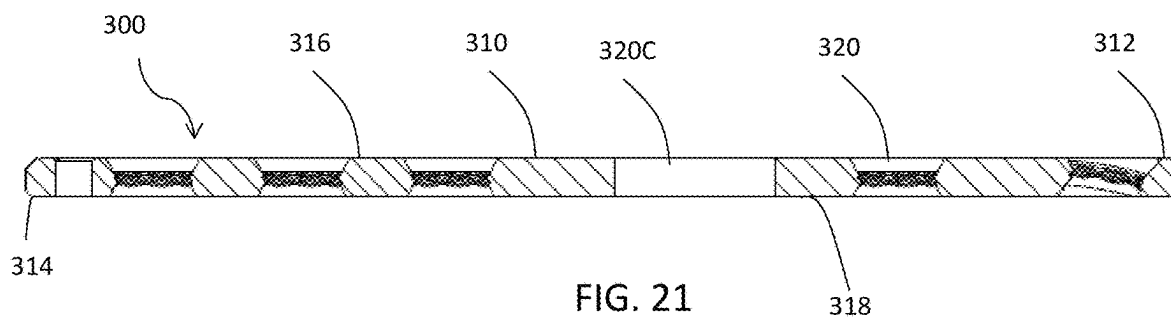


FIG. 21

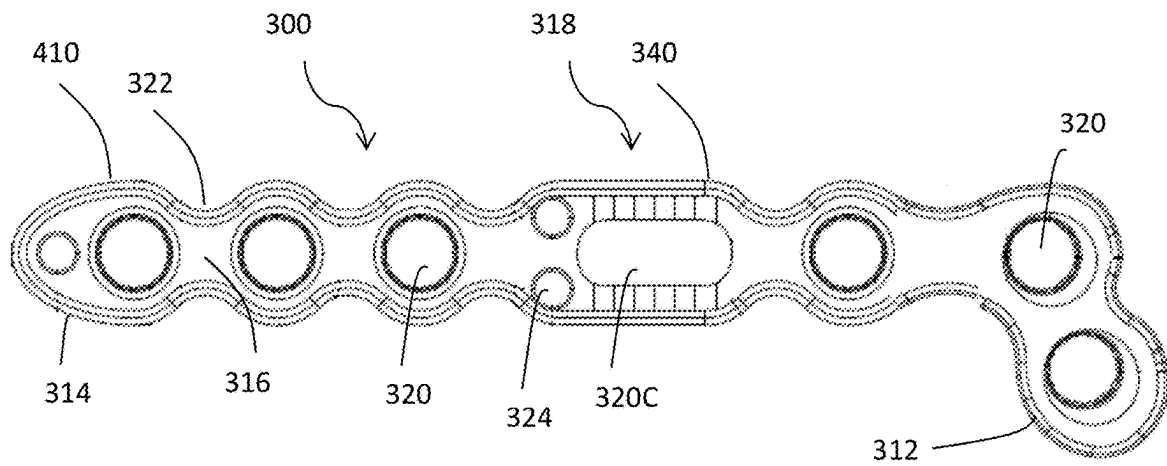


FIG. 22

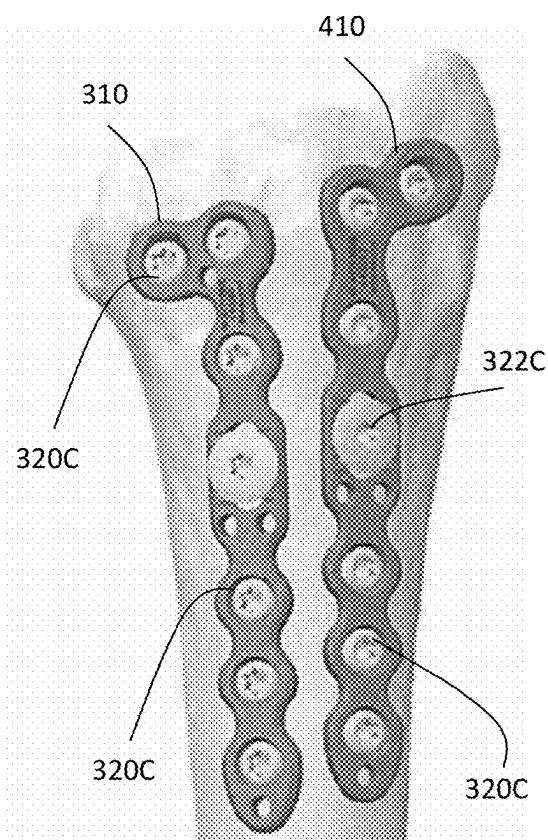


FIG. 23

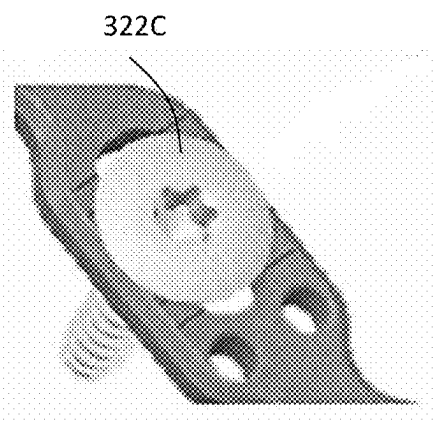


FIG. 24

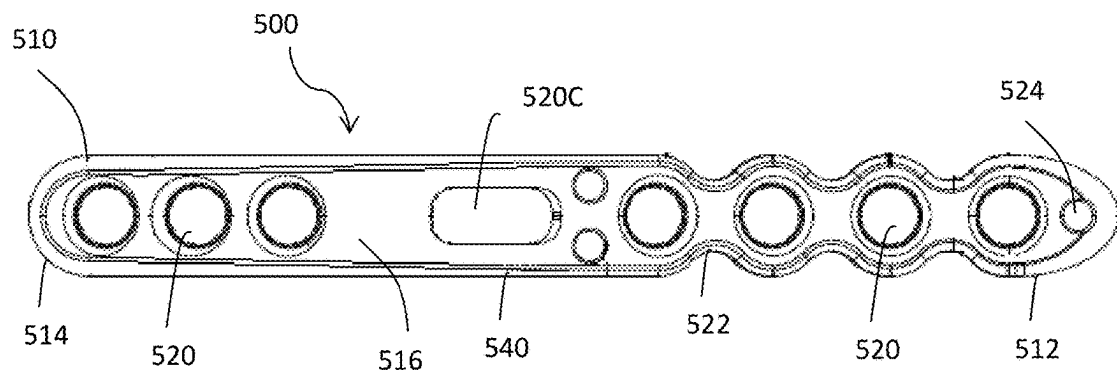


FIG. 25

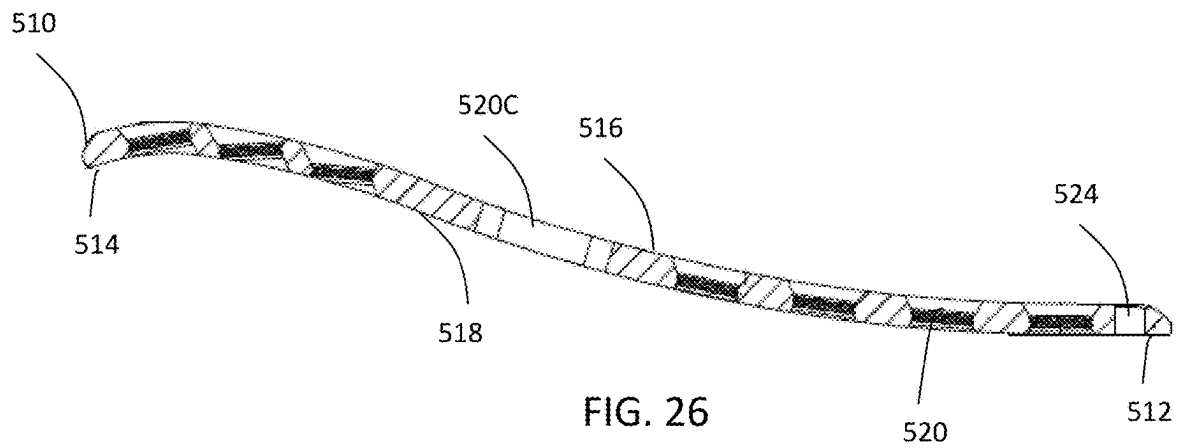


FIG. 26

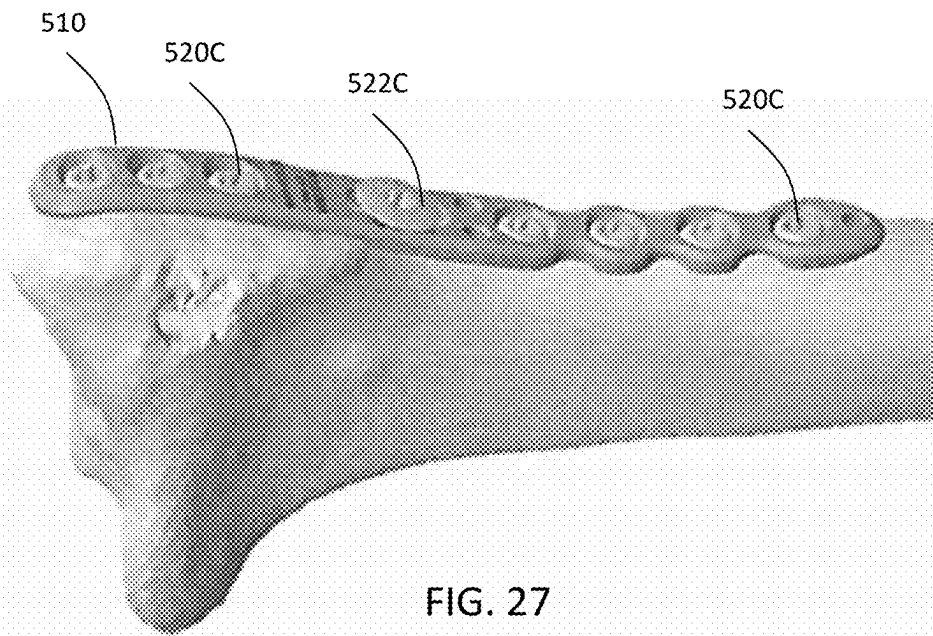


FIG. 27

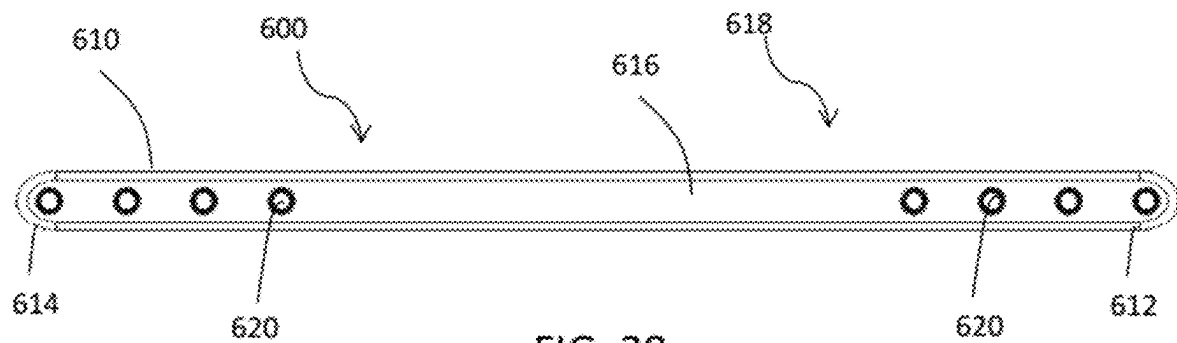


FIG. 28

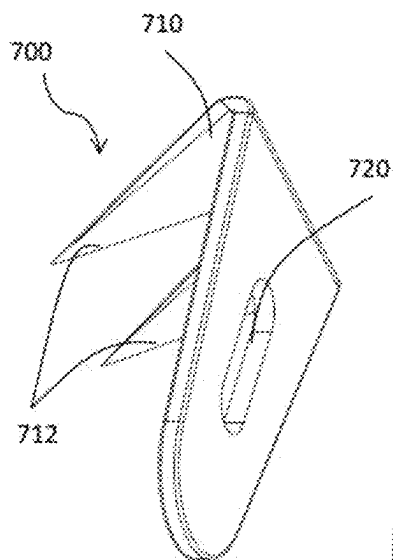


FIG. 29

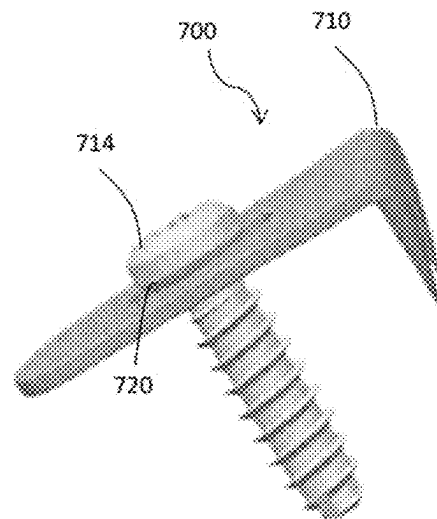


FIG. 30

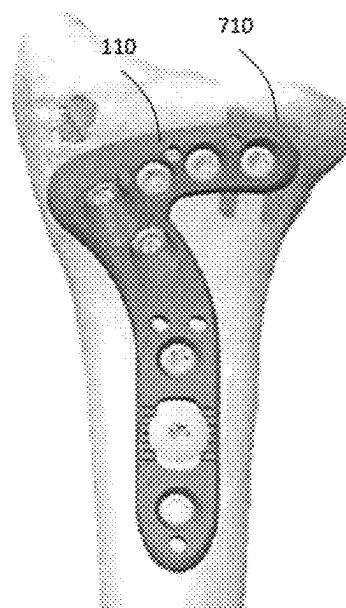


FIG. 31

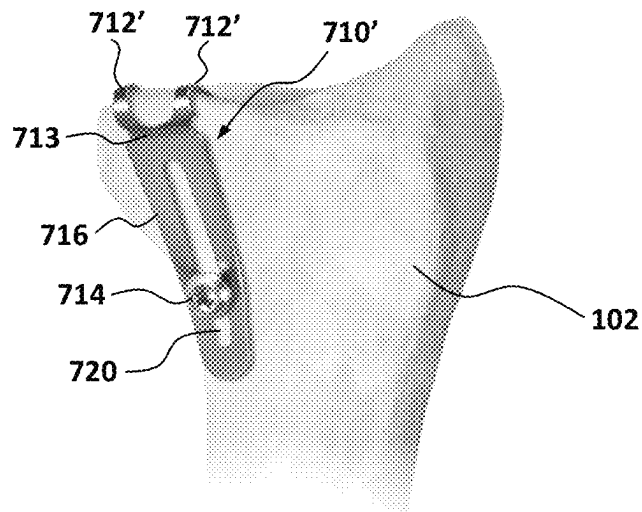


FIG. 32

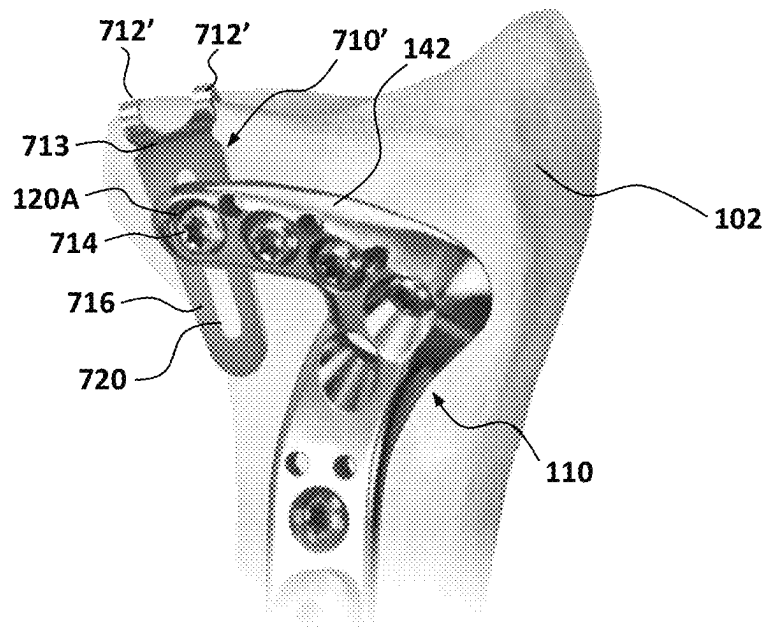


FIG. 33

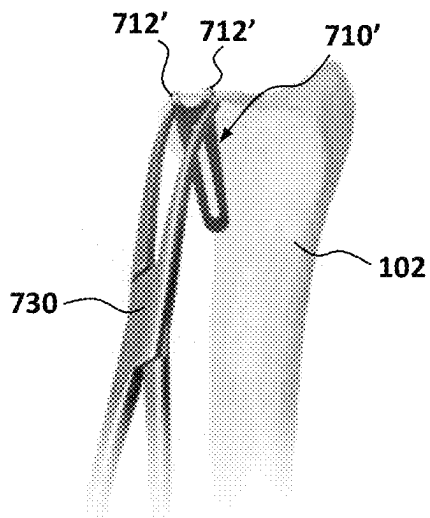


FIG. 34A

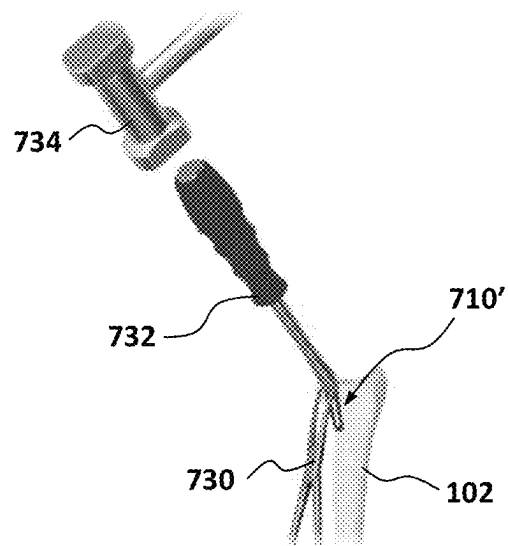


FIG. 34B

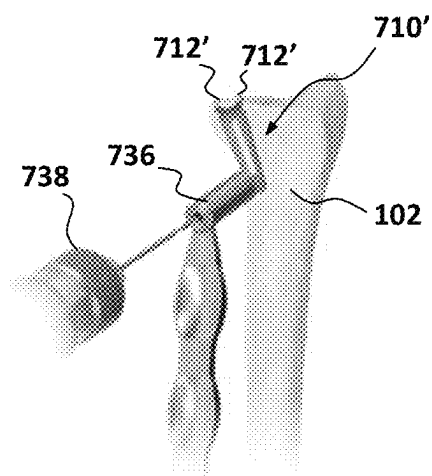


FIG. 34C

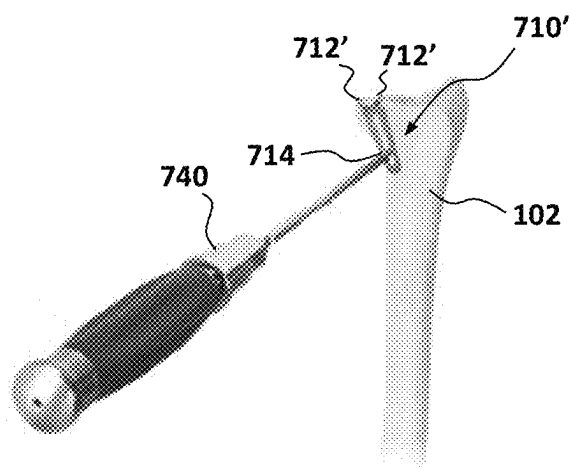


FIG. 34D

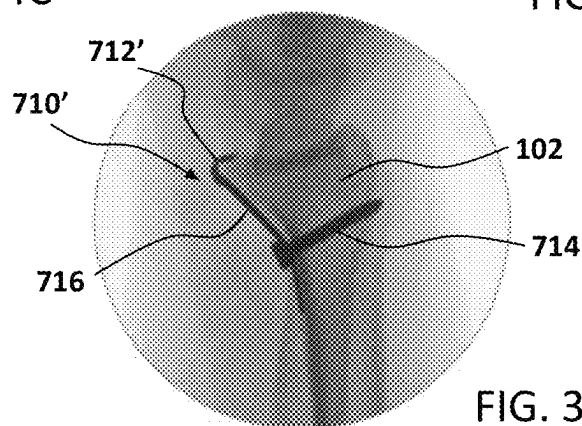


FIG. 34E

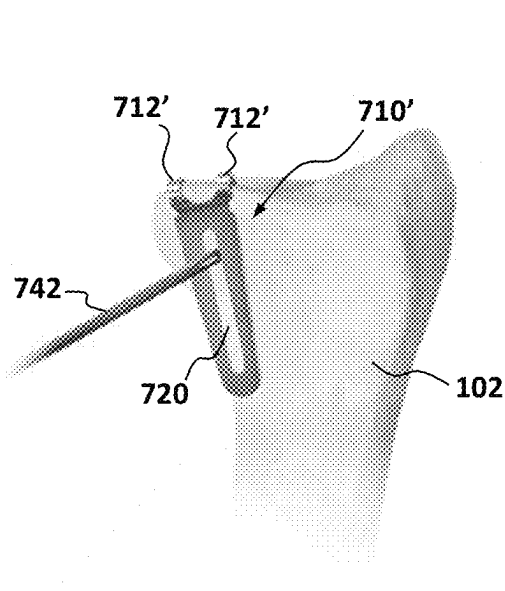


FIG. 35A

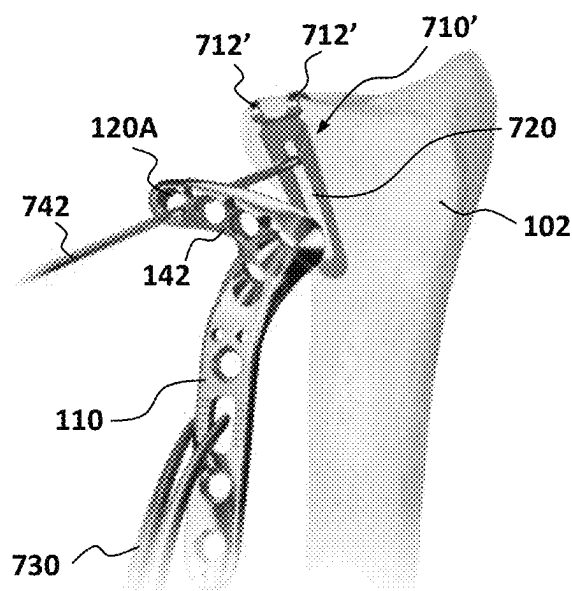


FIG. 35B

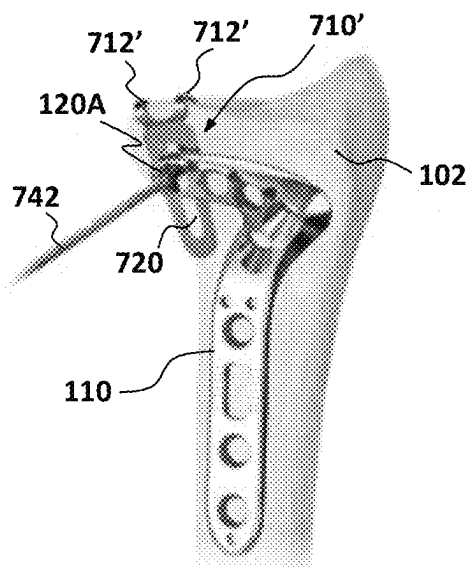


FIG. 35C

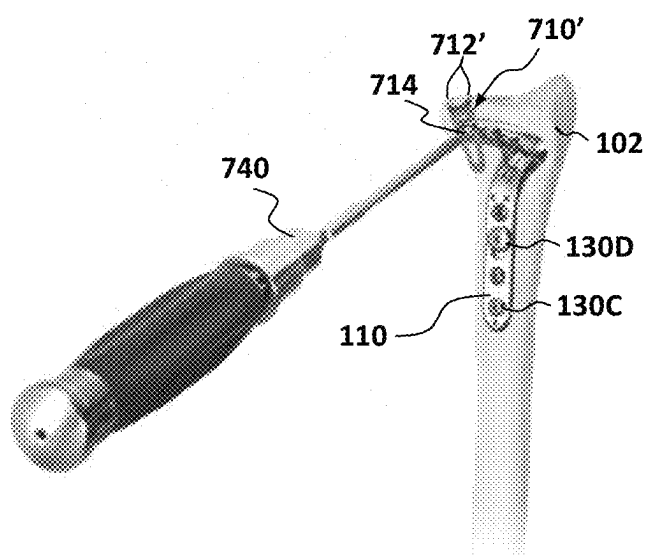


FIG. 35D

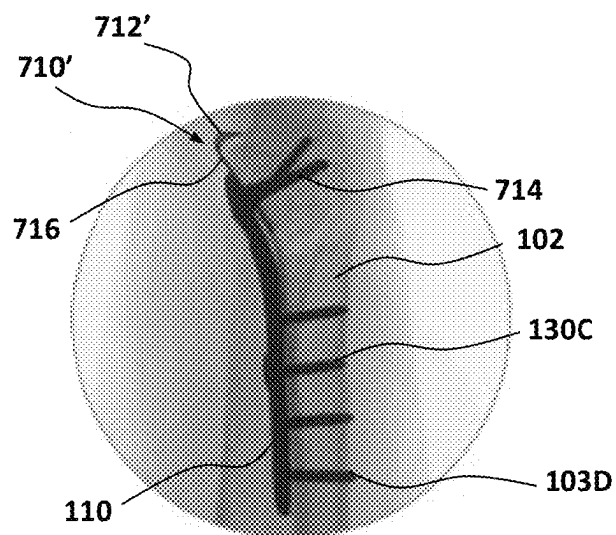


FIG. 35E

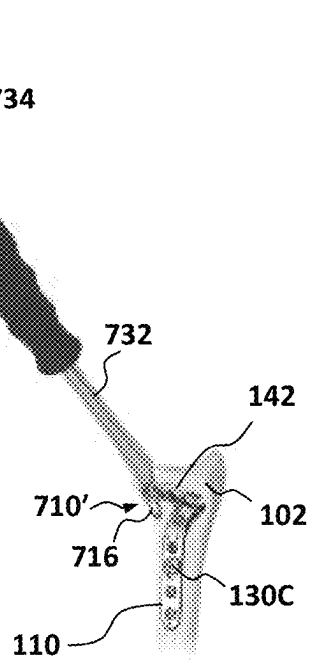


FIG. 36A

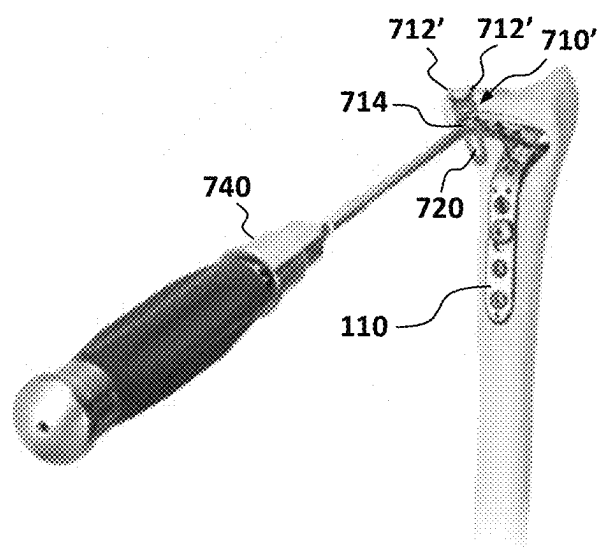


FIG. 36B

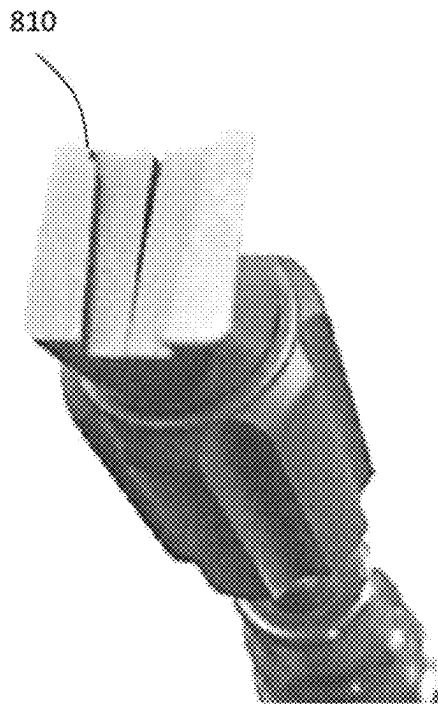


FIG. 37

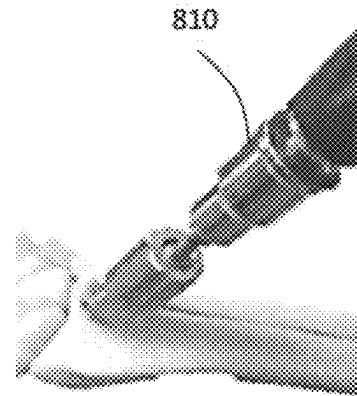


FIG. 38

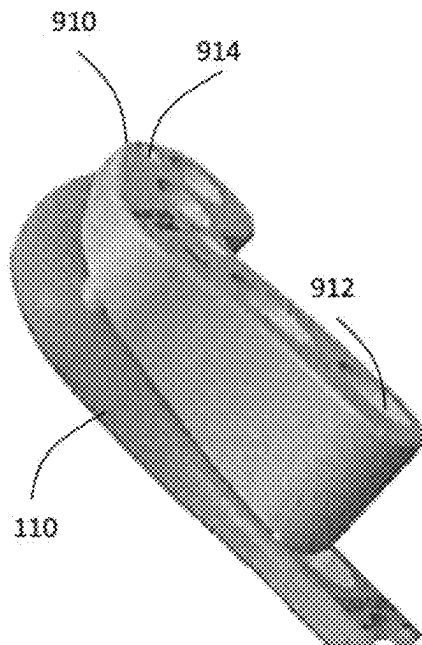


FIG. 39

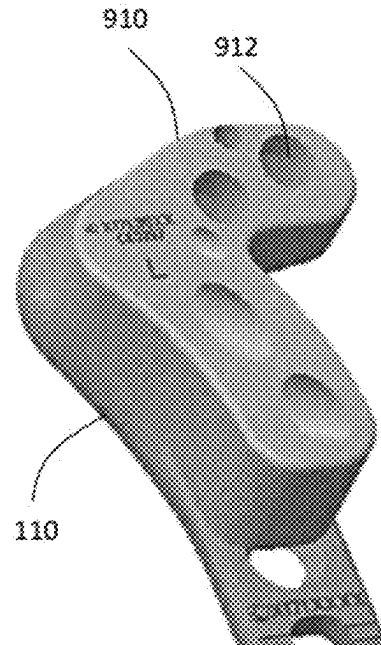


FIG. 40

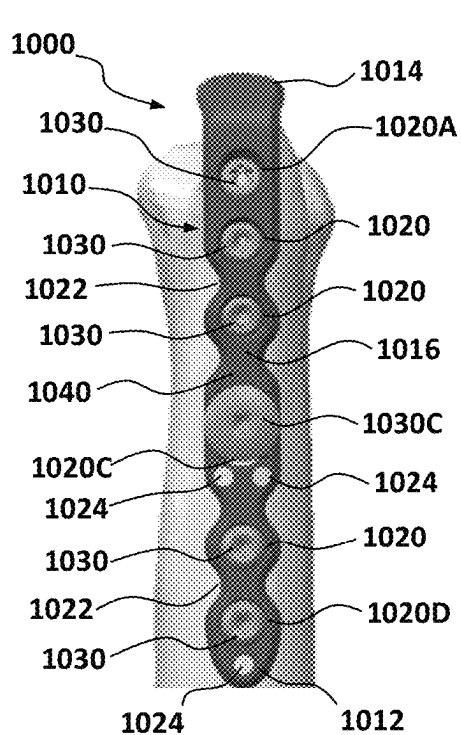


FIG. 41A

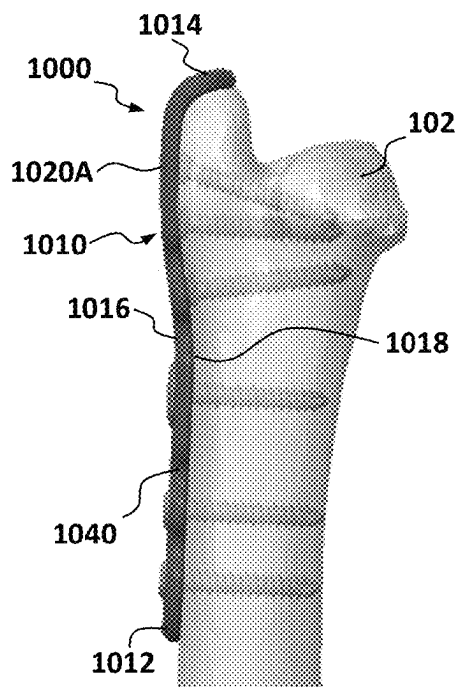


FIG. 41B

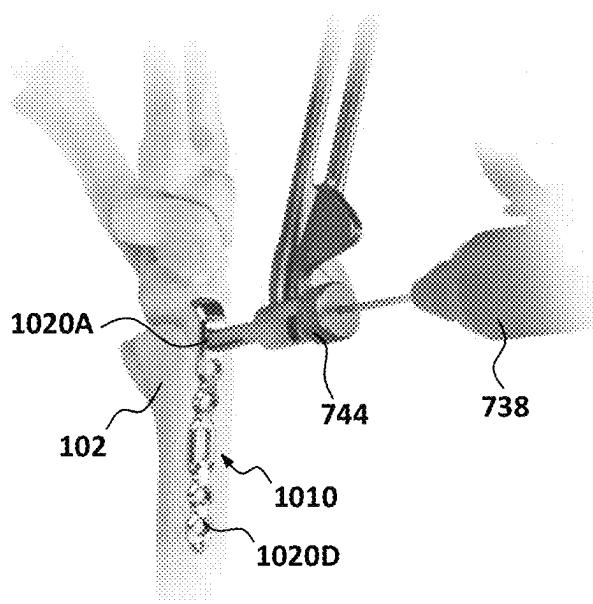


FIG. 42A

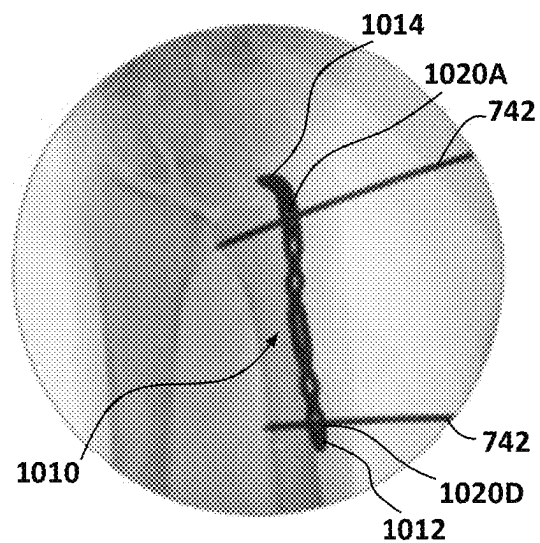


FIG. 42B

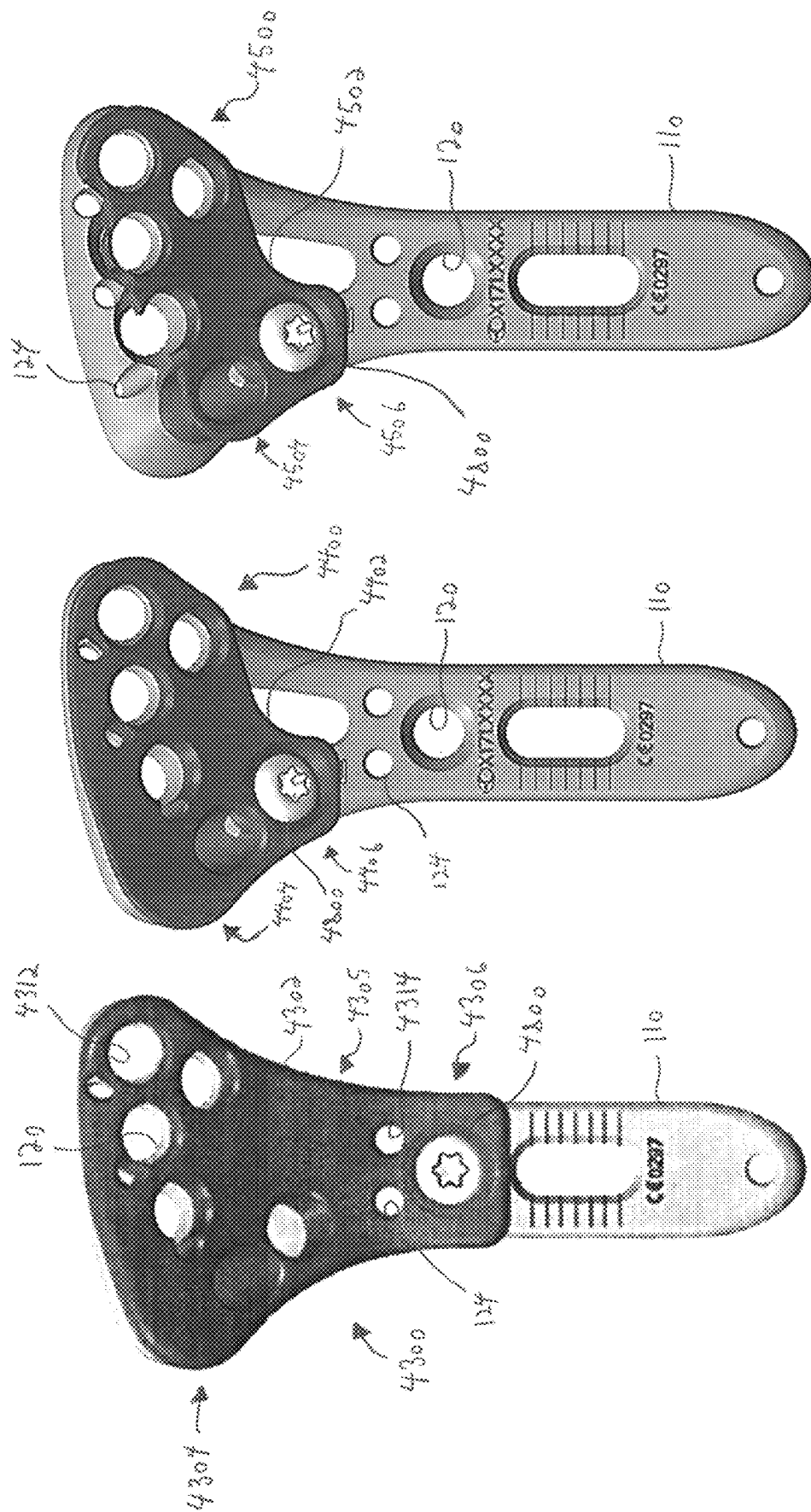


FIG. 43

FIG. 44

FIG. 45

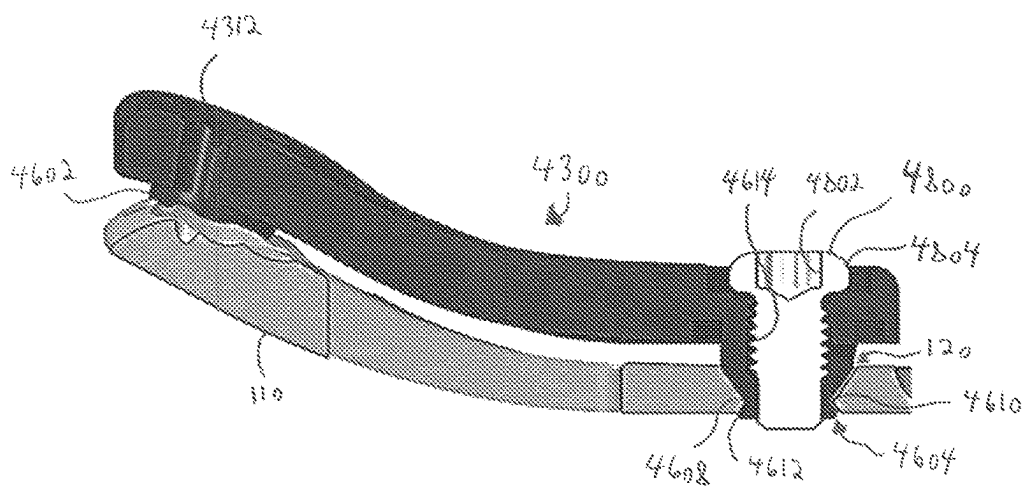


FIG. 46

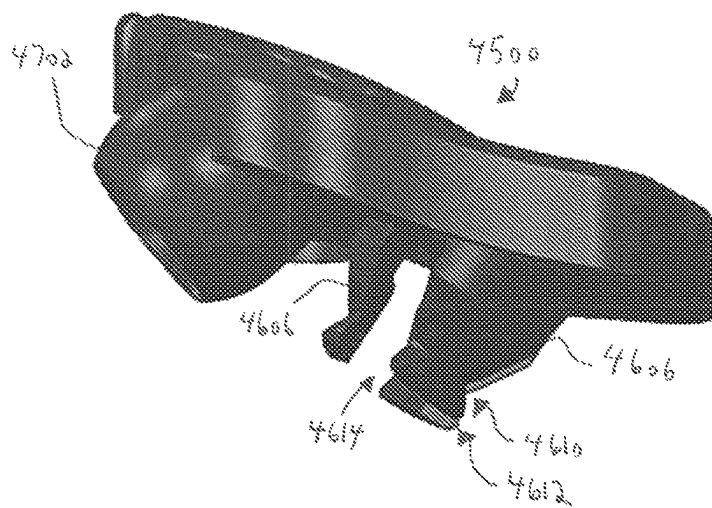


FIG. 47

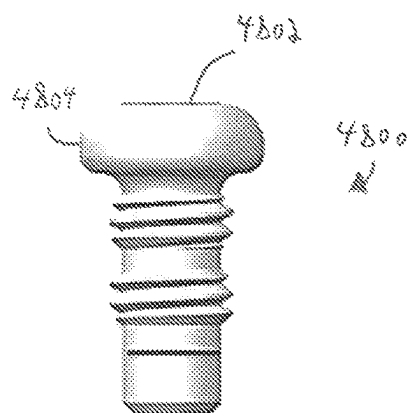


FIG. 48

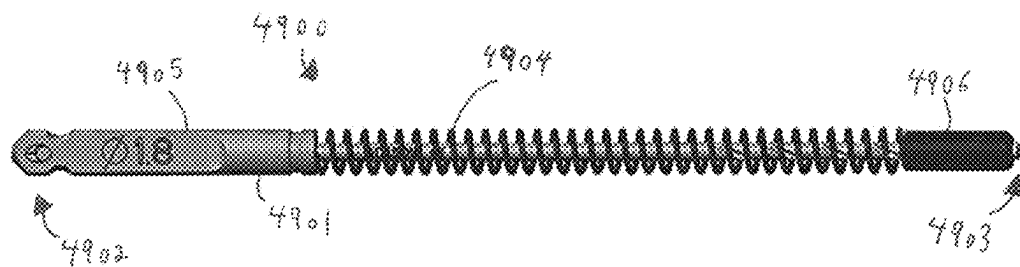


FIG. 49

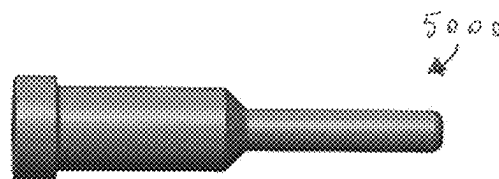


FIG. 50

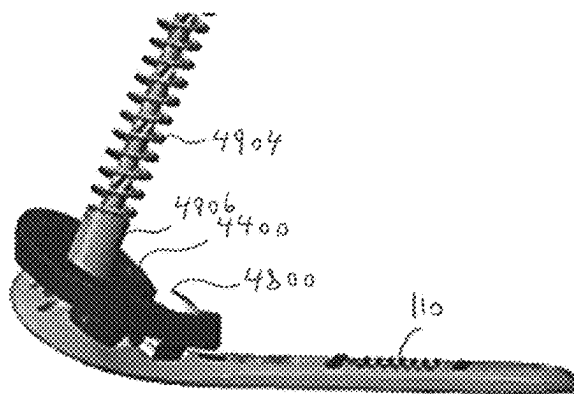


FIG. 51A

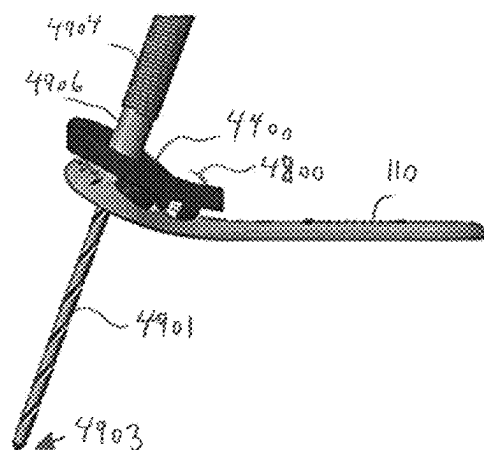


FIG. 51B

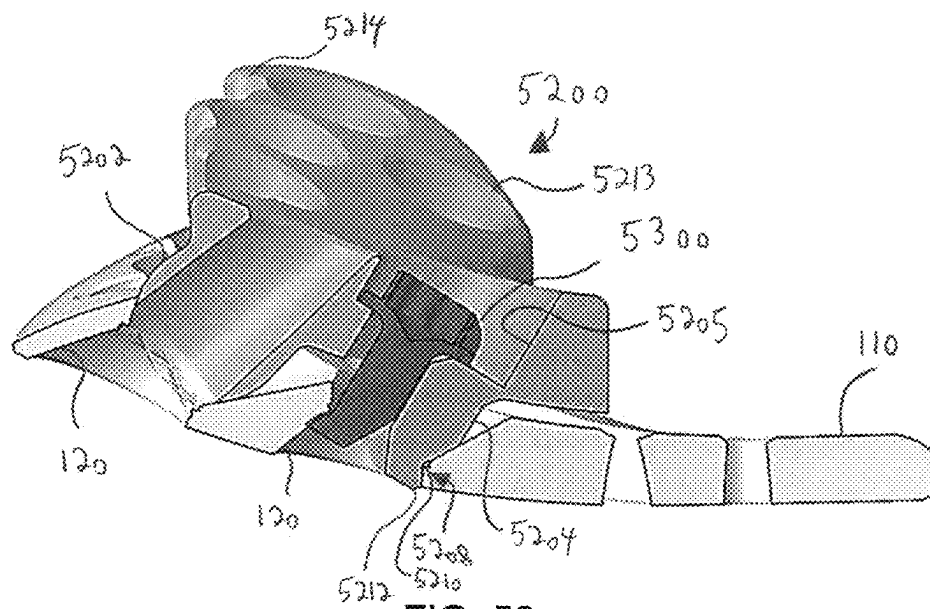


FIG. 52

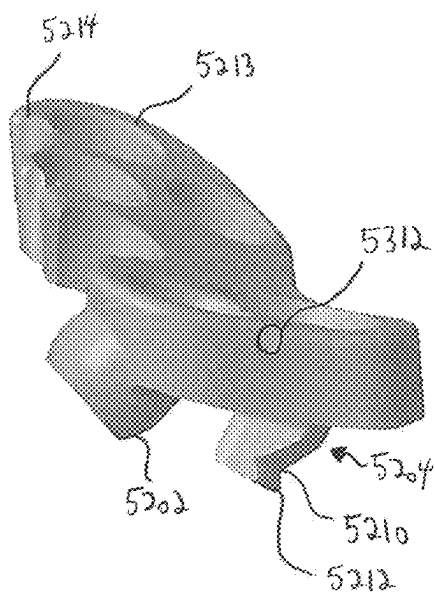


FIG. 53A

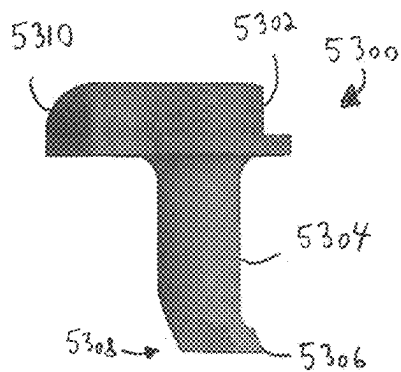


FIG. 53B

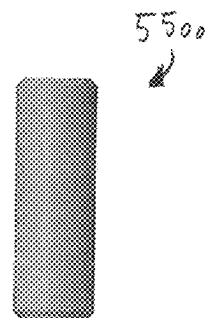


FIG. 53C

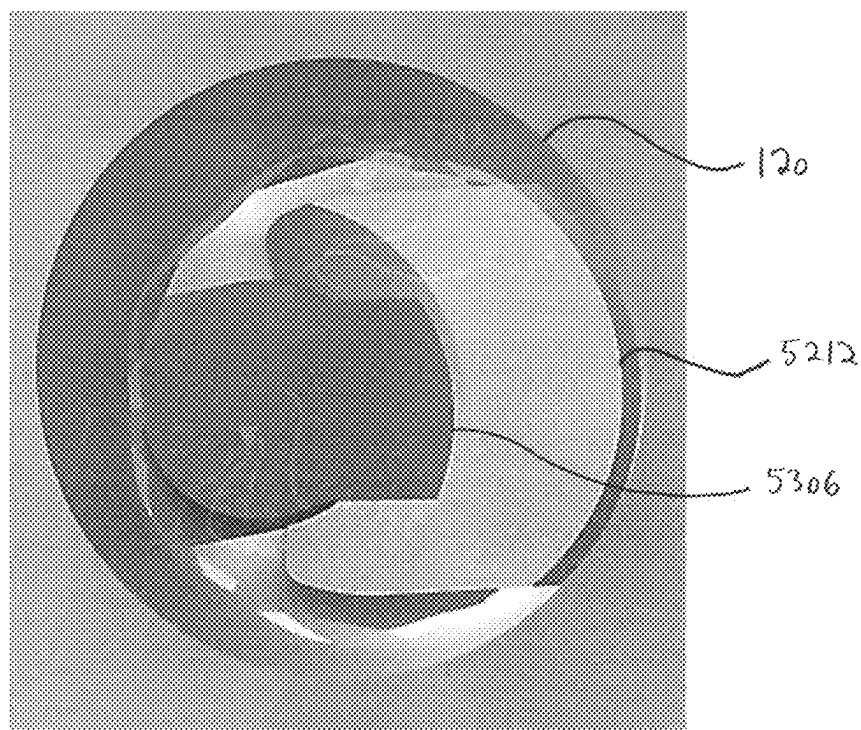


FIG. 54A

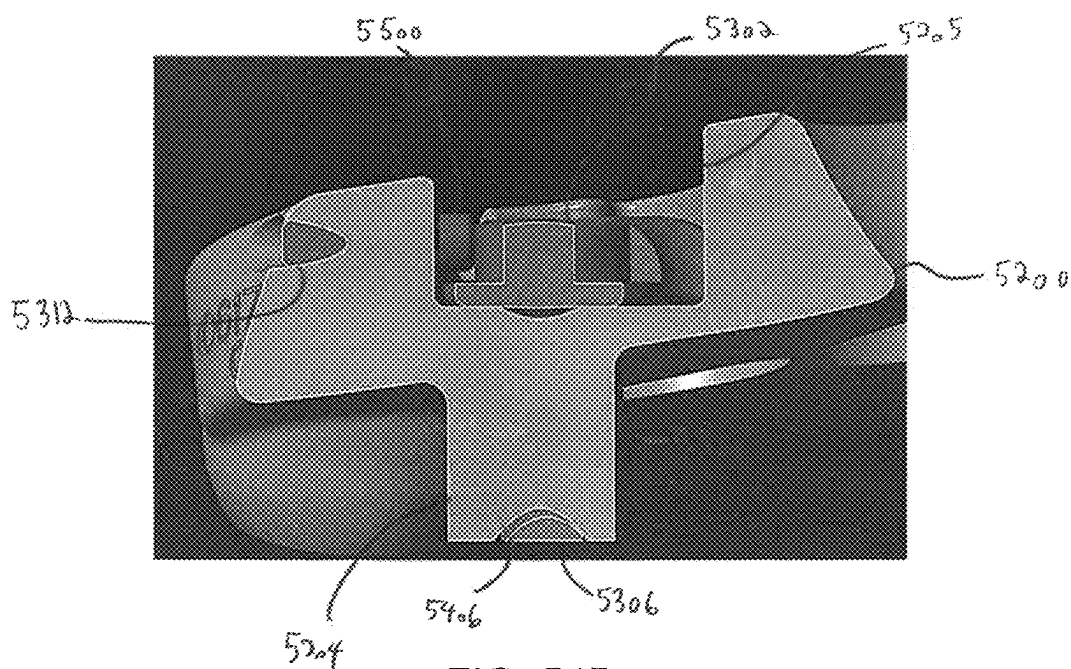


FIG. 54B

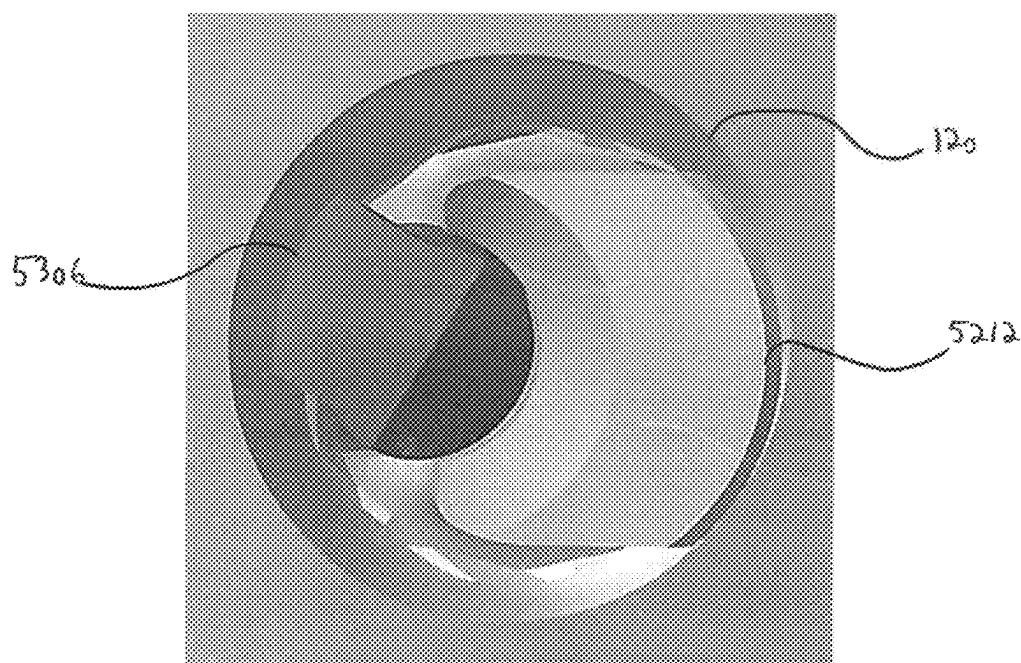


FIG. 54C

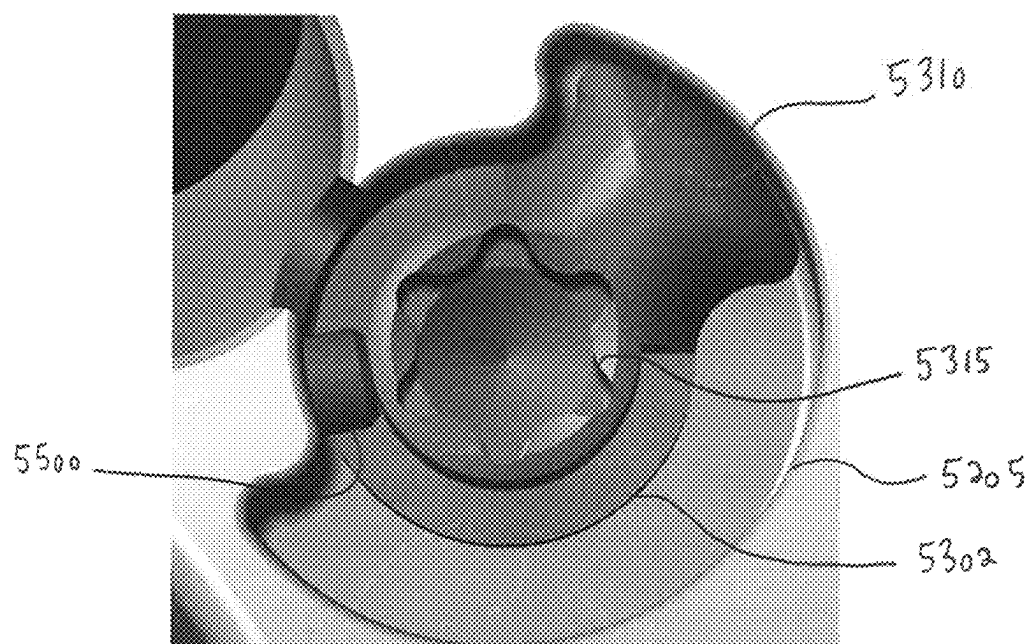


FIG. 55

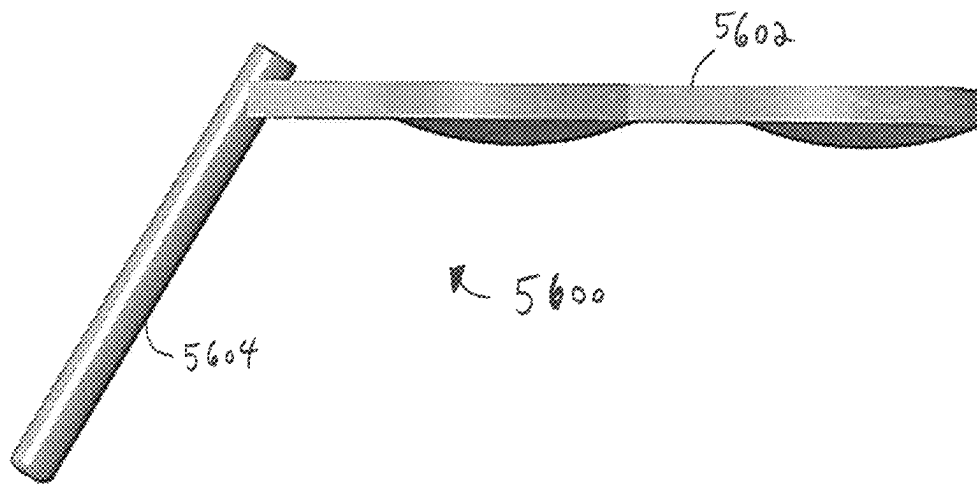


FIG. 56

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DISTAL RADIUS STABILIZATION SYSTEM**CROSS REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation of U.S. patent application Ser. No. 17/660,035 filed on Apr. 21, 2022, which is a continuation of U.S. patent application Ser. No. 16/795,640 filed on Feb. 20, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/440,409, filed Jun. 13, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 15/871,183, filed Jan. 15, 2018, which is a continuation-in-part of U.S. patent application Ser. No. 15/456,642, filed Mar. 13, 2017, which is a continuation-in-part of U.S. patent application Ser. No. 15/238,773, filed on Aug. 17, 2016. This application also claims the benefit of U.S. Provisional Application Ser. No. 62/554,700, filed on Sep. 6, 2017. The contents of each of these applications are incorporated herein by reference in their entirety for all purposes.

FIELD

The present disclosure relates to surgical devices and stabilization systems, for example, for trauma applications, and more particularly, for stabilization of distal radius and ulna fractures.

BACKGROUND

Bone fractures are often repaired by internal fixation of the bone, such as diaphyseal bone, using one or more plates. The plate is held against the fractured bone with screws, for example, which engage the bone and heads which provide a compressive force against the plate. The plate and bone are thus forced against each other in a manner that transfers load primarily between a bone contacting surface of the plate and the bone surface to reinforce the fractured bone during healing. This manner of plating generally creates relatively low stress concentration in the bone, as there may be a large contact area between the plate and the diaphyseal bone surface permitting transfer of load to be dispersed. There may be a desire to use locking screws, non-locking screws, or a combination of both that are able to dynamically compress the bone. Of course, the designs of the plates, types of screws, and locking and/or non-locking capabilities may vary based on the location and type of fracture.

The three long bones of the upper extremity are the humerus, radius, and ulna. In the case of radial fracture fixation, a volar approach may be suitable for plating certain fracture types. There remains a need, however, for improved plating systems for anatomical articular reduction and stable fixation of the radius.

SUMMARY

To meet this and other needs, devices, systems, and methods of bone stabilization are provided, for example, for radius or ulna stabilization. The stabilization systems may include one or more plates and one or more fasteners. Although generally described with reference to the radius or ulna, it will be appreciated that the stabilization systems described herein may be used or adapted to be used for the fixation of other long bones as well, such as the humerus, femur, tibia, etc.

According to one embodiment, a stabilization system includes a bone plate and a plurality of fasteners. The bone

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plate comprises an elongated portion extending along a longitudinal axis, an enlarged head portion, and a transition region connecting the elongated portion to the enlarged head portion, wherein the transition region is curved and connect to an end portion of the enlarged head portion, the bone plate comprising a plurality of through holes extending through the enlarged head portion, the transition region, and the elongated portion. The fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone.

The fasteners may include locking fasteners (e.g., configured to lock to the plate), non-locking fasteners (e.g., configured to provide dynamic compression of the bone), polyaxial fasteners (e.g., configured to be inserted at a plurality of angles or trajectories), fixed angle fasteners (e.g., configured to be inserted at a fixed angle or trajectory), or any other suitable fasteners known in the art.

In some instances, the locking fasteners may include fasteners having self-forming threads on a head portion of the fasteners, which are configured to lock to at least one of the plurality of through holes on the plate.

According to another embodiment, a stabilization system configured to stabilize a radius includes a bone plate, a plurality of fixed angle fasteners, a polyaxial fastener, and a fastener. The bone plate comprises an elongated portion extending along a longitudinal axis, an enlarged head portion, and a transition region connecting the elongated portion to the enlarged head portion, wherein the transition region is curved and connect to an end portion of the enlarged head portion, the bone plate comprising a plurality of fixed angle holes positioned in general alignment along the elongated portion, a polyaxial hole positioned proximate to the end portion of the enlarged head portion connected to the transition region, and an elongated slot on the elongated portion. The fixed angle fasteners are configured to be received in the fixed angle holes, the plurality of fixed angle fasteners configured to be aimed at a radio-carpal joint and a distal radio-ulnar joint. The polyaxial fastener is configured to be received in the polyaxial hole, the polyaxial fastener configured to be aimed at a radial styloid. The fastener is configured to be received in the elongated slot, wherein the elongated slot allows for proximal-distal and medial-lateral adjustment of the plate.

According to another embodiment, a stabilization system for stabilizing a bone includes a bone plate and a plurality of fasteners. The bone plate has an upper surface and a lower surface configured to contact the bone, wherein the lower surface comprises one or more recesses configured to reduce contact between the plate and a surface of the bone. The bone plate comprises an elongated portion extending along a longitudinal axis, an enlarged head portion, and a transition region connecting the elongated portion to the enlarged head portion, wherein the transition region is connect to an end portion of the enlarged head portion and the other end portion of the enlarged head portion is a free end, the bone plate comprising a plurality of through holes extending through the enlarged head portion, the transition region, and the elongated portion. The plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone.

According to another embodiment, the stabilization system may include a bone plate having an elongated portion extending along a longitudinal axis between a proximal end and a distal end. The bone plate defines a plurality of through holes extending through the elongated portion. A plurality of

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fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone. The proximal end of the elongate portion has an arcuate configuration.

According to another embodiment, the stabilization system may include a bone plate having an upper surface and a lower surface configured to contact the bone. The bone plate includes a plurality of through holes extending through the elongated portion. At least one of the through holes is a locking through hole which defines an upper tapered portion extending from the upper surface and a lower tapered portion extending from the lower surface with a deformation area defined between the upper tapered portion and the lower tapered portion. A plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone. At least one of the fasteners includes self-forming threads on a head portion of the fastener which are configured to deform the deformation area and lock to one of the locking through holes on the plate.

According to another embodiment, the stabilization system may include a bone plate having an upper surface and a lower surface configured to contact the bone. The bone plate further has a first portion and a second portion with the second portion extending at an angle relative to the first portion. The first portion of the bone plate includes an opening for receiving a fixation member and the second portion of the plate includes at least one hook having an arcuate configuration.

According to yet another embodiment, one or more methods of installing a stabilization system may include aligning a bone plate against the volar side of the radial bone, and inserting one or more fasteners through the bone plate and into the bone to stabilize the radius and repair the fracture.

Also provided are kits for the stabilization systems including bone plates of varying sizes and orientations, fasteners including locking fasteners, non-locking, compression fasteners, polyaxial fasteners, fixed angle fasteners, or any other suitable fasteners, drill guides, k-wires, and other components for installing the same.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:

FIGS. 1A-1K depict stabilization systems according to embodiments including volar distal radius bone plates and a plurality of bone fasteners;

FIG. 2 is a top perspective view of two fasteners engaged with combination holes according to an embodiment;

FIG. 3 is a close-up view of an alternative version of a combination hole according to another embodiment;

FIGS. 4A-4C show a perspective view, top view, and cross-section view, respectively, of an another embodiment of a combination hole;

FIGS. 5A-5C show a perspective view, top view, and cross-section view, respectively, of an another embodiment of a combination hole;

FIGS. 6A-6C show a perspective view, top view, and cross-section view, respectively, of an another embodiment of a hole for receiving a fastener;

FIGS. 7A-7C show a perspective view, top view, and cross-section view, respectively, of an another embodiment of a combination hole;

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FIGS. 8A-8C show a perspective view, top view, and cross-section view, respectively, of an another embodiment of separate locking and non-locking holes;

FIGS. 9A-9D show a perspective view, a top view, a cross-section view, and a perspective view with a locking fastener, respectively, according to another embodiment of a plate including three overlapping locking and non-locking holes;

FIGS. 10A-10B show perspective views of a plate according to another embodiment with locking and non-locking functionality;

FIGS. 11A-11E shows alternative locking screw and openings in plates according to yet another embodiment;

FIGS. 12A and 12B depict a perspective view and cross-section view of an alternative version of a plate with blocking screws;

FIGS. 13A and 13B depict a fastener according to another embodiment with self-forming threads configured to form threads in the opening of a plate;

FIG. 13C illustrates an opening in a plate which is usable with the fastener;

FIGS. 14A and 14B depict an opening in a plate according to one embodiment having a windswept cut configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;

FIGS. 15A and 15B depict an opening in a plate according to another embodiment having a knurled cut configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;

FIGS. 15C and 15D depict an opening in a plate according to another embodiment having triangular cuts configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;

FIGS. 16A and 16B depict an opening in a plate according to another embodiment having a polygonal cut configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;

FIG. 17A depicts an alternative opening in a plate according to another embodiment;

FIG. 17B depicts another alternative opening in a plate according to yet another embodiment;

FIGS. 18A-18D depict a plate assembly according to one embodiment where a locking or non-locking fastener may be positioned at an angle or perpendicular to the plate;

FIG. 19 depicts a stabilization system according to one embodiment including a volar distal radius dia-meta bone plate;

FIG. 20 depicts a stabilization system according to one embodiment including an acute dorsal bone plate.

FIG. 21 is a cross sectional view of the dorsal bone plate of FIG. 20;

FIG. 22 depicts a stabilization system according to one embodiment including an oblique dorsal bone plate;

FIG. 23 depicts dorsal bone plates of FIGS. 20 and 22 with fixation screws;

FIG. 24 is a close up view of a portion of the dorsal bone plates of FIGS. 20 and 22;

FIG. 25 depicts a stabilization system according to one embodiment including a lateral bone plate;

FIG. 26 is a cross sectional view of the lateral bone plate of FIG. 25;

FIG. 27 depicts the lateral bone plate of FIG. 25 with fixation screws;

FIG. 28 depicts a stabilization system according to one embodiment including a bridge bone plate;

FIG. 29 depicts a stabilization system according to one embodiment including a lunate facet hook plate;

FIG. 30 depicts the lunate facet hook plate of FIG. 29 with a fixation screw;

FIG. 31 depicts the volar distal radius plate of FIG. 1 used with the lunate facet hook plate of FIG. 30;

FIG. 32 depicts an alternative stabilization system including a lunate facet hook plate;

FIG. 33 depicts the volar distal radius plate of FIG. 1 used with the lunate facet hook plate of FIG. 32;

FIGS. 34A-34E depict an illustrative method of installing the lunate facet hook plate of FIG. 32;

FIGS. 35A-35E depict an illustrative method of installing the lunate facet hook plate of FIG. 32 with the volar distal radius plate of FIG. 1;

FIGS. 36A-36B depict another illustrative method of installing the lunate facet hook plate of FIG. 32 with the volar distal radius plate of FIG. 1;

FIGS. 37 and 38 depict a lunate facet hook plate reduction instrument;

FIGS. 39 and 40 depict a backpack drill guide for use with at least the volar distal radius plates of FIG. 1;

FIGS. 41A and 41B depict a stabilization system according to one embodiment including a neck and head bone plate;

FIGS. 42A and 42B depict an illustrative method of installing the bone plate of FIGS. 41A and 41B;

FIG. 43 depicts a backpack drill guide in accordance with embodiments of the present disclosure for use with at least the volar distal radius plates of FIG. 1;

FIG. 44 depicts a backpack drill guide in accordance with embodiments of the present disclosure for use with at least the volar distal radius plates of FIG. 1;

FIG. 45 depicts a backpack drill guide in accordance with embodiments of the present disclosure for use with at least the volar distal radius plates of FIG. 1;

FIG. 46 depicts a cross-sectional view of the backpack drill guide of FIG. 43;

FIG. 47 depicts a side view of the backpack drill guide of FIG. 45;

FIG. 48 depicts a side view of a screw for use with a backpack drill in accordance with embodiments of the present disclosure;

FIG. 49 depicts a side view of a spring drill guide in accordance with embodiments of the present disclosure;

FIG. 50 depicts a side view of a targeting sleeve in accordance with embodiments of the present disclosure;

FIGS. 51A and 51B depict side views of deployment of a backpack drill guide and spring drill guide with a volar distal radius plate in accordance with embodiments of the present disclosure;

FIG. 52 depicts a cross-sectional view of a backpack drill guide in accordance with embodiments of the present disclosure;

FIG. 53A depicts a side view of the backpack drill guide of FIG. 52;

FIG. 53B depicts a side view of a cam screw for use with the backpack drill of FIG. 52;

FIG. 53C depicts a dowel pin for use with the cam screw of FIG. 53B and the backpack drill of FIG. 52;

FIGS. 54A and 54B depict the cam screw of FIG. 53B in an unlocked position in the backpack drill guide;

FIG. 54C depicts the cam screw of FIG. 53B in a locked position in the backpack drill guide;

FIG. 55 depicts a top view of a cam screw disposed in a backpack drill guide in accordance with embodiments of the present disclosure; and

FIG. 56 depicts a side view of a targeting sleeve in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

Embodiments of the disclosure are generally directed to devices, systems, and methods for bone stabilization, especially radius stabilization. Specifically, embodiments are directed to volar distal radius stabilization systems including a bone plate configured to sit against the volar side of the radial bone. The fasteners may be configured to secure the bone plate to the radius. Still other embodiments are directed to different types of holes and fasteners configured to provide locking and/or compression to the bone.

The bone plate may be comprised of titanium, stainless steel, cobalt chrome, carbon composite, plastic or polymer—such as polyetheretherketone (PEEK), polyethylene, ultra high molecular weight polyethylene (UHMWPE), resorbable polylactic acid (PLA), polyglycolic acid (PGA), combinations or alloys of such materials or any other appropriate material that has sufficient strength to be secured to and hold bone, while also having sufficient biocompatibility to be implanted into a body. Similarly, the fasteners may be comprised of titanium, cobalt chrome, cobalt-chrome-molybdenum, stainless steel, tungsten carbide, combinations or alloys of such materials or other appropriate biocompatible materials. Although the above list of materials includes many typical materials out of which bone plates and bone fasteners are made, it should be understood that the bone plates and fasteners comprised of any appropriate material are contemplated.

The embodiments of the disclosure and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments and examples that are described and/or illustrated in the accompanying drawings and detailed in the following description. The features of one embodiment may be employed with other embodiments as the skilled artisan would recognize, even if not explicitly stated herein. Descriptions of well-known components and processing techniques may be omitted so as to not unnecessarily obscure the embodiments of the disclosure. The examples used herein are intended merely to facilitate an understanding of ways in which the disclosure may be practiced and to further enable those of skill in the art to practice the embodiments of the disclosure. Accordingly, the examples and embodiments herein should not be construed as limiting the scope of the disclosure, which is defined solely by the appended claims and applicable law. Moreover, it is noted that like reference numerals represent similar features and structures throughout the several views of the drawings.

Volar Distal Radius Plate System

Referring now to the drawing, FIGS. 1A-1K depict embodiments of a volar distal radius stabilization system 100 including a bone plate 110 configured to sit against the volar side of the radial bone and one or more bone fasteners 130 configured to be received in the bone plate 110 and secured to the radius 102. The radius 102 or radial bone is one of the two large bones of the forearm, the other being the ulna. The radius 102 extends from the lateral side of the elbow to the thumb side of the wrist and runs parallel to the ulna, which exceeds it in length and size. Near the wrist, the distal end 104 of the radius 102 is large and of quadrilateral form. Although generally described with reference to the radius 102, it will be appreciated that the stabilization systems described herein may be used or adapted to be used for the fixation of other long bones as well, such as the humerus, femur, tibia, etc.

The bone plate 110 extends from a first end 112 configured to be positioned proximate to a shaft portion of radius

102 to a second end 114 configured to be positioned proximate to the distal end 104 of the radius 102. The plate 110 includes a top surface 116 and an opposite, bottom surface 118 configured to contact adjacent bone. The top and bottom surfaces 116, 118 are connected by opposite side surfaces extending from the first to second ends 112, 114 of the plate 110. The bottom surface 118 of the plate 110 includes an anatomic contour configured to follow the best approximation of average distal radial anatomy, flaring up slightly along the radial column and more significantly along the intermediate column of the plate 110. The plate 110 is designed to sit low and have a generally low profile proximal portion. The thickness of the plate 110 may generally be about 2 mm along the shaft and distal intermediate column, tapering to a thickness of 2.5 mm along the distal radial column which allows for the severe angle of the radial styloid fastener. The watershed line of the volar distal radius defines the border between the radiocarpal (RC) joint and the volar surface of the radius 102. A chamfer at the second end 114 on the distal radius column of the plate 110 may help to ensure minimal tendon disruption, for example of the flexor pollicis longus and flexor carpi radialis, by maintaining a lower profile over the tendon sites.

The bone plate 110 includes an elongated portion 140 extending along a longitudinal axis L, having a length greater than its width. The elongated portion 140 is configured to contact the shaft of the radius 102. The elongated portion 140 may terminate at the first end 112 with a taper such that it has a width and/or thickness less than the remainder of the elongated portion 140. A transition region 144 may connect the elongated portion 140 to an enlarged head portion 142. The transition region 144 may extend along an axis T which is generally angled relative to the axis L of the elongated portion 140. The transition region 144 may extend at an angle X relative to the elongated portion 140. The angle X of the transition region 144 relative to the elongated portion 140 may range from about 10-60°, about 20-50°, about 30-40°, about 40-50°, or another appropriate angle. The transition region 144 may generally form a curve from the elongated portion 140 to an end of the enlarged head portion 142.

The transition region 144 may connect to an end portion of the enlarged head portion 142 and the other end portion of the enlarged head portion 142 may be a free end. In other words, the opposite end portion of the enlarged head portion 142, not connected to the transition region 144, is not connected to any other portion of the plate 110. The free end of the enlarged head portion 142 may be separated a distance from the transition region 144 and the elongated portion 140 of the plate 110.

The enlarged head portion 142 or a portion thereof is configured to contact the distal end 104 of the radius 102. The enlarged head portion 142 has a width greater than the width of the elongated portion 140. The enlarged head portion 142 extends along an axis A at an angle Y relative to the transition region 144. The angle Y of the head portion 142 relative to the transition region 144 may range from about 10-60°, about 20-50°, about 30-40°, about 40-50°, or another appropriate angle. Accordingly, the axis A of the enlarged head portion 142 may be transverse to the axis L of the elongated portion 140. In some embodiments, the axis A of the enlarged head portion 142 may be generally perpendicular to the axis L of the elongated portion 140. As best seen in FIG. 1C, the bone plates 110 may be available in a variety of lengths and sizes based on the anatomy of the patient. The plates 110 are configured to sit against the volar side of the radial bone 102. The plates 110 are configured in

both left and right designs, in a mirrored configuration, in order to address the anatomy of both the left and right arms of the patient.

As best seen in FIGS. 11 and 1J, the bottom surface 118 of the plate 110 may include a plurality of recesses 119 located along the elongated portion 140 between the fastener openings 120. In the embodiment shown in FIG. 1I, the recesses 119 are in the form small partial bores in the lateral surface, which are configured to facilitate bending of the plate 110. The recesses 119 remove material such that the plate 110 shield stress from the fastener openings 120, discouraging hole warping effect during recontouring of the plate 110. The recesses 119 may also provide attachment points for plate placement instrumentation (not shown). In the embodiment shown in FIG. 1J, the recesses 119 are in the form of scallop cuts having partially cylindrical valleys cut around a periphery of the bottom surface 118 of the plate 110. This again shields stress from the fastener openings 120 during bending, discouraging hole warping effects while recontouring the plate 110. This also reduces contact between the plate 110 and the bone surface, thereby helping to preserve blood supply to the bone and prevent osteonecrosis. In addition to or in place of the recesses 119, a plurality of dimples, best seen in FIG. 1K, may be positioned along the bottom surface 118 of the plate 110 (e.g., along the entire bottom surface 118 or a portion thereof) to further reduce contact between the plate 110 and bone surface, further helping to preserve blood supply and prevent osteonecrosis.

The plate 110 includes one or more through openings 120 configured to receive one or more bone fasteners 130. The openings 120 extend through the body of the plate 110 from the top surface 116 to the bottom surface 118. The openings 120 may include cylindrical openings, conical openings, elongated openings, threaded openings, textured openings, non-threaded and/or non-textured openings, and the like. The openings 120 may allow for locking of the fastener 130 to the plate 110 or may allow for movement and dynamic compression of the bone. The plate 110 may comprise any suitable number of openings 120 in any suitable configuration. These openings 120 allow surgeons more flexibility for fastener placement, based on preference, anatomy, and fracture location. Surgeons may have differing opinions as to the number, location, and types of fasteners 130. Further, complexity of fracture location and shape makes having as many locations for fasteners 130 as possible necessary. This design offers surgeons a versatile method to achieve higher accuracy in placement of the fasteners 130.

The openings 120 may be configured to receive one or more bone fasteners 130. The fasteners 130 may include locking fasteners, non-locking fasteners, or any other fasteners known in the art. The fasteners 130 may comprise bone screws or the like. The fasteners 130 may also include other fasteners or anchors configured to be secured or engaged with bone, such as nails, spikes, staples, pegs, barbs, hooks, or the like. The fasteners 130 may include fixed and/or variable angle bone screws. The fastener 130 may include a head portion 132 and a shaft portion 134 configured to engage bone. For a locking fastener 130, the shaft portion 134 may be threaded such that the fastener 130 may be threaded into the bone. The head portion 132 may include a textured area, such as threads, around its outer surface sized and configured to engage with the opening 120, for example, and corresponding threads in the opening 120 in order to lock the fastener 130 to the plate 110. In the

alternative, for a non-locking fastener **130**, the head portion **132** may be substantially smooth to allow for dynamic compression of the bone.

In one embodiment, the enlarged head portion **142** of the plate **110** includes a plurality of holes **120A** are aligned so that their nominal trajectories follow the articular surfaces of both the radio-carpal joint and the distal radio ulnar-joint. This allows the fasteners **130A** to buttress and support the articular surfaces during fracture reconstruction. As shown in the embodiment in FIGS. 1A-1C, the plate **110** may have a single row of holes **120A** generally in alignment and a secondary hole **120A** positioned on the transition region **144**. FIG. 1C depicts one embodiment of the plate **110** (right most plate **110**) having a first, distal row of holes **120A** generally in alignment and a second row of holes **120A** generally in alignment. The second row of holes **120A** may receive fasteners **130A** with trajectories converging with the distal row screw trajectories. In an alternative version of the stabilization system **100**, shown in FIGS. 1D and 1E, the elongated portion **140** directly transitions into the enlarged head portion **142** and the enlarged head portion **142** is increased in dimension in order to receive the second row of the fasteners **130A**.

The holes **120A** may be fixed openings configured to accept fixed angle fasteners **130A** that can be secured into the distal end **104** of the radius **102**. The screw holes **120A** and screw heads **132** may have mating conical threads that lock the screw **130A** in both angular and axial alignment to prevent collapse and backout. The fasteners **130A** may have predetermined trajectories based on the orientations of the openings **120A**. An upper portion of the holes **120A** may be tapered **128** to allow for the proper positioning of each of the fasteners **130A**. Each of the fasteners **130A** may be angled along a different trajectory than the other respective fasteners **130A**. Some of the fasteners **130A** may have a greater angulation than other respective fasteners **130A**.

The enlarged head portion of the plate **110** further include a hole **120B** configured to receive fastener **130B** with a trajectory having the severe angle necessary to reach the tip of the radial styloid. An upper portion of the hole **120B** may be tapered **128** and a portion of the plate **110** around the hole **120B** may be enlarged or increased in thickness to allow for the proper angle of the fasteners **130B** to be achieved. The fastener **130B** may be in the form of a polyaxial bone screw, which may be generally larger (e.g., in length and/or diameter) than the other fasteners **130** securing the plate **110** to the bone. The fasteners **130A**, **130B** are optionally cannulated to allow for precise placement with a k-wire (not shown) if desired by the surgeon. In some embodiments, the fasteners **130A**, **130B** may include polyaxial screws having self-forming threads that work by displacement of the plate material, which are described in more detail herein.

The plate **110** also include one or more holes **120C** present along the elongated portion **140** of the plate **110** and configured to accommodate a compression fastener **130C**. As best seen in FIG. 1G, the holes **120C** may offer a sliding slot for proximal-distal adjustment of the plate **110** during provisional placement. The slot **120C** may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plate **110**. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The hole or holes **120C** may be elongated along the longitudinal axis **L** of the elongated portion **140** as well as elongated, relative to the fastener **130C**, from lateral side to lateral side. The elongated hole or holes **120C** may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot **120C**. In the alternative

embodiment of the slot **120C'** illustrated in FIG. 1H, adjustment markings **125** are provided along each longitudinal side of the slot **120C'**. The adjustment markings **125** may include be applied via ink, etching, for example, via laser etching, or other suitable means. The adjustment markings **125** are spaced in increments, for example, equally spaced increments, such as 1 mm increments, to assist with accurate proximal/distal adjustment of the plate **110**.

The hole **120C** may be configured to accommodate non-locking, compression screws **130C**, the heads of which have a spherical underside, so the screw **130C** may be placed at varying angles. The compression screw **130C** can be inserted and preliminarily tightened to secure the plate **110** to the bone. As the screw **130C** is inserted eccentrically in to the hole **120C**, the screw **130C** slides down the slot **120C**, displacing the plate **110** and the bone as well. The compression screw **130C** may have a shorter length and/or a smaller diameter than the screws **130A** and/or **130B**. If the plate **110** needs to be adjusted later, the screw **130C** can be loosened and the plate **110** can be shifted in the proximal, distal, and/or medial-lateral directions. This slot **120C** also accommodates reduction of the radius **102** by inserting a longer compression screw **130C** and pulling the bone to the plate **110**.

The plate **110** may include one or more holes **120D** present along the elongated portion **140** of the plate **110** configured to secure the plate **110** to the shaft of the radius **102**. The holes **120D** may be configured to accommodate fixed and/or variable angle fasteners **130D**. For locking fasteners **130D**, the screw holes **120D** and screw heads **132** may have mating conical threads that lock the screw **130D** in both angular and axial alignment to prevent collapse and backout. An upper portion of the holes **120D** may be tapered **128**, for example, around the perimeter of the hole **120D**, to allow for the proper positioning of each of the fasteners **130D**. For non-locking fasteners **130D**, the head portion **132** may be substantially smooth to allow for dynamic compression of the bone.

The plate **110** including head portion **142** and/or the elongated portion **140** may further comprise a plurality of openings **124** configured to receive one or more k-wires (not shown). The k-wire holes **124** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **120**). The k-wire holes **124** may allow preliminary placement of the plate **110** against the bone and/or to aid in reduction of the fracture. The distal k-wire holes **124** on the head portion **142** may ensure a trajectory to follow the RC joint and provide direction during insertion of the distal locking screws. The proximal k-wire holes in the elongated portion **140** of the plate **110** are arranged between fastener openings **120** and may be angled relative to the surface of the plate **110** to avoid intrusion into areas where instrumentation must pass during screw insertion.

In the embodiment shown in FIGS. 1D and 1E, the plate **110** may also comprise a window **126**. The window **126** may provide visualization of the plate **110** with respect to the radius **102** in the operating environment and on imaging (e.g., fluoroscopy). The window **126** is shown as generally an asymmetrical triangle, but it is envisioned that the window **126**, if present, may be of any suitable shape, size, and dimension.

The bone plate **110** may be attached to a proximal humerus to fixate one or more bone fractures or fragments and thereby promote healing of the bone. In one embodiment, the plate **110** further restores the anatomic alignment of the radius **102**. The plate **110** may be positioned against the volar side of the radial bone. One or more k-wires may

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be supplied through the k-wire holes **124** to assist with preliminary placement of the plate **110**. Pilot holes may be drilled through the fastener openings **120** to prepare to receive the respective fasteners **130A**, **130B**, **130C**, **130D** may be positioned through the respective openings **120A**, **120B**, **120C**, **120D** and into the radius **102**. The fasteners **130** may be affixed to the bone in any suitable order, number, and orientation depending on the anatomy of the bone and the fracture.

Alternative Hole Configurations

The fixed and variable angle, locking and non-locking openings **120**, **220** (e.g., including openings **120A**, **120B**, **120C**, **120D**) and respective fasteners **130**, **230** (e.g., including **130A**, **130B**, **130C**, **130D**) described herein may be substituted with or include one or more of the following openings **20** and/or fasteners **30**, **40**. The openings **20** and/or fasteners **30**, **40** are generally described with reference to a generic plate **10**, which may include plate **110**, **210**, **310**, **410**, **510**, **610**, or any other suitable plate design.

Referring now to the drawing, FIGS. **2-18** depict alternative openings **20** in plate **10**. The openings **20** extending through the plate **10** are configured to accept locking fasteners **30**, non-locking fasteners **40**, or a combination of both locking and non-locking fasteners **30**, **40** that are able to dynamically compress the bone and/or affix the plate **10** to the bone. When plating diaphyseal bone, surgeons may use a combination of both locking and non-locking fasteners **30**, **40** that are able to dynamically compress bone and to connect the bone and the plate **10**. Dynamic compression may also be desirable to create interfragmental compression while tightening the fasteners **30**, **40**.

The plate **10** includes a top surface **16** and an opposite, bottom surface **18** configured to contact adjacent bone. The plate **10** includes one or more through openings **20** configured to receive one or more bone fasteners **30**, **40**. The openings **20** extend through the body of the plate **10** from the top surface **16** to the bottom surface **18**. In the embodiments depicted in FIGS. **2-3**, for example, the openings **20** may be in the form of a combination opening that has at least two overlapping holes. As shown in FIG. **2**, the combination opening **20** includes a first hole **22** overlapping a second hole **24**. One of the holes **22** may be configured to be the locking hole **22**, thereby able to receive and secure the locking fastener **30** to the plate **10**, and the other of the holes **24** may be configured to be the dynamic compression hole **24**, thereby allowing the non-locking fastener **40** to freely move in the hole **24** and apply dynamic compression. The locking hole **22** may have one or more locking features designed to engage with a locking fastener **30**, and the dynamic compression hole **24** may be elongated, for example, along the central longitudinal axis of the plate **10**. The screw holes **22**, **24** are not constrained to parallel axes. This hole geometry may be used in bone plates **10** to utilize either fixed angle or variable angle locking screws **30** and/or polyaxial non-locking screws **40** that can achieve dynamic compression.

These openings **20** allow surgeons more flexibility for fastener placement, based on preference, anatomy, and fracture location. Surgeons may have differing opinions as to whether non-locking or locking screws **30**, **40** (or some combination of the two) should be used in diaphyseal bone. Further, complexity of fracture location and shape makes having as many locations for fasteners **30**, **40** as possible necessary. This design offers surgeons a versatile method to achieve higher accuracy in placement of locking and/or non-locking screws **30**, **40**.

As best seen in FIG. **2**, the locking and non-locking fasteners **30**, **40** are shown. The locking and non-locking

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fasteners **30**, **40** may include traditional fasteners known in the art. The locking and non-locking fasteners **30**, **40** may comprise bone screws or the like. The fasteners **30**, **40** may also include other fasteners or anchors configured to be secured or engaged with bone, such as nails, spikes, staples, pegs, barbs, hooks, or the like. The fasteners **30**, **40** may include fixed and/or variable angle bone screws.

The locking fastener **30** may include a head portion **32** and a shaft portion **34** configured to engage bone. The shaft portion **34** may be threaded such that the fastener **30** may be threaded into the bone. The head portion **32** of the locking fastener **30** includes a textured area **36** around its outer surface sized and configured to engage with the locking hole **22** of the combination opening **20**. The textured area **36** may include threads, ridges, bumps, dimples, serrations, or other types of textured areas. As shown, the texture area **36** preferably includes a threaded portion extending substantially from the top of the head portion **32** to the bottom of the head portion **32** proximate to the shaft portion **34**. Thus, when the textured area **36** engages the locking hole **22**, the locking fastener **30** is thereby locked to the plate **10**.

The non-locking fastener **40** includes a head portion **42** and a shaft portion **44** configured to engage bone. The shaft portion **44** may be threaded such that the fastener **40** may be threaded into the bone. The head portion **42** of the non-locking fastener **40** is substantially smooth around its outer surface such that it is able to slide along the elongated compression hole **24**. Thus, the non-locking fastener **30** may be coupled to the plate **10**, but not locked thereto to enable dynamic compression of the bone. It will be recognized that the head portions **32**, **42** of the fasteners **30**, **40** may include a recess configured to receive a driver or the like.

The locking hole portion **22** of the combination opening **20** includes a textured portion **26**. The textured portion **26** may include threads, ridges, bumps, dimples, serrations, knurls, or other types of textured areas. The textured portion **26** may be of the same type (e.g., mating surfaces) or different from the textured area **36** of the locking fastener **30**. As shown, the textured portion **26** is serrated or knurled along an inner portion of the hole **22**. The knurled surface may include straight, angled, or crossed lines cut or rolled into the material. In the embodiment shown in FIG. **2**, the textured portion **26** extends along substantially the entire inner surface of the hole **22**. With reference to the embodiment shown in FIG. **3**, the combination hole **20** is substantially the same as that shown in FIG. **2** except that the textured portion **26** the locking hole **22** now includes a thin centralized textured ribbon of material. For example, the textured portion **26** takes up about half or less of the surface area of the hole **22**. In this instance, only a portion of the textured area **36** of the head portion **32** of the locking fastener **30** engages with and locks to the textured portion **26** of the hole **22**.

An upper portion of the hole **22** may be tapered **28**, without texturing, for example, to facilitate alignment of the fastener **30** with the opening **20**. As shown in FIG. **3**, this tapered portion **28** is enlarged in area relative to the embodiment in FIG. **2**. The hole **22** may be configured to receive a fixed or variable angle fastener **30**. The hole **22** may be generally conical in shape such that it is wider near the top surface **16** of the plate **10** and narrower toward the bottom surface **18** of the plate **10**. The tapered portion **28** and/or the textured area **26** may be conical in shape. In this embodiment, the locking hole **22** is a textured fixed angle conical hole configured to receive locking fastener **30**. The textured holes **22** may deform as the fastener head **32** interferes with

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the textured portion 26 of the hole 22, thereby providing a positive lock between the fastener 30 and the plate 10.

The second hole portion 24 of the combination opening 20 may be an elongated dynamic compression hole. The dynamic compression hole 24 may be elongated such that it has a length greater than its width. The hole 24 may be elongated along the longitudinal axis of the plate 10. In the alternative, the hole 24 may be generally cylindrical such that the hole 24 only permits polyaxial movement of the fastener 40. The inner surface of the hole 24 may be substantially smooth such that the non-locking fastener 40 is able to freely pivot and/or slide along the hole 24. This provides for at least two directions of compressive force (e.g., along the longitudinal axis and perpendicular to the longitudinal axis of the plate 10). The head portion 42 of the non-locking fastener 40 may be substantially smooth around its outer surface. The head portion 42 is sized and configured to engage with and be retained within the hole portion 24 of the combination opening 20. The hole 24 may be configured to receive a fixed or variable angle fastener 40. In one embodiment, the hole 24 may be generally conical in shape and/or tapered such that it is wider near the top surface 16 of the plate 10 and narrower toward the bottom surface 18 of the plate 10. In this embodiment, the hole 24 is a smooth variable angle conical hole configured to receive the non-locking fastener 40. The hole 24 may receive the fastener head 42 allowing movement of the fastener 40, for example, in a polyaxial fashion and/or along the length of the hole 22, thereby providing dynamic compression of the bone.

Turning now to FIGS. 4-7, alternative types of openings 20A-20G, which provide for locking and/or non-locking, dynamic compression are provided. As many of the features of these openings are similar to the combination openings 20 described already for FIGS. 2-3, only the different features will be further explained.

With reference to FIGS. 4A-4C, the combination opening 20A is similar to combination opening 20 except that the dynamic compression hole 24A has the same general diameter as the locking hole 22A, and the locking hole 22A includes a different type of textured portion 26A. In this embodiment, the locking hole 22A has a first diameter D1, and the dynamic compression hole 24A has a second diameter D2. Unlike the elongated hole 24 described earlier, dynamic compression hole 24A has substantially same diameter as the locking hole 22A. Thus, the first and second diameters D1, D2 are substantially the same. The hole 24A may be formed by milling or drilling a sphere out of the plate 10 in the center of the circle with tapers or ramps on either side. The hole 24A is not elongated, but is generally circular and the non-locking fastener 40 will be allowed to translate in the hole 24A because the diameter of the head portion 42 and/or shaft (e.g., bone thread) will be smaller than the size of the hole 24A in the plate 10. With respect to hole 22A, the textured portion 26A of the hole 22A may be in the form of a tapered thread. This tapered thread may generally correspond to a similar tapered thread on the locking fastener 30. This hole 22A also does not include a tapered portion, and the textured portion 26A begins at the intersection with the top surface 16 of the plate 10. This alternative opening 20A also provides for the use of both locking and non-locking fasteners 30, 40 that are able to dynamically compress bone and/or lock the plate 10 to the bone.

Turning now to FIGS. 5A-5C, the combination opening 20B is similar to other combination openings except that the locking hole 22B includes a different type of textured portion 26B. The textured portion 26B includes a series of alternating recesses and protrusions around a central portion

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of the hole 22B. The recesses may be in form of a wave of alternating cutouts extending around the inner perimeter of the hole 22B. The textured portion 26B may lock the fastener 30 with a friction fit or may be modified during insertion of the fastener 30 to form a lock in situ. In this embodiment, the locking hole may allow for polyaxial locking. The plate 10 and the locking fastener 30 may be made of dissimilar materials having dissimilar hardness values. For example, the fastener 30 may have a higher hardness (e.g., on the Rockwell scale) relative to the plate 10, which may be formed of a material having a lower relative hardness value. Due to the increased hardness, the head portion 32 of the locking fastener 30 may create a thread in the plate 10 as the fastener 30 is inserted (e.g., threaded) into the hole 22B, thereby locking the fastener 30 to the plate 10.

With reference to FIGS. 6A-6C, the opening 20C includes locking hole 22C and dynamic compression hole 24C with a more open configuration. The locking portion 22C has a textured portion 26C in the form of a tapered thread. This tapered thread may generally correspond to a similar tapered thread on the locking fastener 30. The opposite portion 24C of the opening 20C is oblong with a ramp 25C milled into the top surface 16 of the plate 10 to allow for dynamic compression. As best seen in FIG. 6C, the ramp may be partially spherical in shape and extend from the top surface 16 of the plate 10 and connect to the textured portion 26C. When viewed from above in FIG. 6B, the ramp 25C creates a square-like, key-hole, and/or non-hole geometry that sweeps into the tapered threaded locking hole 22C. This alternative opening 20C also provides for the use of both locking and non-locking fasteners 30, 40 that are able to dynamically compress bone and/or lock the plate 10 to the bone.

Turning now to FIGS. 7A-7C, the opening 20D includes locking hole 22D and dynamic compression hole 24D. These holes 22D, 24D are connected and close together but are not overlapping. The holes 22D, 24D are separated by a small portion or sliver of plate material proximate to the lower portion of the holes 22D, 24D (e.g., at bottom surface 18 of the plate 10 and partially extending between the holes 22D, 24D). The locking portion 22D has a textured portion 26D in the form of a tapered thread. The textured portion 26D extends around almost the entire circumference of the hole 22D except where connected to hole 24D. The dynamic compression hole 24D is elongated and has ramped portions 25D on opposite sides of the hole 24D to receive fastener 40. This configuration allows for a very close population of holes 22D, 24D on the plate 10 while giving structural stability at the holes 22D, 24D.

With reference to FIGS. 8A-8C, locking hole 22E and dynamic compression hole 24E are adjacent, but separate from one another. The holes 22E, 24E are completely separated from one another by a wall 56 of plate material. The locking portion 22E has a textured portion 26E in the form of a tapered thread extends around the entire perimeter of the hole 22E. The dynamic compression hole 24E is elongated and has ramped portions 25E on opposite sides of the hole 24E. This configuration also allows for a very close population of holes 22E, 24E on the plate 10 while giving options for both locking and/or dynamic compression.

Turning now to FIGS. 9A-9D, an alternative version of opening 20F is provided. In this embodiment, the hole construct 20F is comprised of at least three overlapping conical threaded holes in the plate 10. The opening 20F includes a first, locking hole 22F, a second hole 24F, and a third hole 23F arranged along a longitudinal axis of the plate

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10. The third hole 23F is the mirror image of hole 24F across the first locking hole 22F. The conically threaded holes 22F, 23F, 24F may or may not have parallel axes. Each hole 22F, 23F, 24F may include a textured portion 26F, for example, in the form of one or more threaded portions. Thus, the locking fastener 30 may lock to any of the holes 22F, 23F, 24F. Although each of the holes 22F, 23F, 24F are shown in with the textured portion 26F, it will be appreciated that one or more of the holes 22F, 23F, 24F may have a substantially smooth inner portion instead of the textured portion 26F. The upper part of the hole construct at the first and second ends of the hole 20F each have a ramped feature 25F (e.g., adjacent to holes 23F and 24F) to allow for dynamic compression of the plate 10. In addition, the ramped feature 25F may span the three or more conical holes 22F, 23F, 24F (e.g., around the entire perimeter of the opening 20F).

The non-locking compression fasteners 40 may have a major bone thread diameter such that the fastener 40 can translate between overlapping holes 22F, 24F, 23F without interference. As best seen in FIG. 9D, the locking fastener 30 may include a textured area 36, for example, in the form of a thread, configured to engage with the textured portion 26F of any of the holes 22F, 23F, 24F. The hole geometry of opening 20F can be applied to bone plates 10 to utilize either fixed angle and/or variable angle locking screws 30 and/or polyaxial non-locking screws 40 that can achieve dynamic compression. This allows surgeons more flexibility for screw placement, based on preference, anatomy, and fracture location.

Turning now to FIGS. 10A-10B, another embodiment of opening 20G is provided. This opening 20G may be comprised of one elongate hole or slot extending from the top surface 16 to the bottom surface 18 of the plate 10. A locking portion 22G of the opening 20G may include a textured portion 26G having straight machine threads. The threads may extend more than 180 degrees to retain the locking fastener 30. A non-locking portion 24G of the opening 20G may be positioned opposite the locking portion 22G to complete the opening 20G. The upper part of the opening 20G may have one or more ramped features 25G to allow for dynamic compression of the plate 10. The ramp 25G may span along the entire upper perimeter of the elongated slot 20G or a portion thereof. The compression screws 40 may have a major bone thread diameter such that the screws 40 are able to translate along the opening 20G without interference.

With reference to FIGS. 11A-11E, alternative embodiments of the locking fastener 30 may be used with any plate 10. The head portion 32 of the fastener 30 may include a textured area 36 in the form of a thread, for example, to lock the fastener 30 to the plate 10. The fastener 30 and/or plate 10 may also include one or more mechanisms to prevent back out of the fastener 30 from the plate 10. In FIG. 11A, the head portion 32 includes a threaded portion 36A (e.g., having straight threads) that interface with the plate 10 and the top of the head extends larger than the threads. The head portion 32 bottoms out when the fastener 30 is fully inserted and creates preload in the fastener 30, thus locking the fastener 30 rotationally. In FIG. 11B, the head portion 32 includes threaded portion 36B. The head portion 32 has a constant major diameter while the minor diameter is tapered. The thread depth may go to zero at the top of the head portion 32 of the screw 30. The first few turns smoothly insert, but as the tapered portion of the male thread engages with the plate 10, interference occurs, jamming and/or locking the screw 30 and preventing backout. In FIG. 11C, a screw thread 36C on the head portion 32, similar to the

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design in FIG. 11B, except the minor diameter of the screw 30 stays constant while the major diameter of the head portion 32 gets larger toward the top of the screw 30. A similar jamming and locking mechanism results through tightening of the screw 30 in the plate 10. In FIG. 11D, the threaded portion 36D has areas of varying pitch. In particular, a straight screw thread on the head portion 32 of the screw 30 has a similar pitch to that of the plate 10 at the bottom of the head portion 32 of the screw 30. The pitch then increases or decreases towards the top of the head portion 32, which thereby results in jamming of the threads and preventing unwanted backout of the screw 30. In an alternative variation of the concept of FIG. 11D, shown in FIG. 11E, the opening in the plate 10 is provided with areas of varying pitch while the pitch of the threaded portion 36D remains constant. For example, the head portion 32 may include a straight thread with a constant pitch. The upper surface of the plate 10 may include a thread pitch is similar to that of the screw 10, but towards the bottom surface of the plate 10, the thread pitch would either increase or decrease to lock the screw 30 to the plate 10.

Turning now to FIGS. 12A and 12B, the plate 10 includes an additional anti-backout feature. In this embodiment, the plate 10 includes cylindrical holes or openings 20H configured to accept either the compression fastener 40 or the locking fastener 30. Each opening 20H may include a ramped portion 25H extending around a portion or the entire perimeter of the opening 20H to allow for dynamic compression with a compression fastener 40. Each opening 20H may include a cylindrical feature to provide angular stability with a locking fastener 30. The opening 20H may also include an angular taper 28 to cause compressive tightening between the locking fastener 30 and the cylindrical opening 20H. Each opening 20H has an accompanying blocking screw 46 that can be actuated to block the fastener 30, 40 from backing out. The blocking screw 46 may extend from a first end at the top surface 16 to a second end at the bottom surface 18 of the plate 10. The first end of the blocking screw 46 may include a recess sized to receive an instrument to rotate the blocking screw 46 from an unblocked position to a blocked position. The blocked position may include a portion of the blocking screw 46 covering a portion of the head portion 42 of the fastener 40, thereby further preventing backout of the fastener 40 from the plate 10.

According to yet another embodiment, the plate 10 may include one or more openings 20 configured to receive the locking fastener 30 having self-forming threads that work by displacement of the plate material to lock the fastener 30 to the plate 10. Turning now to FIGS. 13-18, the locking fastener 30 and alternative embodiments of the openings 20 in the plate 10 are shown. In these embodiments, the locking mechanism of the fastener 30 (e.g., bone screw) to the internal fixation plate 10 may allow for variable angle screw insertion. The fastener 30 may be inserted within an angular cone where the force required to dislodge the head portion 32 of the fastener 30 is substantially equivalent to the force required when the fastener 30 is inserted perpendicular to the plate 10. The holes or openings 20 in the plate 10 may be shaped such that the fastener 30 may be inserted at different angles. The geometry of the opening 20 is conducive to catching the threads on the head portion 32 of the fastener 30 and to reduce the axial force necessary to initiate the thread formation.

The locking mechanism includes a fastener 30 having a head portion 32 with self-forming threads that displace the plate material. The plate 10 may be made of a material softer than the fastener 30 to facilitate displacement. For example,

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the plate 10 may be comprised of titanium, alloys, polymers, or other materials having a lower material hardness (e.g., Rockwell hardness). The fastener 30 may be made of a harder relative material, for example, comprised of cobalt chrome, tungsten, alloys, or other materials having a higher material hardness. Preferably, the fastener 30 is comprised of a material having a strong, stiff, and high surface hardness which facilitates the thread forming process. The forming mechanism works by displacement of material rather than removal of the material of the plate 10, thereby minimizing fragments or chips which are created from tapping.

In FIGS. 13A-13B, the locking fastener 30 includes a head portion 32 and a shaft portion 34 configured to engage bone. Although not shown, the shaft portion 34 may be threaded such that the fastener 30 may be threaded into the bone. The head portion 32 may be tapered (e.g., at an angle of about 20°) such that the fit within the opening 20 in the plate 10 becomes tighter as the fastener 30 is advanced in to the bone. The head portion 32 of the locking fastener 30 includes a textured area 36 around its outer surface sized and configured to engage an opening 20 in the plate 10. The textured area 36 may include threads, ridges, bumps, dimples, serrations, or other types of textured areas. As shown, the textured area 36 preferably includes a threaded portion extending substantially from the top of the head portion 32 to the bottom of the head portion 32 proximate to the shaft portion 34. The threads 36 may run generally perpendicular to the conical surface of the head portion 32. The threaded portion 36 is in the form of self-forming threads configured to displace the plate material and create threads in the opening 20 of the plate 10. The threaded portion has an exaggerated sharp thread peak to facilitate cutting or forming of the plate material.

Turning now to FIGS. 13C-17, alternative versions of the openings 20 are shown before being tapped with the fastener 30. Once the fastener 30 is inserted, these openings 20 are modified based on the self-forming threads. The geometry of the openings 20 are conducive to catching the threads 36 and designed to reduce the axial force necessary to initiate the thread formation. An upper portion of the hole 20 may be tapered 28, for example, with a conical straight tapered surface cut through the top surface 16 of the plate 10 for clearance of the head portion 32 of the fastener 30 during off angle insertion. A lower portion of hole 20 may further be tapered 29, for example, with a conical straight tapered surface cut through the bottom surface 18 of the plate 10 for clearance of the shaft portion 34 during off angle insertion. The upper tapered portion 28 may be larger, for example, with a larger degree of taper than the lower tapered portion 29. For example, the upper tapered portion 28 may have a taper in a range from about 60-90°, 70-80°, or 72-78°, preferably about 70°, 75°, or 80° whereas the lower tapered portion 29 may have a taper in a range from about 50-70°, 55-65°, or 57-63°, preferably about 55°, 60°, or 65°. The upper and/or lower tapered portions 28, 29 may be substantially conical (e.g., FIGS. 14B, 15B, 15C, 16B) or may be segmented with more than one section, such as two separate conical sections having different diameters or degrees of taper (e.g., FIGS. 17A and 17B).

At the intersection between the upper tapered portion 28 and the lower tapered portion 29, a narrowed central portion, as indicated by the area 31 within the dashed lines of FIG. 13C defines the area where thread forming takes place. The area 31 provides a concentric ring of material for material displacement and thread forming. The area 31 may have the untextured surface illustrated in FIG. 13C or may have a textured portion 26. As described herein, the textured por-

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tion 26 may include threads, ridges, bumps, dimples, serrations, or other types of textured areas. In the embodiment shown in FIGS. 14A-14B, the textured portion 26 includes a windswept cut design comprised of a plurality of shallow cuts where each cut overlaps the next. For example, the windswept design may include a plurality of threadlike helical cut sweeps. Each cut has a smooth transition into the inner diameter of the hole 20 (e.g., into the upper and lower tapered portions 28, 29). The windswept cuts provide a positive surface for the self-forming threads to cut into, thereby helping to prevent peeling of the newly formed threads into the plate 10.

In FIGS. 15A-15D, the textured portion 26 includes a knurled cut design. In the embodiment illustrated in FIGS. 15A-15B, a rounded transition between the upper tapered portion 28 and the lower tapered portion 29 (e.g., the two conical cuts) provides a workable surface for the knurling process as well as a surface for the head portion 32 to be able to roll over during off-axis locking. The knurled design may include a plurality of shallow knurled grooves set in a diamond pattern (e.g., about 45°) where each cut overlaps the next. The knurled grooves allow for the self-forming threads to cut more deeply into the material and reduce the necessary axial force to begin the thread forming process. In the embodiment illustrated in FIGS. 15C-15D, the area 31 has a textured portion 26 defined by a plurality of 360° circular swept cuts. Additionally, a series of 60° triangular cuts is made at a 17° trajectory from a plane normal to hole axis, with the same number of cuts being applied in both a clockwise and counter-clockwise fashion, creating a pattern on the inside ring of material. The cuts create "plateaus" of material protruding into the hole, as shown. While specific angles are described, the disclosure is not limited to the specific angles. The resultant geometry from the cuts provides positive surfaces to cut into, dramatically reducing the axial force necessary to lock the screw to the plate. As such, the mechanism does not rely on bone purchase to engage the threads in the head of the screw. Secondly, the material removed by the cuts allow the head threads to cut deeper by reducing the amount of material which must be formed, and reducing friction between the screw 30 and plate 10 during the forming process.

FIGS. 16A-16B depict a polygon form cut design. In this design, there is no textured portion at the transition between the upper tapered portion 28 and the lower tapered portion 29. Instead, the narrowed central region has an overall polygonal form such that the hole 20 is neither cylindrical nor conical. The polygonal shape includes a number of sides with distinct linear section of material and rounded corners around which the form cut is allowed to sweep. For example, the polygonal shape may be substantially hexagonal (6-sided), heptagonal (7-sided), octagonal (8-sided), etc. The hole 20 may also be represented without lobe cuts, as a single concentric ring with the same geometry.

In FIG. 17A, the upper tapered portion 28 includes a conical straight tapered surface cut for clearance of the head portion 32 of the fastener 30 during off angle insertion. The upper tapered portion 28 is segmented to have an upper area with a larger area relative to a lower area proximate the transition to the lower tapered portion 29 having a narrower diameter. The central area between the upper and lower tapered portions 28, 29, where the thread forming process occurs, includes two peaks or concentric rings of material (e.g., a superficial ring 60 and a deep ring 62) with a groove 27 being located in between for material removal and thread forming relief. The groove 27 between the rings 60, 62 may be angled, for example, in the range of about 40-80°,

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about 50-70°, or about 60°. The superficial ring **60** is of a slightly smaller inner diameter than the deep ring **62**, as the superficial ring **60** is responsible for supporting a majority of the cantilever loads. The deep ring **62** provides additional fixation and support during off-angle insertion as well as additional support during nominal trajectory insertion. The lower tapered portion **29** includes a straight tapered surface that provides clearance for the shaft **34** of the fastener **30** when inserted off angle.

The embodiment of the opening **20** in FIG. **17B** is similar to FIG. **17A**, but further includes textured portion **26** in the form of a plurality of helical swept cuts at the transition between the upper tapered portion **28** and the lower tapered portion **29**. The shallow helical cuts or windswept cuts may include a series of cuts at a steep pitch. The windswept cuts may be angled, for example, at about 50-70°, or about 60°. The same number of cuts may be made in both a clockwise and counter-clockwise fashion. The cuts may create plateaus of material protruding into the opening **20**. The resultant geometry provides positive surfaces for the fastener **30** to cut into, which can dramatically reduce the axial force necessary to lock the fastener **30** to the plate **10**. This mechanism does not need to rely on bone purchase in order to engage the threads in the head portion **32** of the fastener **30**. The material removed during insertion of the fastener **30** allows the self-forming threads to cut deeper by removing material which much be formed and reducing friction between the fastener **30** and the plate **10** during the forming process.

FIGS. **18A-18D** depict a screw-plate assembly. The assembly, in FIG. **18C**, shows the locking fastener **30** placed at an angle, other than perpendicular, to the upper surface **16** of the plate **10**. In FIG. **18D**, a non-locking fastener **40** is placed generally perpendicular to the plate **10**. It will be appreciated that the locking fastener **30** and non-locking fastener **40** may be oriented at any appropriate angle relative to the plate **10**. The section view in FIG. **18C** shows the thread engagement with the plate **10** in which material of the plate **10** is displaced around the threads of the fastener **30**. By using the self-forming threads, the fastener **30** is able to be inserted into the plate **10** at variable angles and engages with the plate **10** with one-step locking requiring no additional steps to lock the fastener **30** to the plate **10**. The section view in FIG. **18D** show the compressive, non-locking screw **40** received in the opening **20**, without threaded locking thereto. The non-locking screw **40** may provide for dynamic compression of the bone. Accordingly, the fasteners and openings described herein provide a wide variety of options for the surgeon, thereby providing appropriate locking and/or unlocking capability for dynamic compression depending on the desired treatment of the fracture and the bone.

Dia-Meta Volar Distal Radius Plate System

FIG. **19** depicts embodiments of a dia-meta volar distal radius stabilization system **200** including a bone plate **210** configured to sit against the volar side of the radial bone and one or more bone fasteners are configured to be received in the bone plate **210** and secured to the radius and radial shaft of a bone. Although generally described with reference to the radius and radial shaft, it will be appreciated that the stabilization system **200** described herein may be used or adapted to be used for the fixation of other long bones as well, such as the humerus, femur, tibia, etc.

The bone plate **210** extends from a first end **212** configured to be positioned on a shaft portion of radial bone to a second end **214** configured to be positioned proximate to the distal end of the radius. The plate **210** includes a top surface

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216 and an opposite, bottom surface **218** configured to contact adjacent bone. The top and bottom surfaces **216**, **218** are connected by opposite side surfaces extending from the first to second ends **212**, **214** of the plate **210**. The bottom surface **218** of the plate **210** includes an anatomic contour configured to follow the best approximation of average distal radial anatomy, flaring up slightly along the radial column and more significantly along the intermediate column of the plate **210**. The plate **210** is designed to sit low and have a generally low profile proximal portion. The thickness of the plate **210** may generally be about 2 mm along the shaft and distal intermediate column, tapering to a thickness of 2.5 mm along the distal radial column which allows for the severe angle of the radial styloid fastener. The thickness of the plate **210** may generally increase towards the first end **212** when compared to the second end **214**. In addition, the width of the plate **210** proximate the first end **212** and along the elongate portion **240** may be thicker than the width of the plate at the second end **214**. The design of plate **210** allows for an easy transition from the second end **214** of the plate **210** to the elongate portion **240** to the first end **212** of the plate **210** to address fractures proximal to the second end of the plate **214** while also providing adequate support in the radial shat of the bone.

The second end **214** of the bone plate **210** toward the elongate portion **240** of the bone plate **210** is very similar to the bone plate **110**, thus the features and disclosures set forth above relating to the bone plate **110** are equally applicable to bone plate **210** and are incorporated in their entirety herein.

Looking at the elongate portion or dia-meta portion **240** of the plate **210**, the plate **210** includes one or more through openings **220** configured to receive one or more bone fasteners. The openings **220** extend through the body of the plate **210** from the top surface **216** to the bottom surface **218**. The openings **220** may include cylindrical openings, conical openings, elongated openings, threaded openings, textured openings, non-threaded and/or non-textured openings, and the like. The openings **220** may allow for locking of the fastener to the plate **210** or may allow for movement and dynamic compression of the bone. The plate **210** may comprise any suitable number of openings **220** in any suitable configuration. These openings **220** allow surgeons more flexibility for fastener placement, based on preference, anatomy, and fracture location. Surgeons may have differing opinions as to the number, location, and types of fasteners. Further, complexity of fracture location and shape makes having as many locations for fasteners as possible necessary. This design offers surgeons a versatile method to achieve higher accuracy in placement of the fasteners.

The openings **220** may be configured to receive one or more bone fasteners. The fasteners may include locking fasteners, non-locking fasteners, or any other fasteners known in the art. The fasteners may comprise bone screws or the like. The fasteners may also include other fasteners or anchors configured to be secured or engaged with bone, such as nails, spikes, staples, pegs, barbs, hooks, or the like. The fasteners may include fixed and/or variable angle bone screws. The fastener may include a head portion and a shaft portion configured to engage bone. For a locking fastener, the shaft portion may be threaded such that the fastener may be threaded into the bone. The head portion may include a textured area, such as threads, around its outer surface sized and configured to engage with the opening **220**, for example, and corresponding threads in the opening **220** in order to lock the fastener to the plate **210**. In the alternative, for a

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non-locking fastener, the head portion may be substantially smooth to allow for dynamic compression of the bone.

The plate 210 may further comprise a plurality of openings 224 configured to receive one or more k-wires (not shown). The k-wire holes 224 may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings 220). The k-wire holes 224 may allow preliminary placement of the plate 210 against the bone and/or to aid in reduction of the fracture. The distal k-wire holes 224 on the head portion 242 may ensure a trajectory to follow the RC joint and provide direction during insertion of the distal locking screws. The proximal k-wire holes in the elongated portion 240 of the plate 210 are arranged between fastener openings 220 and may be angled relative to the surface of the plate 210 to avoid intrusion into areas where instrumentation must pass during screw insertion.

Dorsal Plate System

FIGS. 20-24 depict embodiments of a dorsal stabilization system 300 including bone plates 310, 410 which are configured to sit against the dorsal portion of bone. One or more bone fasteners 320C are configured to be received in the bone plates 310, 410 to secure the plates 310, 410 to the dorsal portion of a bone. Although generally described with reference to the dorsal portion of bone, it will be appreciated that the stabilization system 300 described herein may be used or adapted to be used for the fixation of other bones as well, such as other portions of the identified bones. It should be noted that the same reference numerals are being used for plates 310, 410 because the plates are similar except for their respective first ends 312 which show different opening 320 configurations. FIG. 20 shows an acute configuration and FIG. 22 shows an oblique configuration.

As shown in FIGS. 20-22, the plates 310, 410 each have a body that extends from a first end 312 to a second end 314. The plates 310, 410 each include a top surface 316 and an opposite, bottom surface 318 configured to contact adjacent bone. The top and bottom surfaces 316, 318 are connected by opposite side surfaces extending from the first to second ends 312, 314 of the plate 310. Although the plate 310, 410 are shown having a generally longitudinal body, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

The bone plates 310, 410 include one or more openings 320. The openings 320 extend through the plate 310, 410 from the upper surface 316 to the bottom surface 318 and are configured to accept locking fasteners and non-locking fasteners 320C. When using the plates 310, 410 with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plates 310, 410. The openings 320 may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the dia-meta plate system, and the alternative hole configurations.

The plates 310, 410 also include one or more slots 320C present along the elongated portion 340 of the plates 310, 410 and configured to accommodate a sliding fastener 322C, shown in FIGS. 23 and 24. As best seen in FIGS. 20-24, the slot 320C may offer a sliding slot for proximal-distal adjustment of the plates 310, 410 during provisional placement. The slot 320C may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plates 310, 410. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The slot 320C may be elongated along a longitudinal axis of the elongated portion 340 as well as elongated, perpendicular to

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the longitudinal axis, from lateral side to lateral side. The elongated slot 320C may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot 320C. The plates 310, 410 may include etch lines adjacent to slot 320C for more accurate adjustment of the plate 310 when being positioned on bone.

As best seen in FIGS. 20 and 22, plates 310, 410 also may include a plurality of side relief cuts or scalloped edging 322 along the length of the plates 310, 410 which allows the plates 310, 410 to be bent, for example, in three dimensions. The side relief cuts or scalloped edges 322 may be in the form of one or more curves having a widened portion along the sides of the plates 310, 410 and a narrowed portion towards the center of the plates 310, 410. The side relief cuts or scalloped edges 322 may be positioned between consecutive openings 320. The plurality of relief cuts or scalloped edges 322 may form a scalloped or wavy profile along the side edges of the plates 310, 410. As a result, the plates 310, 410 are able to be shaped to a multi-contour surface without warping the openings 320.

The plates 310, 410 may further comprise a plurality of openings 324 configured to receive one or more k-wires (not shown). The k-wire holes 324 may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings 320). The k-wire holes 324 may allow preliminary placement of the plates 310, 410 against the bone and/or to aid in reduction of the fracture.

Lateral Plate

FIGS. 25-27 depict embodiments of a lateral stabilization system 500 including bone plate 510 which is configured to sit against the lateral portion of bone to address fractures on the side of the radius. One or more bone fasteners 520C are configured to be received in the bone plate 510 to secure the plate 510 to the lateral portion of a radius of a bone. Although generally described with reference to the lateral portion of the radius of the bone, it will be appreciated that the stabilization system 500 described herein may be used or adapted to be used for the fixation of other bones, such as long bones, as well as other portions of the identified bones.

The plate 510 has a body that extends from a first end 512 to a second end 514. The plate 510 includes a top surface 516 and an opposite, bottom surface 518 configured to contact adjacent bone. The top and bottom surfaces 516, 518 are connected by opposite side surfaces extending from the first to second ends 512, 514 of the plate 510. Although the plate 510 is shown having a generally longitudinal body, that contours or radius upwardly to accommodate distal radius bony anatomy, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

The bone plate 510 includes one or more openings 520. The openings 520 extend through the plate 510 from the upper surface 516 to the bottom surface 518 and are configured to accept locking fasteners and non-locking fasteners 520C. When using the plate 510 with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plate 510. The openings 520 may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the dia-meta plate system, the dorsal plates and the alternative hole configurations.

The plate 510 also includes one or more slots 520C present along the elongated portion 540 of the plate 510 and configured to accommodate a sliding fastener 522C, shown in FIG. 27. As best seen in FIGS. 25-26, the slot 520C may offer a sliding slot for proximal-distal adjustment of the plate 510 during provisional placement. The slot 520C may allow

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for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plate **510**. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The slot **520C** may be elongated along a longitudinal axis of the elongated portion **540** as well as elongated, perpendicular to the longitudinal axis, from lateral side to lateral side. The elongated slot **520C** may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot **520C**. The plate **510** may include etch lines adjacent to slot **520C** for more accurate adjustment of the plate **510** when being positioned on bone.

As best seen in FIGS. **25** and **27**, plate **510** also may include a plurality of side relief cuts or scalloped edging **522** along a portion of the length of the plate **510** which allows that portion of the plate **510** to be bent, for example, in three dimensions. The side relief cuts or scalloped edges **522** may be in the form of one or more curves having a widened portion along the sides of the plate **510** and a narrowed portion towards the center of the plate **510**. The side relief cuts or scalloped edges **522** may be positioned between consecutive openings **520**. The plurality of relief cuts or scalloped edges **522** may form a scalloped or wavy profile along the side edges of the plate **510**. As a result, a portion of the plate **510** is able to be shaped to a multi-contour surface without warping the openings **520**.

The plate **510** may further comprise a plurality of openings **524** configured to receive one or more k-wires (not shown). The k-wire holes **524** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **520**). The k-wire holes **524** may allow preliminary placement of the plate **510** against the bone and/or to aid in reduction of the fracture.

Bridge Plate

FIG. **28** depicts an embodiment of a stabilization system **600** including bone plate **610** which acts as an internal fixator for high energy comminuted distal radius fractures. The plate **610** is placed dorsally and extends from the third or second metacarpal to approximately a third to half way down the radius. One or more bone fasteners are configured to be received in the bone plate **610** to secure the plate **610** to the desired portions of bone. Although generally described with reference to the radius and metacarpals, it will be appreciated that the stabilization system **600** described herein may be used or adapted to be used for the fixation of other bones, such as long bones, as well as other portions of the identified bones.

The plate **610** has a body that extends from a first end **612** to a second end **614**. The plate **610** includes a top surface **616** and an opposite, bottom surface **618** configured to contact adjacent bone. The top and bottom surfaces **616**, **618** are connected by opposite side surfaces extending from the first to second ends **612**, **614** of the plate **610**. Although the plate **610** is shown having a generally longitudinal body that is generally planar, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

The bone plate **610** includes one or more openings **620**. The openings **620**, which are located proximate the first end **612** and the second end **614**, extend through the plate **610** from the upper surface **616** to the bottom surface **618** and are configured to accept locking fasteners and non-locking fasteners. When using the plate **610** with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plate **610**. The openings **620** may be in the form of any of the openings discussed above with respect to the volar

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distal radial plate system, the dia-meta plate system, the dorsal plates, the lateral plates and the alternative hole configurations.

Lunate Facet Hook Plate

FIGS. **29-36** depict embodiments of a stabilization system **700** including hook plate **710**, **710'** which is designed for fracture patterns that involve the volar ulnar corner of the distal radius. The plate **710**, **710'** may be used as a stand-alone stabilization plate, as shown in FIGS. **32** and **34** or may be used in combination with a volar distal radius plate **110**, as shown in FIGS. **31**, **33** and **35-36**.

When the plate **710** is used alone, the hooks **712** of the plate are embedded or tapped into bone to prevent the shifting of the plate in a lateral or medial direction. It is contemplated that there may be one, two, or more hooks **712**. The plate **710** also includes an opening **720** to receive a fixation screw **714**, which may aid in further fixation of the plate **710** the bone and the fracture site.

When the plate **710** is used with the volar distal radius plate, the plate **710** is configured and dimensioned such that it can be slidably placed under a pre-positioned volar distal radius plate **110**. The opening **720** will align with an opening **120** on the volar distal radius plate **110** such that a fastener will pass through the opening **120** on the volar distal radius plate **110** and the opening **720** on the plate **710**. The opening **720** can accept a locking screw or a non-locking screw.

Referring to FIGS. **32** and **34A-34E**, the plate **710'** is similar to the previous embodiment and includes a pair of hooks **712'** extending from the body **716** of the plate **710'** with a transition area **713** therebetween. The hooks **712'** have an arcuate configuration such that the hooks **712'** complement the curvature of the rim of the lunate facet (see FIGS. **32** and **34E**). The transition area **713** narrows between the elongate body **716** and the hooks **712'**. An elongate slot **720** extends through the elongate body **716** and is configured to receive a fixation screw **714**. As seen in FIG. **34E**, the elongate body **716** may have an initial curvature or be bent to complement the contour of the bone **102**.

Referring to FIGS. **34A-34E**, an illustrative method of installing the plate **710'** as a stand-alone stabilization plate will be described. As shown in FIG. **34A**, an inserter **730** is utilized to hold plate **710'** and apply the plate **710'** to the bone **102** with the hooks **712'** positioned over the lunate facet. A tamp **732** is struck with a mallet **734** or the like to tamp the hooks **712'** in or over the facet as shown in FIG. **34B**. With the plate **710'** in position, a hole is drilled through the slot **720**, for example, utilizing a drill **738** with a bit passing through a soft tissue protector **736** as illustrated in FIG. **34C**. It may be desired to position the hole in a proximal portion of the slot **720**. After the hole is drilled, the desired screw length may be determined utilizing a depth gauge or the like (not shown). The desired fixation screw **714** may then be inserted into the hole using a driver **740** or the like, as shown in FIG. **34D**. Non-locking screws lag the plate **710'** to the bone **102** and help to maintain fracture compression and avoiding last translation of the plate **710'** distally. FIGS. **32** and **34E** illustrate the final construct of the stabilization plate. Fluoroscopy, as shown in FIG. **34E**, may be utilized to confirm proper screw **714** placement.

Turning to FIGS. **35A-35E**, a first illustrative method of installing the plate **710'** along with a volar distal radius plate **110** will be described. In this illustrative method, the plate **710'** is positioned prior to the volar plate insertion. Initially, the plate **710'** is positioned relative to the bone **102** and the hooks **712'** are tamped into position in a manner similar to that described with respect to FIGS. **34A** and **34B**. With the plate **710'** in position, a k-wire **742** is inserted through the

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slot 720 and into the bone 102 as shown in FIG. 35A. The volar plate 110 is then positioned such that the k-wire 742 extends through one of the screw holes 120A in the head portion 142 of the volar plate 110 as shown in FIG. 35B. The inserter 730 may be utilized to hold and manipulate the volar plate 110. The volar plate 110 is slid along the k-wire 742 such that the volar plate 110 is properly positioned on the bone 102 and is covering the hook plate 710' as shown in FIG. 35C. When positioned correctly, the ulnar-most subchondral locking screw hole 120A aligns with the slot 720 of the hook plate 710'. With the volar plate 110 positioned properly, a fixation screw 714 is inserted through the screw hole 120A and the slot 720 and secured within the bone. FIGS. 33 and 35E illustrate the final construct of the stabilization assembly. Fluoroscopy, as shown in FIG. 35E, may be utilized to confirm proper screw 714 placement and proper placement of the hook plate 710'.

Referring to FIGS. 36A and 36B, another illustrative method of installing the plate 710' along with a volar distal radius plate 110 will be described. In this illustrative method, the volar plate 110 has already been installed and thereafter it is determined that a hook plate 710' is desired, for example, when an unstable lunate facet fracture is identified after the volar plate fixation. Initially, at least one of the fixation screws securing the head portion of the volar plate 110 is removed (not shown). Thereafter, the elongate body 716 of the hook plate 710' is slid behind the volar plate 110 and the hooks 712' are tamped into position as shown in FIG. 36A. With the plate 710' in position, a k-wire 742 is inserted through the slot 720 and into the bone 102 as shown in FIG. 35A. The volar plate 110 is then positioned such that the k-wire 742 extends through one of the screw holes 120A in the head portion 142 of the volar plate 110 as shown in FIG. 35B. Once the hook plate 710' is positioned properly, a fixation screw 714 is inserted through the screw hole 120A and the slot 720 and secured within the bone as illustrated in FIG. 36B. The final construct of the stabilization assembly will be as shown in FIGS. 33 and 35E. Again, fluoroscopy, as shown in FIG. 35E, may be utilized to confirm proper screw 714 placement and proper placement of the hook plate 710'.

FIGS. 37 and 28 show a lunate facet hook plate reduction instrument 810. The instrument is capable of being connected to any quick connect handle known in the industry, such as the AO quick-connect handle. The reduction instrument 810 utilizes a two-piece contact surface that is capable of capturing a lunate facet hook plate and releasing the hook plate when it is positioned in the desired location and orientation.

FIGS. 39 and 40 depict a drill guide 910 that can be attached to second end 114 of the volar distal radius plate 110. The drill guide 910 may include a plurality of cannulated openings 912 which correspond to each of the respective openings 120 in the plate 110. The drill guide 910 openings 912 may be configured in order to drill the pilot holes at the appropriate trajectories for each opening 120, and subsequently receive the respective fasteners at the correct trajectories. The drill guide 910 may also include a plurality of k-wire openings 914 which match with the k-wire openings in the plate 110. The drill guide 910 may be secured to the plate 110 with one or more fasteners or may be secured to the plate 110 through an integrated connection system such as a thumb screw, an interference fit, etc. The fastener may thread into the plate 110 or otherwise temporarily secure the drill guide 910 to the plate 110. The drill guide 910 may be pre-assembled to the plate 110 or may be attached at any other suitable time before or during the

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surgery. The fastener may be secured, for example, in the operating room, via thumb or hexalobular fastener, to attach the drill guide 910 to the plate 110. After the pilot holes are drilled, the drill guide 910 may then be removed and the fasteners positioned through the respective openings 120. The drill guide 910 may be relatively slim in thickness, for example, not protruding more than 10 mm above the plate 110, to prevent impinging on soft tissue.

Neck and Head Plate

FIGS. 41A and 41B depict an embodiment of a lateral stabilization system 1000 including bone plate 1010 which is configured to treat fractures of the ulnar neck and head. One or more bone fasteners 1030 are configured to be received in the bone plate 1010 to secure the plate 1010 to the neck and head portion of the ulna 102. The plate 1010 has a body that extends from a first end 1012 to a second end 1014. The plate 1010 includes a top surface 1016 and an opposite, bottom surface 1018 configured to contact adjacent bone. The top and bottom surfaces 1016, 1018 are connected by opposite side surfaces extending from the first to second ends 1012, 1014 of the plate 1010. The first end 1012 of the plate 1010 preferably has a chamfer which allows distal placement of the plate 1010. The second end 1014 of the plate 1010 has an arcuate configuration which complements the curvature of the distal portion of the ulna 102. The plate 1010 has a generally longitudinal body, that contours or radius upwardly slightly to accommodate distal radius bony anatomy, however, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

The bone plate 1010 includes one or more openings 1020. The openings 1020 extend through the plate 1010 from the upper surface 1016 to the bottom surface 1018 and are configured to accept locking fasteners and non-locking fasteners 1030. When using the plate 1010 with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plate 1010. The openings 1020 may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the dia-meta plate system, the dorsal plates and the alternative hole configurations. The proximal most opening 1020A preferably is a polyaxial screw hole which is angled distally. Such a configuration helps to prevent screw impingement on the articular surface.

The plate 1010 also includes one or more slots 1020C present along the elongated portion 1040 of the plate 1010 and configured to accommodate a sliding fastener 1030C. The slot 1020C has a configuration similar to the slot 120C' illustrated in FIG. 1H and may offer a sliding slot for proximal-distal adjustment of the plate 1010 during provisional placement. The slot 1020C may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plate 1010. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The slot 1020C may be elongated along a longitudinal axis of the elongated portion 1040 as well as elongated, perpendicular to the longitudinal axis, from lateral side to lateral side. The elongated slot 1020C may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot 1020C. The plate 1010 may include etch lines adjacent to slot 1020C for more accurate adjustment of the plate 1010 when being positioned on bone.

As best seen in FIG. 41A, plate 1010 also may include a plurality of side relief cuts or scalloped edging 1022 along a portion of the length of the plate 1010 which allows that

portion of the plate **1010** to be bent, for example, in three dimensions. The side relief cuts or scalloped edges **1022** may be in the form of one or more curves having a widened portion along the sides of the plate **1010** and a narrowed portion towards the center of the plate **1010**. The side relief cuts or scalloped edges **1022** may be positioned between consecutive openings **1020**. The plurality of relief cuts or scalloped edges **1022** may form a scalloped or wavy profile along the side edges of the plate **1010**. As a result, a portion of the plate **1010** is able to be shaped to a multi-contour surface without warping the openings **1020**.

The plate **1010** may further comprise a plurality of openings **1024** configured to receive one or more k-wires (not shown). The k-wire holes **1024** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **1020**). The k-wire holes **1024** may allow preliminary placement of the plate **1010** against the bone and/or to aid in reduction of the fracture. Alternatively, as illustrated in FIG. **42A**, a drill guide **744** may be utilized to direct a drill bit of a drill **738** through the openings **1020** through the plate **1010**. For example, the proximal most opening **1020A** and the distal most opening **1020D** may be utilized for positioning of k-wires **742** to provisionally hold the plate **1010** in place. The placement of the k-wires **742** may be confirmed utilizing fluoroscopy as illustrate in FIG. **42B**.

According to one embodiment, the ulna plate **1010** may be used for fixation of an unstable ulna following distal radius repair. Using a subcutaneous ulnar approach, the patient's arm may be positioned on a hand table with the elbow flexed. The forearm may be positioned to expose the subcutaneous border of the ulna. A longitudinal incision may be created distally and proximally. The interval between the extensor carpi ulnaris (ECU) and the flexor carpi ulnaris (FCU) may be split to expose the ulnar shaft. The plate **1010** may be applied dorsally if desired. The fracture may be reduced and the reduction may be confirmed, for example, with fluoroscopy. The speed locking drill guide **744** may be used to place k-wires **742** in the distal and proximal screw holes to provisionally hold the plate in position. A hole may be drilled through the center of the positioning slot, and a screw may be positioned therein, thereby allowing for adjustment of the plate **1010** proximal-distal and/or medial-lateral for optimal placement. The remaining screws may be predrilled and placed and the k-wires may be replaced with locking screws.

Targeting Guide and Spring Drill Guide

FIG. **43** depicts a targeting guide **4300** in accordance with embodiments of the present disclosure. The targeting guide **4300** includes a body **4302** having an upper portion **4304** configured to be attached to the second end **114** of the volar distal radius plate **110** (e.g., the volar distal plate **110** depicts in FIGS. **1D** and **1E**) and a lower portion **4306** having a collet (shown in FIGS. **46** and **47**) configured to couple the targeting guide **4300** to the plate **110**, and a neck portion **4305** extending between the upper and lower portions **4304**, **4306**. The targeting guide **4300** may include a plurality of cannulated openings **4312** which correspond to each of the respective openings **120** in the plate **110**. The openings **4312** may be configured in order to drill the pilot holes at desired trajectories for each opening **120**, and subsequently receive the respective fasteners at the desired trajectories. The targeting guide **4300** may also include a plurality of k-wire openings **4314** which match with the k-wire openings **124** in the plate **110**.

FIG. **44** depicts an alternative targeting guide **4400** in accordance with embodiments of the present disclosure. The

targeting guide **4400** is substantially similar to the targeting guide **4300** except that the targeting guide **4400** has a lower profile design than the targeting guide **4300**. This lower profile design is achieved by eliminating the neck portion such that a lower portion **4406** having the collet is adjacent an upper portion **4404** of the body **4402**. FIG. **45** depicts another embodiment of a targeting guide **4500** according to the present disclosure. The targeting guide **4500** is also substantially similar to the targeting guides **4300** and **4400** except that the targeting guide **4500** has an even lower profile than the targeting guide **4400** to advantageously minimize disruption of soft tissue, maximize visualization of the surgical field, and enable insertion and removal of the targeting guide without removal of any k-wires extending through the distal radius plate **110**. To facilitate such functionality, a body **4502** of the targeting guide **4500** having an upper portion **5404** and a lower portion **4506** is configured so that it does not surround the k-wire openings **124** of the volar distal radius plate **110**.

FIGS. **46-48** illustrate how the targeting guide (i.e., the targeting guides **4300**, **4400**, **4500**) is coupled to the volar distal radius plate **110** in accordance with embodiments of the present disclosure. FIG. **46** is a cross-sectional view of the targeting guide **4300** coupled to the distal radius plate **110**. FIG. **47** is a side view of the targeting guide **4500**. Although specific targeting guides are illustrated in FIGS. **46** and **47**, the following description of the coupling of the targeting guide applies to the other embodiments of the targeting guide discussed herein. As shown more clearly in FIGS. **46** and **47**, each targeting guide (**4300**, **4400**, and **4500**) includes a pair of protrusions **4602** (shown in FIG. **46**) and **4702** (shown in FIG. **47**) that extend towards the distal radius plate to serve as standoffs which provide improved rigidity when the targeting guide is coupled to the distal radius plate. Also shown in FIGS. **46** and **47** is a collet **4604** which is configured to be inserted in one of the openings **120** to couple the targeting guide to the distal radius plate **110**.

The collet **4604** includes a plurality of arms **4606**. In some embodiments, the plurality of arms **4606** may include 3 to 8 arms. The arms **4606** are configured to be deflected radially inward by the walls of the opening **120** during insertion of the collet **4604** into the opening **120** and subsequently return to their initial undeflected position after the collet **4604** is pushed beyond a portion **4608** of the opening **120** having a reduced diameter. In some embodiments, the collet **4604** includes a reduced diameter portion **4610** configured to receive and mate with the portion **4608** of the opening **120** and a flange **4612** disposed distal to the reduced diameter portion **4610** and extending radially outward from the collet **4604**. The reduced diameter portion **4610** and the flange **4612** are configured to lock the collet **4604** (and the targeting guide) to the distal radius plate **110**. The collet **4604** advantageously provides a rigid method of coupling the targeting guide to the distal radius plate without the need for threaded holes.

In some embodiments, the targeting guide may further include a screw **4800** configured to be inserted into a central opening **4614** of the collet **4604** to prevent inadvertent deflection of the collet **4604** radially inward and thus, removal of the targeting guide from the distal radius plate **110**. In some embodiments, a distal end of the screw **4800** may be capped to prevent binding of the screw in the central opening **4614**. As shown more clearly in FIGS. **43-45**, the screw **4800** includes an engagement feature **4802** (e.g., a hexalobular shaped feature) formed in the head **4804** of the screw **4800** to facilitate the engagement of a driver (not shown) with the screw **4800**. The targeting guides **4300**,

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4400, 4500 advantageously allow for attachment of the guide to the distal radius plate on a back table or intra-operatively with the plate already fixed to the bone.

FIG. 49 depicts a spring drill-guide 4900 in accordance with embodiments of the present disclosure. The spring drill-guide 4900 includes a drill bit 4901 extending from a proximal end 4902 to a distal end 4903. The drill bit 4901 includes an attachment portion 4905 at the proximal end 4902 that facilitates attachment of the drill bit 4901 to a drill (not shown). The attachment portion 4905 has a larger diameter than the remainder of the drill bit 4901. The spring drill-guide 4900 further includes a compression element such as, for example, a helical spring 4904 and a sleeve 4906 coupled to a distal end of the helical spring 4904. The spring 4904 extends from a distal end of the attachment portion to the distal end 4903 of the drill bit 4901. The spring 4904 has a diameter smaller than a diameter of the attachment portion such that an axial force applied to the spring 4904 compresses the spring. The drill bit 4901 extends through the helical spring 4904 and the sleeve 4906 such that in an initial state where the spring is not compressed, the distal end 4903 extends through the sleeve 4906. The sleeve 4906 is configured such that when the spring drill-guide 4900 is inserted into one of the openings (e.g., 4310) of the targeting guide (e.g., 4300), the sleeve 4906 remains in the opening as the drill bit 4901 is inserted further through the opening, thus resulting in the compression of the spring 4904. FIG. 51A depicts the initial insertion of the spring drill-guide 4900 into one of the openings of the targeting guide. FIG. 51B depicts the further insertion of the drill bit 4901 through the opening while the sleeve 4906 remains in the opening and the spring 4904 is compressed. Because the sleeve 4906 is part of the spring drill-guide 4900, the number of steps and hands needed to place screws using a targeting guide is advantageously reduced. Furthermore, the spring drill-guide 4900 also advantageously modulates the speed at which the drill bit 4901 is plunged into the bone (due to the spring forces), thus reducing the likelihood of a surgeon accidentally plunging the drill bit too far. FIG. 50 illustrates a targeting sleeve 5000 which may alternatively be used with a conventional drill bit (not shown) and any one of the targeting guides 4300, 4400, 4500.

FIG. 52 depicts a cross-sectional view of a targeting guide 5200 coupled to the volar distal radius plate 110 in accordance with embodiments of the present disclosure. FIG. 53A is a side view of the targeting guide 5200. Similar to the targeting guide 4300 discussed above, the targeting guide 5200 includes a pair of protrusions 5202 (one shown in FIG. 52) that extend towards the distal radius plate to serve as standoff which provide improved rigidity when the targeting guide is coupled to the distal radius plate. Also shown in FIGS. 52 and 53A is a retaining arm 5204 disposed adjacent to a counterbored hole 5205 and extending downward from the targeting guide 5200 and configured to be inserted in one of the openings 120 to couple the targeting guide to the distal radius plate 110. Similar to the targeting guides discussed above, the targeting guide 5200 may include a plurality of cannulated openings 5213 which correspond to each of the respective openings 120 in the plate 110. The openings 5213 may be configured in order to drill the pilot holes at desired trajectories for each opening 120, and subsequently receive the respective fasteners at the desired trajectories. The targeting guide 5200 may also include a plurality of k-wire openings 5214 which match with the k-wire openings 124 in the plate 110.

Although only one retaining arm is shown, the targeting guide 5200 may alternatively include a plurality of arms. In

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some embodiments, the retaining arm 5204 is configured to be deflected radially inward by the walls of the opening 120 during insertion of the retaining arm 5204 into the opening 120 and subsequently return to its initial undeflected position after the retaining arm 5204 is pushed beyond a portion 5208 of the opening 120 having a reduced diameter. In some embodiments, the retaining arm 5204 includes a detent 5210 configured to receive and mate with the portion 5208 of the opening 120 and a flange 5212 disposed distal to the detent 5210 and extending radially outward from the retaining arm 5204.

FIG. 53B depicts a side view of a cam screw 5300 for use with the targeting guide 5200 in accordance with embodiments of the present disclosure. FIG. 53C depicts a dowel pin 5500 for use with the targeting guide 5200 in accordance with embodiments of the present disclosure. To lock the targeting guide 5200 to the plate 110, a cam screw 5300 is inserted into the counterbored hole 5205 of the targeting guide 5200. In some embodiments, the cam screw 5300 includes a head 5302 and a shaft 5304 extending from the head 5302. Similar to the retaining arm 5204, the shaft 5304 includes a radially outwardly extending flange 5306 extending from a distal end 5308 of the shaft 5304. As shown in FIG. 53B, the head 5302 includes a cam surface 5310 that extends radially from the head 5302 in a direction opposite to that of the flange 5306. During initial placement of the targeting guide 5200 onto the plate 110, the cam screw 5300 is disposed in the counterbored hole 5205 such that the flange 5306 is adjacent the flange 5212 of the retaining arm 5204, as shown in FIG. 54A. In some embodiments, the flange 5212 includes a detent 5406 configured to receive the flange 5306 of the cam screw 5300, as shown in FIG. 54B. As a result, the retaining arm 5204 is easily inserted into the opening 120 of the plate. To lock the targeting guide 5200 to the plate 110, the cam screw 5300 is rotated so that the flange 5306 is disposed on an opposite side of the opening 120 beneath the portion 4608 of the opening 120 having a reduced diameter, as shown in FIG. 54C. The extension of the flange 5306 of the cam screw 5300 and the flange 5212 of the retaining arm 5204 in substantially opposite directions beneath the portion 4608 of the opening 120 lock the targeting guide 5200 to the plate 110 in a similar manner to the collet 4604 described above.

As shown in FIG. 55, in some embodiments, the cam screw 5300 may be retained in the counterbored hole 5205 by a dowel pin 5502. The dowel pin 5502 extends through a hole 5312 (shown in FIG. 53A) that extends transversely to the counterbored hole 5205. To prevent the cam screw 5300 from backing out, the dowel pin 5502 extends above the cam screw 5300. The cam screw 5300 also includes an engagement feature 5315 configured to be engaged by a driver to rotate the cam screw 5300.

Once the targeting guide 5200 is locked onto the plate 110, holes may be drilled into a bone to which the plate 110 is to be coupled using, for example, the spring drill-guide 4900. FIG. 56 illustrates a targeting sleeve 5600 which may alternatively be used with a conventional drill bit (not shown) and any one of the targeting guides 4300, 4400, 4500, 5200. In some embodiments, the targeting sleeve 5600 includes a handle portion 5602 configured to be grasped by a user's hand and a guide portion 5604 configured to allow a drill bit to pass therethrough.

Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention

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covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that all ranges broadly recited in this document include within their scope all narrower ranges which fall within the broader ranges. It is also intended that the components of the various devices disclosed above may be combined or modified in any suitable configuration.

What is claimed is:

1. A method for positioning a volar distal plate on bone, comprising the steps of:

providing a targeting guide having:

a body having an upper portion configured to be disposed on a head portion of the volar distal radius plate and a lower portion configured to couple the targeting guide to the volar distal radius plate;

a counterbored hole extending through the body;

a retaining arm disposed adjacent the counterbored hole and extending from the lower portion of the body and into an opening in the volar distal radius plate, wherein the retaining arm includes a detent configured to receive and mate with a portion of the opening of the volar distal radius plate and a flange extending radially outward from a distal end of the retaining arm; and

a cam screw disposed through the counterbored hole, wherein the cam screw includes a head and a shaft extending from the head, wherein the shaft includes a radially outwardly extending flange extending from a distal end of the shaft;

drilling pilot holes and inserting fasteners at desired trajectories through a plurality of cannulated openings and through corresponding respective openings of the volar distal radius plate;

securing the volar distal radius plate to bone.

2. The method of claim 1, further comprising:

a plurality of k-wire openings corresponding to a plurality of k-wire openings in the volar distal radius plate.

3. The method of claim 1, wherein in the unlocked position, the radially outwardly extending flange of the cam screw is disposed adjacent the flange of the retaining arm, and wherein in the locked position, the radially outwardly extending flange of the cam screw extends in a substantially opposite direction as the flange of the retaining arm.

4. The method of claim 1, further comprising:

a dowel pin extending through an opening formed in the targeting guide, wherein the opening extends transverse to the counterbored hole, and wherein the dowel pin extends above the cam screw to prevent the cam screw from backing out.

5. The method of claim 1, wherein the body further includes a neck portion extending between the upper portion and the lower portion.

6. The method of claim 1, wherein the upper portion is adjacent the lower portion.

7. The method of claim 1, wherein the body does not surround k-wire holes in the volar distal radius plate.

8. A method for stabilizing a bone, the method comprising:

providing a system comprising:

a bone plate, the bone plate comprising an elongated portion extending along a longitudinal axis, an enlarged head portion, a transition region connecting the elongated portion to the enlarged head portion, and a plurality of through holes extending through the elongated portion and the head portion;

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a targeting guide configured to be coupled to the bone plate, comprising:

a body having an upper portion configured to be disposed on the enlarged head portion of the bone plate and a lower portion configured to couple the targeting guide to the bone plate;

a plurality of cannulated openings corresponding to respective through holes of the bone plate, wherein the cannulated openings are configured to facilitate drilling pilot holes and inserting fasteners at desired trajectories through the through holes of the bone plate;

a counterbored hole extending through the body;

a retaining arm disposed adjacent the counterbored hole and extending from the lower portion of the body and into an opening in the bone plate, wherein the retaining arm includes a detent configured to receive and mate with a portion of the opening of the bone plate and a flange extending radially outward from a distal end of the retaining arm; and

a cam screw disposed through the counterbored hole, wherein the cam screw includes a head and a shaft extending from the head, wherein the shaft includes a radially outwardly extending flange extending from a distal end of the shaft,

a spring drill-guide configured to be coupled to a drill and comprising:

a drill bit extending from a proximal end to a distal end, a compression element; and

a sleeve coupled to a distal end of the compression element, wherein the drill bit extends through the compression element and the sleeve such that, in an initial state where the compression element is not compressed, only the distal end of the drill bit extends through the sleeve,

using the system to stabilize the bone.

9. The method of claim 8, wherein the compression element is a helical spring.

10. The method of claim 8, wherein the targeting guide further comprises:

a plurality of k-wire openings corresponding to a plurality of k-wire openings in the bone plate.

11. The method of claim 8, wherein in the unlocked position, the radially outwardly extending flange of the cam screw is disposed adjacent the flange of the retaining arm, and wherein in the locked position, the radially outwardly extending flange of the cam screw extends in a substantially opposite direction as the flange of the retaining arm.

12. The method of claim 8, wherein the targeting guide further comprises:

a dowel pin extending through an opening formed in the targeting guide, wherein the opening extends transverse to the counterbored hole, and wherein the dowel pin extends above the cam screw to prevent the cam screw from backing out.

13. The method of claim 8, wherein the body of the targeting guide further includes a neck portion extending between the upper portion and the lower portion.

14. The method of claim 8, wherein the upper portion of the targeting guide is adjacent the lower portion of the targeting guide.

15. The method of claim 14, wherein the body of the targeting guide does not surround k-wire holes in the bone plate.

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